INTEGRATED REPORT - 3

Requirements Analysis, Data Design and ETL Design using SSIS

Design and Implementation of a Data Warehouse for a Retail Store with Store-level Data

ISTM 637 - Data Warehousing
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Section 1: Introduction

1.1 About Dominick's Fine Foods

Dominick's Finer Foods was a prominent Chicago-area supermarket chain that operated approximately 100 stores throughout the metropolitan region during the late 1980s and early 1990s. Between 1989 and 1994, the University of Chicago Booth School of Business (James M. Kilts Center for Marketing) partnered with Dominick's to conduct store-level research on shelf management and pricing. Randomized experiments were conducted in over 25 product categories across the chain. The resulting dataset approximately 5 GB of store-level scanner data is one of the most comprehensive publicly available retail datasets (Kilts Center, 2013).

The objective of this project is to design and develop a data warehouse for DFF using their store-level data to help analyze sales performance, promotional effectiveness, and customer traffic patterns across all branches.

1.2 Project Design Specifications

Specification	Choice
Implementation Architecture	Hybrid Data Pipeline
Warehouse Architecture	Independent Data Marts
Modeling Scheme	Dimensional Modeling (Star Schema)
OLAP Style	HOLAP (Hybrid Online Analytical Processing)
Target Infrastructure	SQL Server 2016

1.3 Understanding of the Data

The DFF dataset consists of four categories of OLTP source files:

Category	File Type	Files	Total Rows	Description
General	Customer Count (CCOUNT)	1	327,045	Store traffic and coupon usage by store and department
General	Demographics (DEMO)	1	108	Store-level demographics from 1990 US Census
Category-Specific	Movement (Wxxx)	24	~134.9M	Weekly sales by UPC, store, and week
Category-Specific	UPC (UPCxxx)	28	~14,000	Product master data per category

Movement Files contain weekly sales data at the UPC-store-week grain. Key columns include UPC, STORE, WEEK, MOVE (units sold), QTY, PRICE, SALE (deal flag: B/C/S/blank), PROFIT, and OK (quality flag).

UPC Files contain product master data: COM_CODE (manufacturer), UPC, DESCRIP (product description), SIZE, CASE (pack quantity), and NITEM (item number).

DEMO.csv contains store-level demographic attributes derived from the 1990 U.S. Census, including income levels, education percentages, urban/suburban classification, pricing zone, and population density.

CCOUNT.csv contains daily customer transaction counts by store and department, including department-level traffic (Grocery, Dairy, Frozen, Meat, etc.) and coupon redemption counts.

1.4 Key Data Quality Issues

Issue	File(s)	Impact
DATE column corrupted	CCOUNT	Must use WEEK for time dimension instead
Dots '.' as missing values	CCOUNT, DEMO	Requires cleaning; columns read as text
SALE mostly NULL (~90%)	Movement	Promotion analysis limited to ~10% of rows
PRICE = 0 rows	Movement	Must filter to avoid division errors
Category not a column	Movement, UPC	Must derive from filename
WEEK ↔ calendar date mapping	All	Proprietary IDs; Week 1 ≈ Sept 14, 1989

1.5 DFF Hybrid ETL Pipeline Architecture

The following diagram illustrates the complete data flow from operational source files through the ETL pipeline to the data mart. The DFF source data consists of four entity types with the following relationships:

Entities and Relationships:

- **Movement Files (Wxxx) → UPC Files (UPCxxx)**: Many-to-one on UPC. Each movement record references one product.
- **Movement Files (Wxxx) → DEMO.csv**: Many-to-one on STORE. Each movement record belongs to one store.
- **Movement Files (Wxxx) → CCOUNT.csv**: Many-to-one on STORE + WEEK. Each movement week maps to customer traffic.
- Category code is implicit from the filename (e.g., wsdr.csv → SDR), not stored as a column.

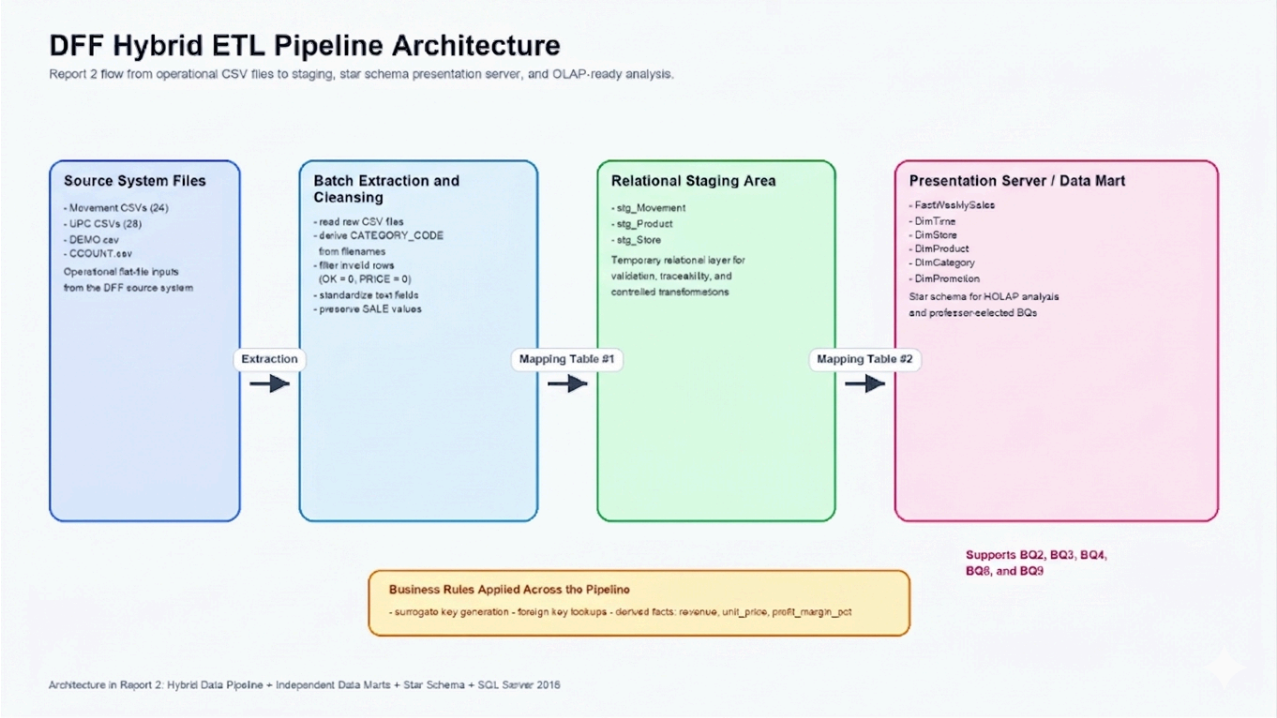

Figure 1 DFF Hybrid ETL Pipeline Architecture Source files → Staging → Data Mart

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Section 2: Subject Area Understanding

2.1 Literature Review

Our research draws on published studies that used the DFF dataset:

1. **Hoch, Drèze, and Purk (1994)** "EDLP, Hi-Lo, and Margin Arithmetic." Demonstrated that DFF's zone-based pricing strategy could increase profits by 3–5% through optimized price tiers, validating the importance of store-level pricing analysis.
2. **Chevalier, Kashyap, and Rossi (2003)** "Why Don't Prices Rise During Periods of Peak Demand?" Used DFF data to show that retailers use loss-leader pricing during peak seasons. Found that promotional pricing follows counter-cyclical patterns, directly relevant to our promotional lift analysis.
3. **Chintagunta (2002)** "Investigating Category Pricing Behavior at a Retail Chain." Analyzed brand-level pricing competition within DFF categories, supporting the need for product-level and brand-level granularity in the data warehouse.
4. **Mehta and Ma (2012)** "A High Dimensional Data Analysis Approach for Retail Data." Used DFF scanner data to demonstrate data mining techniques for category management and demand forecasting at retail chains.

5. **Montgomery (1997)** "Creating Micro-Marketing Pricing Strategies Using Supermarket Scanner Data." Used DFF data to develop store-level pricing models, reinforcing the importance of demographic segmentation in pricing decisions.

2.2 Business Questions (10 Questions with Difficulty Classification)

Based on our analysis of the DFF data, published research, and class slides, we developed 10 OLAP business questions. Each question is classified by difficulty (Easy, Medium, Hard) based on the complexity of the required OLAP operations. The professor selected 5 of these for implementation.

#	Business Question	Difficulty	OLAP Operation	Selected
BQ1	What were the total units sold per product category across the entire dataset period?	Easy	Roll-up	
BQ2	**What were the total weekly unit sales of Soft Drinks across all stores for each week?**	Easy	Roll-up	Selected
BQ3	**How do promotion weeks compare to non-promotion weeks in terms of sales volume?**	Easy	Slice	Selected
BQ4	**Which promotion type (B/C/S) generated the highest incremental unit sales lift in the Canned Soup category?**	Medium	Dice	Selected
BQ5	How did quarterly revenue for Frozen Entrees differ between urban and suburban stores?	Medium	Drill-down	
BQ6	How has the monthly market share of the top 5 Cereal brands changed over time?	Medium	Pivot + Ratio	
BQ7	Which 10 Beer products had the largest price variance across stores in the same week?	Medium	Dice + Rank	
BQ8	**Which stores fall into the top 25%, middle 50%, and bottom 25% of total Toothpaste revenue, and how do their demographics differ?**	Hard	NTILE + Drill-down	Selected
BQ9	**For each week, rank the top 10 Cracker products (UPCs) by unit sales and show**	Hard	RANK + LAG	Selected

	their week-over-week sales change.			
BQ10	For each pricing zone, what is the percentile rank of each store's average weekly Cigarette profit vs. the zone average?	Hard	PERCENT_RANK	

Difficulty Classification Rationale:

- **Easy:** Single-dimension aggregation, simple GROUP BY, one source file, no window functions
- **Medium:** Cross-dimensional joins, conditional aggregation, multiple source files, subqueries
- **Hard:** Window functions (RANK, NTILE, LAG, PERCENT_RANK), complex CTEs, multi-step analysis

2.3 Prioritization of Business Questions

Priority	BQ	Rationale
1	BQ4	Highest ROI understanding which promotion type works best per category directly impacts promotional budget allocation
2	BQ2	High-volume trend monitoring Soft Drinks is the largest category (17.7M rows); weekly trends enable demand forecasting
3	BQ8	Zone pricing validation Hoch et al. showed zone pricing could increase profits 3–5%; quartile analysis reveals miscalibrated zones
4	BQ3	Promotion ROI quantifies the aggregate sales lift from promotions, informing overall promotional strategy
5	BQ9	Product velocity monitoring weekly product rankings enable rapid assortment adjustments
6	BQ1	Category management foundation total volume determines shelf space and supply chain priorities
7	BQ5	Location strategy 30/70 urban-suburban split informs store openings and remodeling
8	BQ6	Competitive intelligence brand share trends help negotiate with manufacturers
9	BQ7	Pricing consistency audit large price variance may indicate pricing errors
10	BQ10	Profit optimization percentile ranking identifies top and bottom performers within pricing zones

2.4 Data Evidence Supporting Selected Business Questions

Our exploratory analysis of sample data confirms the feasibility of each selected BQ:

Promotion Effectiveness (supports BQ3, BQ4):

Category	% Rows Promoted	Avg Units (Promoted)	Avg Units (Non-Promoted)	Sales Lift
Soft Drinks	23.1%	59.25	4.37	13.6×
Canned Soup	7.5%	4.79	1.43	3.4×

Store Demographics (supports BQ8): 107 stores across 15 pricing zones, 30% urban / 70% suburban, weekly volume range 250–875 units.

Product Variety (supports BQ9): Crackers category has 305 unique UPCs across 3.6M records, providing sufficient data for weekly product ranking.

Section 3: Overview of Kimball's Methodology

The data warehouse logical design follows Kimball's bottom-up methodology for building independent data marts. This methodology is critical for this project because it ensures the data mart is driven by specific business questions rather than merely mirroring the OLTP source systems. The six steps are:

Step 1: Requirements Analysis. We gathered 10 business questions from stakeholder analysis and research on the DFF dataset (Section 2). The professor selected 5 BQs (BQ2, BQ3, BQ4, BQ8, BQ9) spanning four product categories (Soft Drinks, Canned Soup, Toothpaste, Crackers).

Step 2: Build the Bus Matrix. A matrix was created identifying the dimensions needed for each business process. FactWeeklySales supports all 5 BQs using five dimensions: DimTime, DimStore, DimProduct, DimCategory, and DimPromotion.

Step 3: Design Fact Tables. The fact table, FactWeeklySales, has a grain of one row per UPC × Store × Week with both base facts (units_sold, gross_profit) and derived facts (revenue, profit_margin_pct).

Step 4: Design Dimension Tables. Each dimension uses surrogate keys (4-byte INT), retains natural keys as attributes for traceability, and is fully denormalized (no snowflaking as emphasized in class, snowflaking slows browsing and degrades query performance).

Step 5: Feedback for the Design. The schema was validated against all 10 BQs; all are answerable. The 5 selected BQs were specifically verified to be fully supported (Section 4.3).

Step 6: Data Sourcing. Source files are identified, mapping tables are prepared (Sections 4.5–4.6), and the ETL plan is developed (Section 5).

Why is this methodology important for independent data marts? Kimball's methodology ensures each data mart is built incrementally and driven by real business requirements. The star schema structure provides high-performance query access, an intuitive format for business users, and ensures that new business processes can be added as separate data marts without modifying existing schemas.

Data Mart / Dimension Bus Matrix

Data Mart (Business Process)	DimTime	DimStore	DimProduct	DimCategory	DimPromotion	BQs Supported
FactWeeklySales	✓	✓	✓	✓	✓	BQ2, BQ3, BQ4, BQ8, BQ9

Section 4: Data Warehouse Logical Design (Star Schema Design)

4.1 Selected Business Questions

The professor selected 5 BQs from our list of 10 for implementation:

BQ	Question	Category	OLAP Op	Difficulty
BQ2	Total weekly unit sales of Soft Drinks across all stores	SDR	Roll-up	Easy
BQ3	Promotion vs non-promotion sales volume comparison	SDR	Slice	Easy
BQ4	Promotion type with highest sales lift in Canned Soup	CSO	Dice	Medium
BQ8	Store quartile tiers by Toothpaste revenue + demographics	TPA	NTILE + Drill-down	Hard
BQ9	Weekly top 10 Cracker products with week-over-week change	CRA	RANK + LAG	Hard

4.2 Star Schema Table Definitions

Implementation scope: The full DFF dataset contains 28 product categories (~14,000 UPCs, ~134.9M movement rows). This implementation is scoped to the 4 categories required by the 5 selected BQs: Soft Drinks (SDR), Canned Soup (CSO), Toothpaste (TPA), and Crackers (CRA), yielding ~3,112 UPCs and ~34.6M movement rows. The schema supports full-scale loading of all 28 categories without structural changes.

FactWeeklySales (Central Fact Table)

Column	Data Type	Description	Source	Additivity
sales_fact_id (PK)	INT	Surrogate key	Generated	
product_key (FK)	INT	→ DimProduct	Mapped from UPC	
store_key (FK)	INT	→ DimStore	Mapped from STORE	
time_key (FK)	INT	→ DimTime	Mapped from WEEK	
category_key (FK)	INT	→ DimCategory	Derived from filename	
promotion_key (FK)	INT	→ DimPromotion	Mapped from SALE	

units_sold	INT	Units moved/sold	MOVE	Additive
unit_price	DECIMAL(8,2)	Price per unit	PRICE / QTY	Non-additive
shelf_price	DECIMAL(8,2)	Listed shelf price	PRICE	Semi-additive
price_qty	INT	Quantity for listed price	QTY	Semi-additive
revenue	DECIMAL(12,2)	Derived: MOVE × (PRICE/QTY)	Computed	Additive
gross_profit	DECIMAL(10,2)	Gross profit	PROFIT	Additive
profit_margin_pct	DECIMAL(5,2)	Derived: (PROFIT/revenue)×100	Computed	Non-additive

Grain: One row per UPC × Store × Week | **Estimated rows:** ~34.6M (4 categories)

DimProduct

Column	Data Type	Description	Source
product_key (PK)	INT	Surrogate key	Generated (IDENTITY)
upc (Unique)	BIGINT	Universal Product Code	UPC from UPC files
description	VARCHAR(100)	Product description	DESCRIP (cleaned)
size	VARCHAR(30)	Package size	SIZE
case_pack	INT	Case pack quantity	CASE
commodity_code	INT	Manufacturer code	COM_CODE
item_number	BIGINT	Internal DFF item number	NITEM

Grain: One row per UPC | **Cardinality:** ~3,112 rows (4 categories)

DimStore

Column	Data Type	Description	Source
store_key (PK)	INT	Surrogate key	Generated (IDENTITY)
store_id (Unique)	INT	Original store number	STORE
store_name	VARCHAR(50)	Store name	NAME
city	VARCHAR(40)	City	CITY
zip_code	VARCHAR(10)	ZIP code	ZIP
zone	INT	Pricing zone	ZONE
is_urban	BIT	Urban flag (0/1)	URBAN
weekly_volume	INT	Avg weekly volume	WEEKVOL
avg_income	DECIMAL(10,2)	Avg area income (log)	INCOME
education_pct	DECIMAL(5,2)	% higher education	EDUC
poverty_pct	DECIMAL(5,2)	% below poverty	POVERTY
avg_household_size	DECIMAL(4,2)	Avg household size	HSIZEAVG
ethnic_diversity	DECIMAL(5,2)	Diversity index	ETHNIC
population_density	DECIMAL(10,2)	Pop density	DENSITY
price_tier	VARCHAR(10)	Low/Medium/High	Derived from PRICLOW/MED/HIGH
age_under_9_pct	DECIMAL(5,2)	% under 9	AGE9
age_over_60_pct	DECIMAL(5,2)	% over 60	AGE60
working_women_pct	DECIMAL(5,2)	% working women	WORKWOM

Grain: One row per store | **Cardinality:** ~107 rows

DimTime

Column	Data Type	Description	Source
time_key (PK)	INT	Surrogate key	Generated (IDENTITY)

week_id (Unique)	INT	DFF week number	WEEK
week_start_date	DATE	Start date	1989-09-14 + (week_id−1)×7
week_end_date	DATE	End date	week_start_date + 6
month	INT	Month (1–12)	Derived
month_name	VARCHAR(15)	Month name	Derived
quarter	INT	Quarter (1–4)	Derived
year	INT	Calendar year	Derived
fiscal_year	INT	DFF fiscal year	Derived
is_holiday_week	BIT	Holiday flag	Engineered

Grain: One row per week | **Cardinality:** ~400 rows

DimCategory

Column	Data Type	Source
category_key (PK)	INT	Generated (IDENTITY)
category_code (Unique)	CHAR(3)	Derived from filename
category_name	VARCHAR(30)	Mapped from codebook
department	VARCHAR(20)	Engineered

Cardinality: 28 rows (4 relevant: SDR, CSO, TPA, CRA)

DimPromotion

Column	Data Type	Source
promotion_key (PK)	INT	Generated (IDENTITY)
deal_code	CHAR(1)	SALE column
deal_type	VARCHAR(20)	Mapped
is_promoted	BIT	Derived

deal_code	deal_type	is_promoted
N	No Promotion	0
B	Bonus Buy	1
C	Coupon	1
S	Sale/Discount	1

Cardinality: 4 rows

4.3 Schema Justification How Each BQ Is Supported

BQ2 (Weekly SDR sales): Query FactWeeklySales joined to DimTime and DimCategory, GROUP BY week_id, SUM(units_sold). All required columns are present.

BQ3 (Promo vs non-promo): Query FactWeeklySales joined to DimPromotion and DimCategory, GROUP BY is_promoted, compare AVG(units_sold). deal_type in DimPromotion provides the segmentation.

BQ4 (Promo lift by type in CSO): Query FactWeeklySales joined to DimPromotion, filtered by DimCategory.category_code = 'CSO'. Compare each deal_type's AVG(units_sold) against the baseline (no promotion).

BQ8 (Store quartile tiers for TPA): Query FactWeeklySales joined to DimStore, filtered by DimCategory.category_code = 'TPA'. Use NTILE(4) on SUM(revenue) per store, then join demographics from DimStore.

BQ9 (Top 10 CRA products with WoW): Query FactWeeklySales joined to DimProduct and DimTime, filtered by 'CRA'. Use RANK() partitioned by week_id and LAG() partitioned by upc.

4.4 Star Schema Diagram

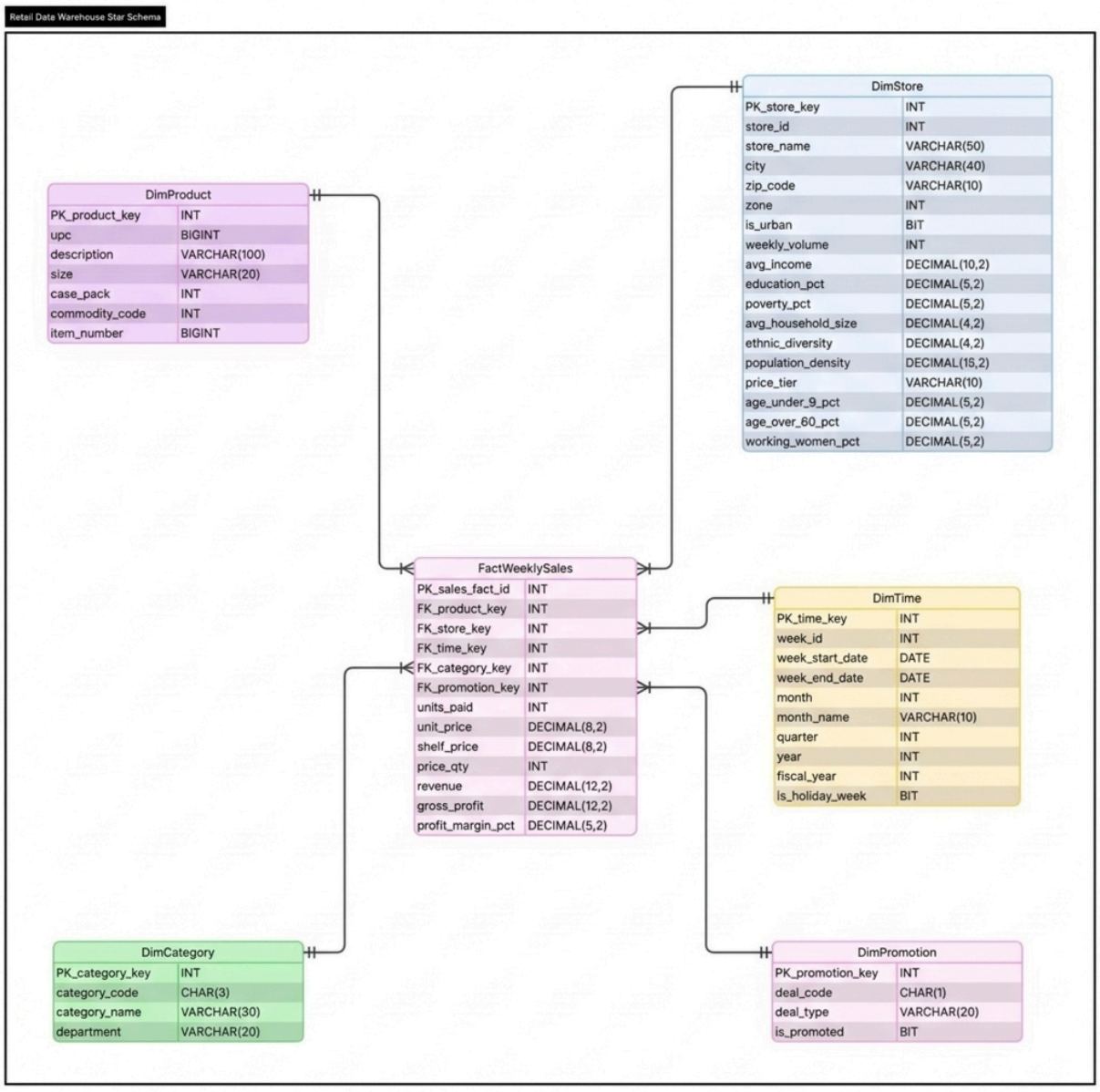


Figure 2.Star Schema ERD for the Dominick's Finer Foods (DFF) Data Warehouse FactWeeklySales with Five Dimensions

4.5 Mapping Table #1: Source Files to Staging Tables

Source File	Source Attribute	Mapping	Staging Table Type	Staging Table	Staging Attribute
wsdr.csv	UPC	Copy	Relation	stg_Movement_SDR	UPC
wsdr.csv	STORE	Copy	Relation	stg_Movement_SDR	STORE
wsdr.csv	WEEK	Copy	Relation	stg_Movement_SDR	WEEK
wsdr.csv	MOVE	Copy	Relation	stg_Movement_SDR	MOVE
wsdr.csv	QTY	Copy	Relation	stg_Movement_SDR	QTY
wsdr.csv	PRICE	Copy	Relation	stg_Movement_SDR	PRICE
wsdr.csv	SALE	Copy	Relation	stg_Movement_SDR	SALE
wsdr.csv	PROFIT	Copy	Relation	stg_Movement_SDR	PROFIT
wsdr.csv	OK	Copy	Relation	stg_Movement_SDR	OK
wsdr.csv	(filename)	Derive: extract 'SDR'	Relation	stg_Movement_SDR	CATEGORY_CODE
WCSO-Done.csv	UPC	Copy	Relation	stg_Movement_CSO	UPC
WCSO-Done.csv	STORE	Copy	Relation	stg_Movement_CSO	STORE
WCSO-Done.csv	WEEK	Copy	Relation	stg_Movement_CSO	WEEK
WCSO-Done.csv	MOVE	Copy	Relation	stg_Movement_CSO	MOVE

WCSO-Done.csv	QTY	Copy	Relation	stg_Movement_CSO	QTY
WCSO-Done.csv	PRICE	Copy	Relation	stg_Movement_CSO	PRICE
WCSO-Done.csv	SALE	Copy	Relation	stg_Movement_CSO	SALE
WCSO-Done.csv	PROFIT	Copy	Relation	stg_Movement_CSO	PROFIT
WCSO-Done.csv	OK	Copy	Relation	stg_Movement_CSO	OK
WCSO-Done.csv	(filename)	Derive: extract 'CSO'	Relation	stg_Movement_CSO	CATEGORY_CODE
WTPA_done.csv	UPC	Copy	Relation	stg_Movement_TPA	UPC
WTPA_done.csv	STORE	Copy	Relation	stg_Movement_TPA	STORE
WTPA_done.csv	WEEK	Copy	Relation	stg_Movement_TPA	WEEK
WTPA_done.csv	MOVE	Copy	Relation	stg_Movement_TPA	MOVE
WTPA_done.csv	QTY	Copy	Relation	stg_Movement_TPA	QTY
WTPA_done.csv	PRICE	Copy	Relation	stg_Movement_TPA	PRICE
WTPA_done.csv	SALE	Copy	Relation	stg_Movement_TPA	SALE
WTPA_done.csv	PROFIT	Copy	Relation	stg_Movement_TPA	PROFIT
WTPA_done.csv	OK	Copy	Relation	stg_Movement_TPA	OK
WTPA_done.csv	(filename)	Derive: extract 'TPA'	Relation	stg_Movement_TPA	CATEGORY_CODE

Done-WCRA.csv	UPC	Copy	Relation	stg_Movement_CRA	UPC
Done-WCRA.csv	STORE	Copy	Relation	stg_Movement_CRA	STORE
Done-WCRA.csv	WEEK	Copy	Relation	stg_Movement_CRA	WEEK
Done-WCRA.csv	MOVE	Copy	Relation	stg_Movement_CRA	MOVE
Done-WCRA.csv	QTY	Copy	Relation	stg_Movement_CRA	QTY
Done-WCRA.csv	PRICE	Copy	Relation	stg_Movement_CRA	PRICE
Done-WCRA.csv	SALE	Copy	Relation	stg_Movement_CRA	SALE
Done-WCRA.csv	PROFIT	Copy	Relation	stg_Movement_CRA	PROFIT
Done-WCRA.csv	OK	Copy	Relation	stg_Movement_CRA	OK
Done-WCRA.csv	(filename)	Derive: extract 'CRA'	Relation	stg_Movement_CRA	CATEGORY_CODE
UPCSDR.csv	COM_CODE	Copy	Relation	stg_Product_SDR	COM_CODE
UPCSDR.csv	UPC	Copy	Relation	stg_Product_SDR	UPC
UPCSDR.csv	DESCRIP	Copy	Relation	stg_Product_SDR	DESCRIP
UPCSDR.csv	SIZE	Copy	Relation	stg_Product_SDR	SIZE
UPCSDR.csv	CASE	Copy	Relation	stg_Product_SDR	CASE_PACK
UPCSDR.csv	NITEM	Copy	Relation	stg_Product_SDR	NITEM

UPCSDR.csv	(filename)	Derive: extract 'SDR'	Relation	stg_Product_SDR	CATEGORY_CODE
UPCCSO.csv	COM_CODE	Copy	Relation	stg_Product_CSO	COM_CODE
UPCCSO.csv	UPC	Copy	Relation	stg_Product_CSO	UPC
UPCCSO.csv	DESCRIP	Copy	Relation	stg_Product_CSO	DESCRIP
UPCCSO.csv	SIZE	Copy	Relation	stg_Product_CSO	SIZE
UPCCSO.csv	CASE	Copy	Relation	stg_Product_CSO	CASE_PACK
UPCCSO.csv	NITEM	Copy	Relation	stg_Product_CSO	NITEM
UPCCSO.csv	(filename)	Derive: extract 'CSO'	Relation	stg_Product_CSO	CATEGORY_CODE
UPCTPA.csv	COM_CODE	Copy	Relation	stg_Product_TPA	COM_CODE
UPCTPA.csv	UPC	Copy	Relation	stg_Product_TPA	UPC
UPCTPA.csv	DESCRIP	Copy	Relation	stg_Product_TPA	DESCRIP
UPCTPA.csv	SIZE	Copy	Relation	stg_Product_TPA	SIZE
UPCTPA.csv	CASE	Copy	Relation	stg_Product_TPA	CASE_PACK
UPCTPA.csv	NITEM	Copy	Relation	stg_Product_TPA	NITEM
UPCTPA.csv	(filename)	Derive: extract 'TPA'	Relation	stg_Product_TPA	CATEGORY_CODE
UPCCRA.csv	COM_CODE	Copy	Relation	stg_Product_CRA	COM_CODE

UPCCRA.csv	UPC	Copy	Relation	stg_Product_CRA	UPC
UPCCRA.csv	DESCRIP	Copy	Relation	stg_Product_CRA	DESCRIP
UPCCRA.csv	SIZE	Copy	Relation	stg_Product_CRA	SIZE
UPCCRA.csv	CASE	Copy	Relation	stg_Product_CRA	CASE_PACK
UPCCRA.csv	NITEM	Copy	Relation	stg_Product_CRA	NITEM
UPCCRA.csv	(filename)	Derive: extract 'CRA'	Relation	stg_Product_CRA	CATEGORY_CODE
DEMO.csv	STORE	Copy	Relation	stg_Store	STORE
DEMO.csv	NAME	Copy	Relation	stg_Store	NAME
DEMO.csv	CITY	Copy	Relation	stg_Store	CITY
DEMO.csv	ZIP	Copy	Relation	stg_Store	ZIP
DEMO.csv	ZONE	Copy	Relation	stg_Store	ZONE
DEMO.csv	URBAN	Copy	Relation	stg_Store	URBAN
DEMO.csv	WEEKVOL	Copy	Relation	stg_Store	WEEKVOL
DEMO.csv	INCOME	Copy	Relation	stg_Store	INCOME
DEMO.csv	EDUC	Copy	Relation	stg_Store	EDUC
DEMO.csv	POVERTY	Copy	Relation	stg_Store	POVERTY
DEMO.csv	HSIZEAVG	Copy	Relation	stg_Store	HSIZEAVG
DEMO.csv	ETHNIC	Copy	Relation	stg_Store	ETHNIC
DEMO.csv	DENSITY	Copy	Relation	stg_Store	DENSITY
DEMO.csv	PRICLOW	Copy	Relation	stg_Store	PRICLOW
DEMO.csv	PRICMED	Copy	Relation	stg_Store	PRICMED
DEMO.csv	PRICHIGH	Copy	Relation	stg_Store	PRICHIGH

DEMO.csv	AGE9	Copy	Relation	stg_Store	AGE9
DEMO.csv	AGE60	Copy	Relation	stg_Store	AGE60
DEMO.csv	WORKWOM	Copy	Relation	stg_Store	WORKWOM

4.6 Mapping Table #2: Staging Tables to Data Mart Tables

Staging Table	Staging Attribute	Mapping	DM Table Type	DM Table	DM Attribute
stg_Movement_SDR/CSO/TPA/CRA	MOVE	Copy	Fact	FactWeeklySales	units_sold
stg_Movement_SDR/CSO/TPA/CRA	PRICE	Copy	Fact	FactWeeklySales	shelf_price
stg_Movement_SDR/CSO/TPA/CRA	QTY	Copy	Fact	FactWeeklySales	price_qty
stg_Movement_SDR/CSO/TPA/CRA	PRICE, QTY	Transform: PRICE/QTY	Fact	FactWeeklySales	unit_price
stg_Movement_SDR/CSO/TPA/CRA	MOVE, PRICE, QTY	Transform: MOVE*(PRICE/QTY)	Fact	FactWeeklySales	revenue
stg_Movement_SDR/CSO/TPA/CRA	PROFIT	Copy	Fact	FactWeeklySales	gross_profit
stg_Movement_SDR/CSO/TPA/CRA	PROFIT, revenue	Transform: (PROFIT/revenue)*100	Fact	FactWeeklySales	profit_margin_pct
stg_Movement_SDR/CSO/TPA/CRA	UPC	Lookup -> DimProduct	Fact	FactWeeklySales	product_key
stg_Movement_SDR/CSO/TPA/CRA	STORE	Lookup -> DimStore	Fact	FactWeeklySales	store_key
stg_Movement_SDR/CSO/TPA/CRA	WEEK	Lookup -> DimTime	Fact	FactWeeklySales	time_key

stg_Movement_SDR/CSO/TPA/CRA	CATEGORY_CODE	Lookup -> DimCategory	Fact	FactWeeklySales	category_key
stg_Movement_SDR/CSO/TPA/CRA	SALE	Lookup -> DimPromotion	Fact	FactWeeklySales	promotion_key
stg_Product_SDR/CSO/TPA/CRA	COM_CODE	Copy	Dimension	DimProduct	commodity_code
stg_Product_SDR/CSO/TPA/CRA	UPC	Copy	Dimension	DimProduct	upc
stg_Product_SDR/CSO/TPA/CRA	DESCRIP	Transform: strip #, ~	Dimension	DimProduct	description
stg_Product_SDR/CSO/TPA/CRA	SIZE	Copy	Dimension	DimProduct	size
stg_Product_SDR/CSO/TPA/CRA	CASE_PACK	Copy	Dimension	DimProduct	case_pack
stg_Product_SDR/CSO/TPA/CRA	NITEM	Copy	Dimension	DimProduct	item_number
stg_Store	STORE	Copy (CAST to INT)	Dimension	DimStore	store_id
stg_Store	NAME	Transform: ISNULL -> 'UNKNOWN'	Dimension	DimStore	store_name
stg_Store	CITY	Transform: ISNULL -> 'UNKNOWN'	Dimension	DimStore	city
stg_Store	ZIP	Copy (CAST to INT)	Dimension	DimStore	zip_code
stg_Store	ZONE	Copy (CAST to INT)	Dimension	DimStore	zone
stg_Store	WEEKVOL	Copy (CAST to INT)	Dimension	DimStore	weekly_volume
stg_Store	URBAN	Copy (CAST to BIT)	Dimension	DimStore	is_urban

stg_Store	INCOME	Copy (CAST to DECIMAL)	Dimension	DimStore	avg_income
stg_Store	EDUC	Copy (CAST to DECIMAL)	Dimension	DimStore	education_pct
stg_Store	POVERTY	Copy (CAST to DECIMAL)	Dimension	DimStore	poverty_pct
stg_Store	HSIZEAVG	Copy (CAST to DECIMAL)	Dimension	DimStore	avg_household_size
stg_Store	ETHNIC	Copy (CAST to DECIMAL)	Dimension	DimStore	ethnic_diversity
stg_Store	DENSITY	Copy (CAST to DECIMAL)	Dimension	DimStore	population_density
stg_Store	AGE9	Copy (CAST to DECIMAL)	Dimension	DimStore	age_under_9_pct
stg_Store	AGE60	Copy (CAST to DECIMAL)	Dimension	DimStore	age_over_60_pct
stg_Store	WORKWOM	Copy (CAST to DECIMAL)	Dimension	DimStore	working_women_pct
stg_Store	PRICLOW, PRICMED, PRICHIGH	Transform: CASE WHEN	Dimension	DimStore	price_tier

4.7 Physical Design Plan

The physical design transforms the logical star schema into a deployable structure on SQL Server 2016. For the **data aggregate plan**, three summary tables are planned: agg_Weekly_Category_Sales (pre-aggregates units_sold and revenue by week and category to accelerate BQ2), agg_Store_Category_Revenue (aggregates total revenue by store and category for BQ8 quartile analysis), and agg_Weekly_Product_Sales (aggregates units_sold by week and product within a category for BQ9 ranking). These aggregate fact tables echo the original FactWeeklySales structure at reduced grain, following Kimball's guidance. For **indexing**, the FactWeeklySales table uses a clustered index on sales_fact_id (primary key) with nonclustered indexes on (time_key, store_key, product_key) for composite lookups and single-column nonclustered indexes on promotion_key and category_key for BQ-specific filtering. Dimension tables use unique nonclustered indexes on surrogate primary keys and additional nonclustered

indexes on frequently filtered columns (deal_code, is_promoted, price_tier, is_urban, department). During bulk ETL loads, indexes are dropped before loading and recreated afterward to avoid performance degradation.

For **data standardization**, all naming follows consistent conventions: dimension tables prefixed with "Dim", fact tables with "Fact", staging tables with "stg_", and aggregate tables with "agg_". Column names use snake_case throughout. All surrogate keys are 4-byte INT, monetary values use DECIMAL, and boolean flags use BIT. For **storage**, the large FactWeeklySales table (~34.6M rows, ~4-5 GB) will be horizontally partitioned by time_key (yearly partitions). Total estimated storage is 8-10 GB including indexes. The data mart architecture ensures new categories or business processes can be added without modifying existing tables.

Section 5: Development of the ETL Plan

This section develops the complete ETL plan following the framework discussed in class.

5.1 All Target Data in the Data Warehouse

The data warehouse consists of 6 tables:

Table	Type	Target Columns	Estimated Rows
DimCategory	Dimension	category_key, category_code, category_name, department	28
DimPromotion	Dimension	promotion_key, deal_code, deal_type, is_promoted	4
DimTime	Dimension	time_key, week_id, week_start_date, week_end_date, month, month_name, quarter, year, fiscal_year, is_holiday_week	~400
DimStore	Dimension	store_key, store_id, store_name, city, zip_code, zone, is_urban, weekly_volume, avg_income, education_pct, poverty_pct, avg_household_size, ethnic_diversity, population_density, price_tier, age_under_9_pct, age_over_60_pct, working_women_pct	~107
DimProduct	Dimension	product_key, upc, description, size, case_pack, commodity_code, item_number	~3,112
FactWeeklySales	Fact	sales_fact_id, product_key, store_key, time_key, category_key, promotion_key, units_sold, unit_price, shelf_price, price_qty, revenue,	~34.6M

		gross_profit, profit_margin_pct	

5.2 All Data Sources

Source	File	Location	Rows	Purpose
Movement – Soft Drinks	wsdr.csv	DFF data/Movement/WSDR/	17.7M	BQ2, BQ3
Movement – Canned Soup	WCSO-Done.csv	DFF data/Movement/WCSO/	7.0M	BQ4
Movement – Toothpaste	WTPA_done.csv	DFF data/Movement/WTPA/	6.3M	BQ8
Movement – Crackers	Done-WCRA.csv	DFF data/Movement/WCRA/	3.6M	BQ9
UPC – Soft Drinks	UPCSDR.csv	DFF data/UPC/	1,746	Product master
UPC – Canned Soup	UPCCSO.csv	DFF data/UPC/	453	Product master
UPC – Toothpaste	UPCTPA.csv	DFF data/UPC/	608	Product master
UPC – Crackers	UPCCRA.csv	DFF data/UPC/	305	Product master
Demographics	DEMO.csv	DFF data/Demographics/	108	Store attributes

5.3 Data Mappings

Two mapping tables have been prepared in Excel format (see Sections 4.5 and 4.6):

- **Mapping Table #1 (Source to Staging):** Documents how each source CSV column maps to a staging table attribute. All mappings at this stage are **Copy** operations, with one exception: CATEGORY_CODE is **Derived** from the source filename.
- **Mapping Table #2 (Staging to Data Mart):** Documents how each staging attribute maps to a data mart column. Mappings include **Copy**, **Transform** (computed columns like revenue, unit_price), and **Lookup** (surrogate key resolution via dimension table joins).

The complete Excel versions of both Mapping Table #1 and Mapping Table #2 are included in Appendix B.

5.4 Comprehensive Data Extraction Rules

Rule	Description
Extraction Technique	Capture of Static Data CSV files are static historical snapshots. No impact on source systems.
Extraction Frequency	One-time initial load (complete historical dataset, no incremental updates).
Time Window	Not applicable batch load of complete files during off-peak hours.
Source Identification	4 Movement CSVs + 4 UPC CSVs + DEMO.csv = 9 source files.
Quality Filter	Extract only rows where OK = 1 from Movement files (quality-validated observations).
Encoding	All CSV files require Windows-1252 (CP1252) encoding, not UTF-8.

File Format	Comma-delimited, no text qualifier, header row present in all files.
Tool	SSIS Data Flow Task with Flat File Source connection manager.

5.5 Data Transformation and Cleansing Rules

#	Rule	Source	Target	Type
T1	Add CATEGORY_CODE column derived from filename	Filename (SDR/CSO/TPA/CRA)	stg_Movement_*.CATEGORY_CODE	Derived value
T2	Replace NULL/blank SALE with 'N'	Movement.SALE	stg_Movement_*.SALE	Default value
T3	Filter rows where OK = 1	Movement.OK	(row selection)	Filter
T4	Filter rows where PRICE > 0	Movement.PRICE	(row selection)	Filter
T5	Strip '#' and '~' from product descriptions	UPC.DESCRIP	stg_Product_*.DESCRIP	String cleaning
T6	Remove '.' placeholder row from demographics	DEMO.STORE	stg_Store	Filter
T7	Replace NULL NAME/CITY with 'UNKNOWN'	DEMO.NAME, CITY	stg_Store.NAME, CITY	Default value
T8	Derive PRICE_TIER from binary flags	DEMO.PRICLOW/MED/HIGH	DimStore.price_tier	Conditional (CASE)
T9	CAST VARCHAR to proper data types	stg_Store	DimStore	Type conversion
T10	Compute unit_price = PRICE / QTY	Movement.PRICE, QTY	FactWeeklySales.unit_price	Derived value
T11	Compute revenue = MOVE × (PRICE / QTY)	Movement.MOVE, PRICE, QTY	FactWeeklySales.revenue	Derived value
T12	Compute profit_margin_pct = (PROFIT / revenue) × 100	Movement.PROFIT, revenue	FactWeeklySales.profit_margin_pct	Derived value
T13	Lookup product_key from DimProduct	Movement.UPC	FactWeeklySales.product_key	Surrogate key lookup
T14	Lookup store_key from DimStore	Movement.STORE	FactWeeklySales.store_key	Surrogate key lookup
T15	Lookup time_key from DimTime	Movement.WEEK	FactWeeklySales.time_key	Surrogate key lookup
T16	Lookup category_key from DimCategory	CATEGORY_CODE	FactWeeklySales.category_key	Surrogate key lookup
T17	Lookup promotion_key from DimPromotion	Movement.SALE	FactWeeklySales.promotion_key	Surrogate key lookup
T18	Generate week_start_date from week_id	DATEADD formula	DimTime.week_start_date	Date calculation
T19	Derive month, quarter, year from date	week_start_date	DimTime.month/quarter/year	Date parts

5.6 Plan for Aggregate Tables

Three pre-computed aggregate tables will be created to improve query performance:

Aggregate Table	Grain	Measures	Accelerates
agg_Weekly_Category_Sales	Week × Category	SUM(units_sold), SUM(revenue), AVG(unit_price)	BQ2
agg_Store_Category_Revenue	Store × Category	SUM(revenue), SUM(gross_profit)	BQ8
agg_Weekly_Product_Sales	Week × Product × Category	SUM(units_sold), RANK	BQ9

These aggregate tables echo the original FactWeeklySales structure at reduced grain, following Kimball's guidance that summary tables should mirror the base fact table's structure.

5.7 Organization of Data Staging Area

```
SQL Server 2016
├── [team1_staging_area]            ← Staging Database
│   ├── dbo.stg_Movement_SDR        (raw wsdr.csv data)
│   ├── dbo.stg_Movement_CSO        (raw WCSO-Done.csv data)
│   ├── dbo.stg_Movement_TPA        (raw WTPA_done.csv data)
│   ├── dbo.stg_Movement_CRA        (raw Done-WCRA.csv data)
│   ├── dbo.stg_Product_SDR         (raw UPCSDR.csv data)
│   ├── dbo.stg_Product_CSO         (raw UPCCSO.csv data)
│   ├── dbo.stg_Product_TPA         (raw UPCTPA.csv data)
│   ├── dbo.stg_Product_CRA         (raw UPCCRA.csv data)
│   ├── dbo.stg_Store               (raw DEMO.csv data)
│   └── dbo.tmp_Product_All  [TEMP] (UNION of 4 UPC tables)
│
└── [team1_dw_area]                 ← Data Mart (Presentation Server)
    ├── dbo.DimCategory             (28 rows)
    ├── dbo.DimPromotion            (4 rows)
    ├── dbo.DimTime                 (~400 rows)
    ├── dbo.DimStore                (~107 rows)
    ├── dbo.DimProduct              (~3,112 rows)
    └── dbo.FactWeeklySales         (~34.6M rows)
```

Temporary tables to be removed after loading:

Temp Table	Purpose	Remove After
tmp_Product_All	UNION of 4 UPC staging tables for loading DimProduct	Loading DimProduct

5.8 Procedures for All Data Extractions and Loadings

The ETL is implemented through **three SSIS packages** executed sequentially:

Package 1: `01_Extract_to_Staging.dtsx` Extracts all 9 CSV source files into staging tables using Data Flow Tasks. Each task uses a Flat File Source (encoding 1252, comma-delimited) connected to an OLE DB Destination targeting the staging database.

Package 2: `02_Transform_Staging.dtsx` Executes transformations T1, T2, and T5–T7 in the staging area using Execute SQL Tasks. This includes adding CATEGORY_CODE columns, replacing NULL values, cleaning descriptions, and creating the tmp_Product_All UNION table.

Package 3: `03_Load_DataMart.dtsx` Creates and populates all dimension and fact tables in the data mart using Execute SQL Tasks. Dimensions are loaded first (DimCategory, DimPromotion, DimTime, DimStore, DimProduct), then the fact table (FactWeeklySales). Completes with DROP TABLE for temporary tables.

5.9 ETL for Dimension Table

Dimensions are loaded **before** the fact table because fact table foreign keys reference dimension surrogate keys. The loading order ensures referential integrity.

DimCategory (28 rows): Loaded via Execute SQL Task with hardcoded INSERT statements. All 28 product categories and their department groupings are inserted directly. No staging table needed.

DimPromotion (4 rows): Loaded via Execute SQL Task with hardcoded INSERT statements. Maps the four possible SALE values (N, B, C, S) to descriptive labels and an is_promoted flag.

DimTime (~400 rows): Generated via Execute SQL Task using a recursive CTE. Week 1 corresponds to September 14, 1989 (per the DFF codebook). Each subsequent week_id adds 7 days via DATEADD. Month, quarter, year, and month_name are derived using DATEPART and DATENAME functions.

DimStore (~107 rows): Loaded from stg_Store using an INSERT INTO...SELECT statement. Transformations applied during loading: CAST string columns to proper types, derive price_tier using CASE WHEN on the three binary pricing flags (PRICLOW/PRICMED/PRICHIGH), and filter out invalid rows (STORE = '.').

DimProduct (~3,112 rows): Loaded from tmp_Product_All (UNION of 4 UPC staging tables) using INSERT INTO...SELECT. Product descriptions are cleaned during the staging transformation phase (strip # and ~ characters).

5.10 ETL for Fact Table

FactWeeklySales (~34.6M rows): This is the largest and most complex load. An INSERT INTO...SELECT statement unions all four movement staging tables (filtered to OK=1 and PRICE>0), then JOINs to all five dimension tables to resolve surrogate keys:

- DimProduct: JOIN on UPC to get product_key
- DimStore: JOIN on STORE to get store_key
- DimTime: JOIN on WEEK to get time_key
- DimCategory: JOIN on CATEGORY_CODE to get category_key
- DimPromotion: INNER JOIN on SALE to get promotion_key

Three derived columns are computed during loading: unit_price (PRICE/QTY), revenue (MOVE × unit_price), and profit_margin_pct (PROFIT/revenue × 100), with NULL guards for division by zero.

Section 6: Implementation of the ETL Plan

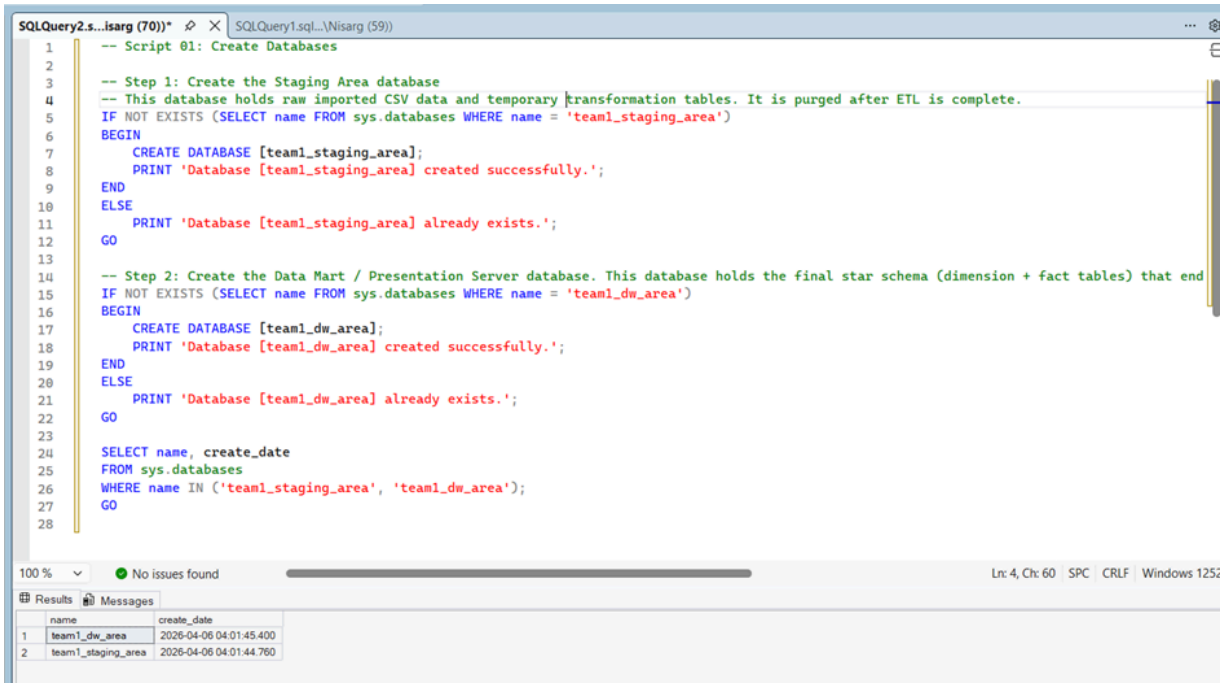
This section documents the actual implementation of the ETL plan using SSIS (SQL Server Integration Services) and SSMS (SQL Server Management Studio) on SQL Server 2016. All SQL statements used for ETL operations are included. Snapshots of before-and-after table contents and ETL runs are provided to demonstrate the effectiveness of each ETL step.

6.1 Database Creation

The two databases were created using the following SQL:

```
CREATE DATABASE [team1_staging_area];
CREATE DATABASE [team1_dw_area];
```

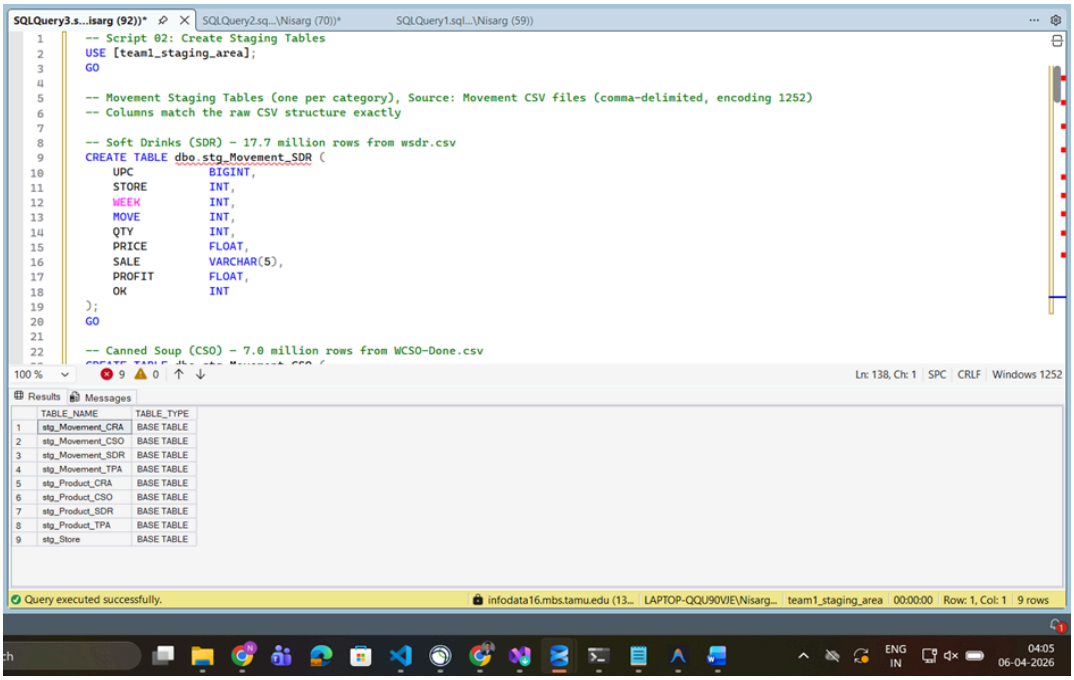
Screenshot 1: SSMS Object Explorer showing both databases

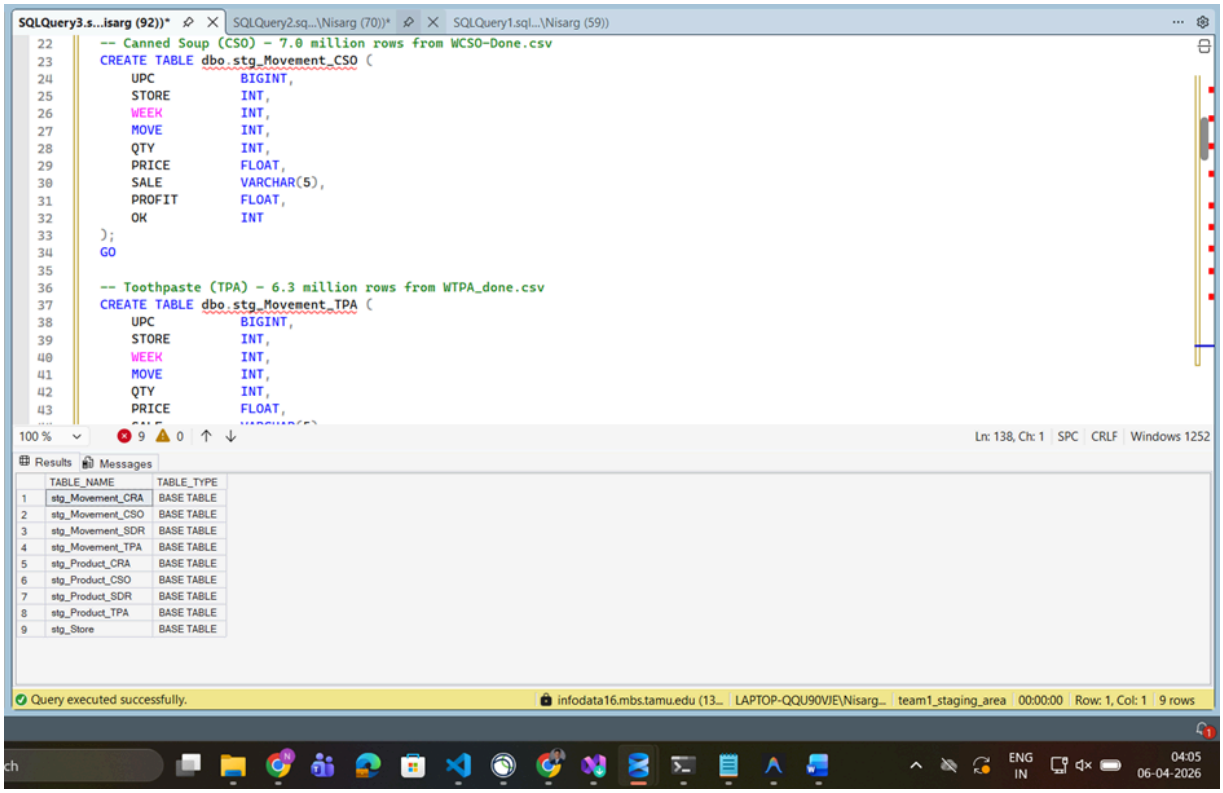


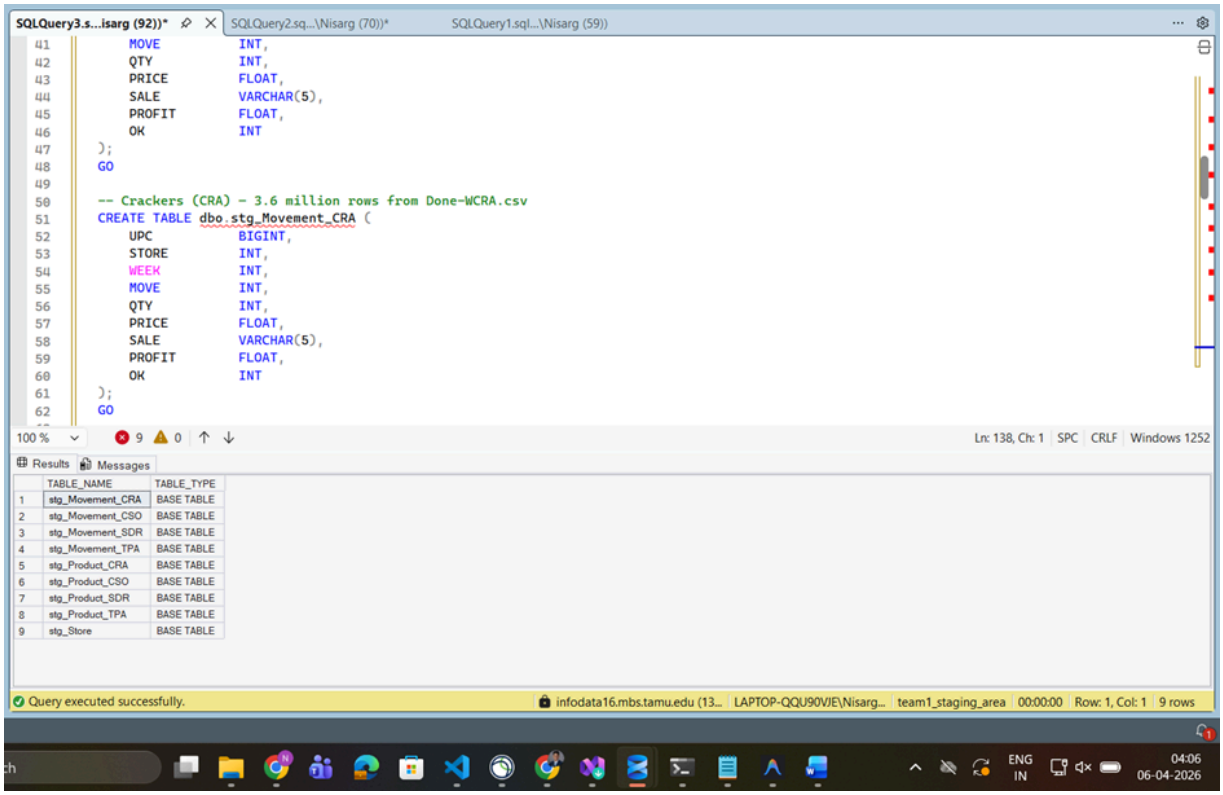
6.2 Staging Table Creation

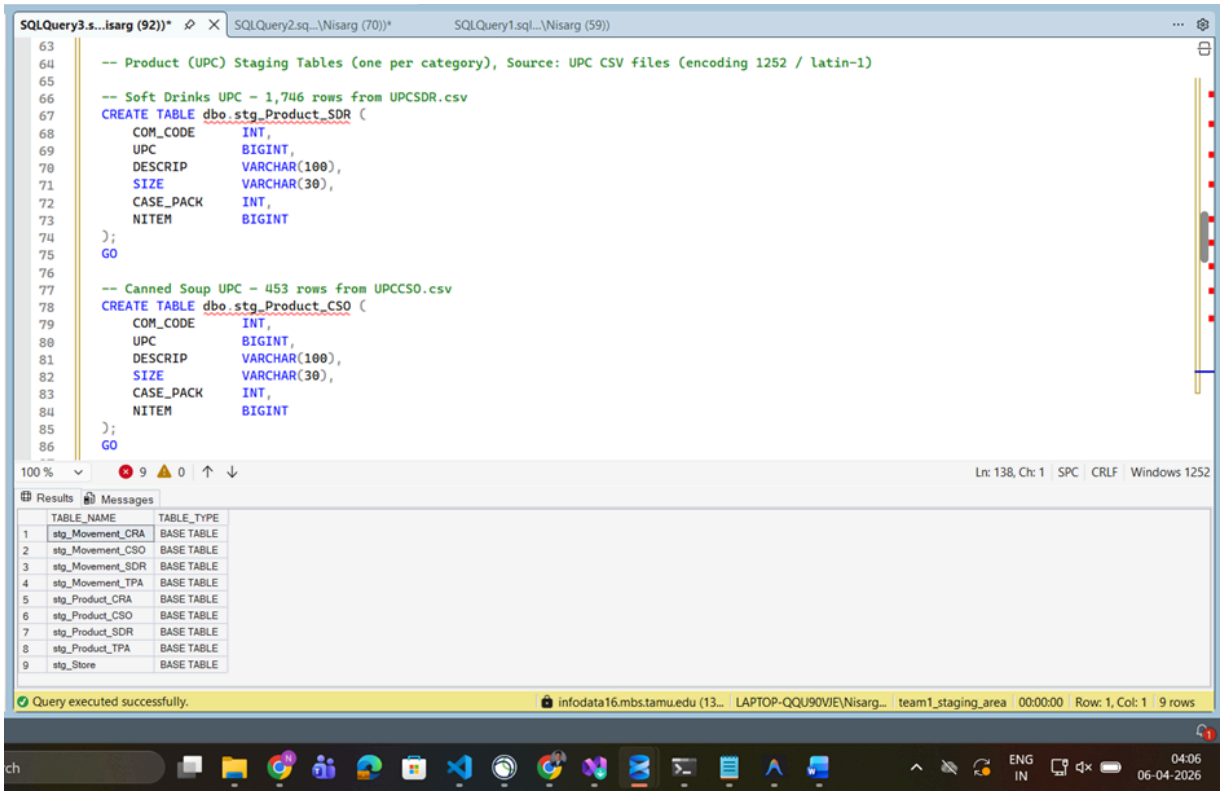
All 9 staging tables and 1 temporary table were created in team1_staging_area. The complete SQL statements are in Appendix A (02_create_staging_tables.sql).

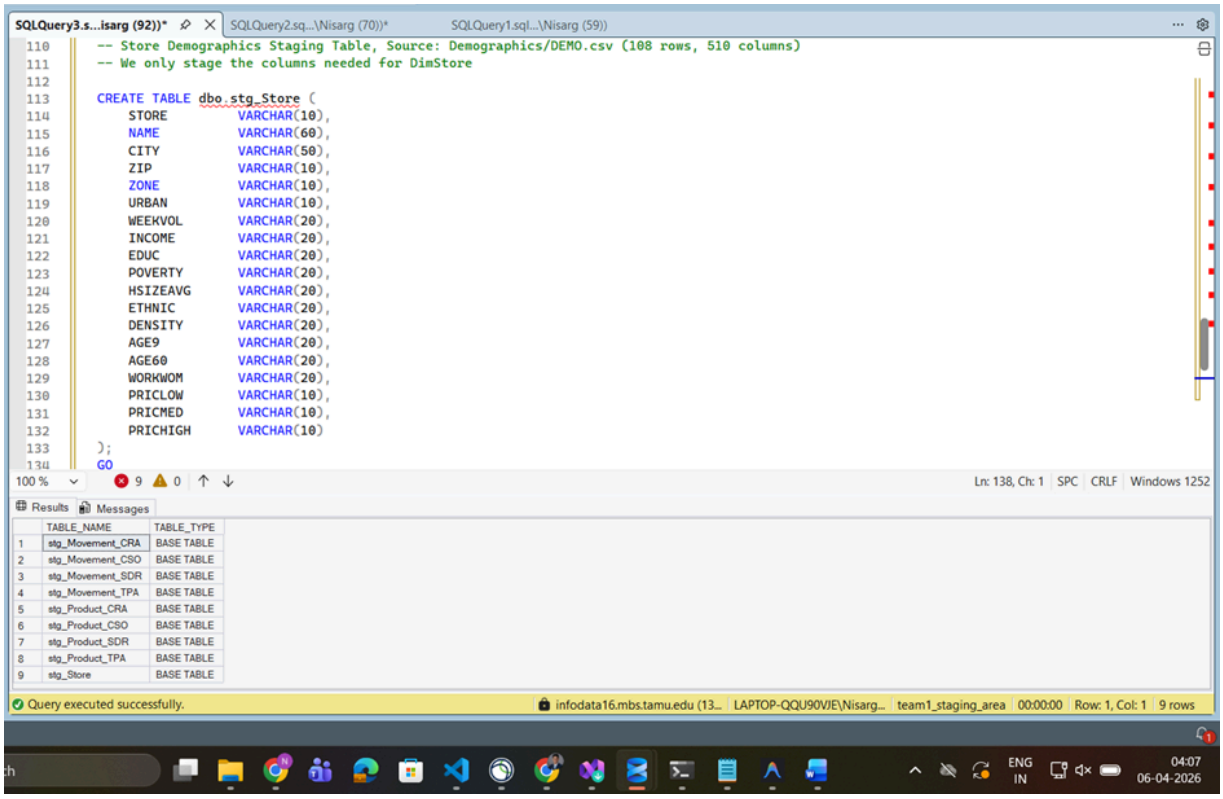
Screenshot 2: SSMS Object Explorer staging tables listed

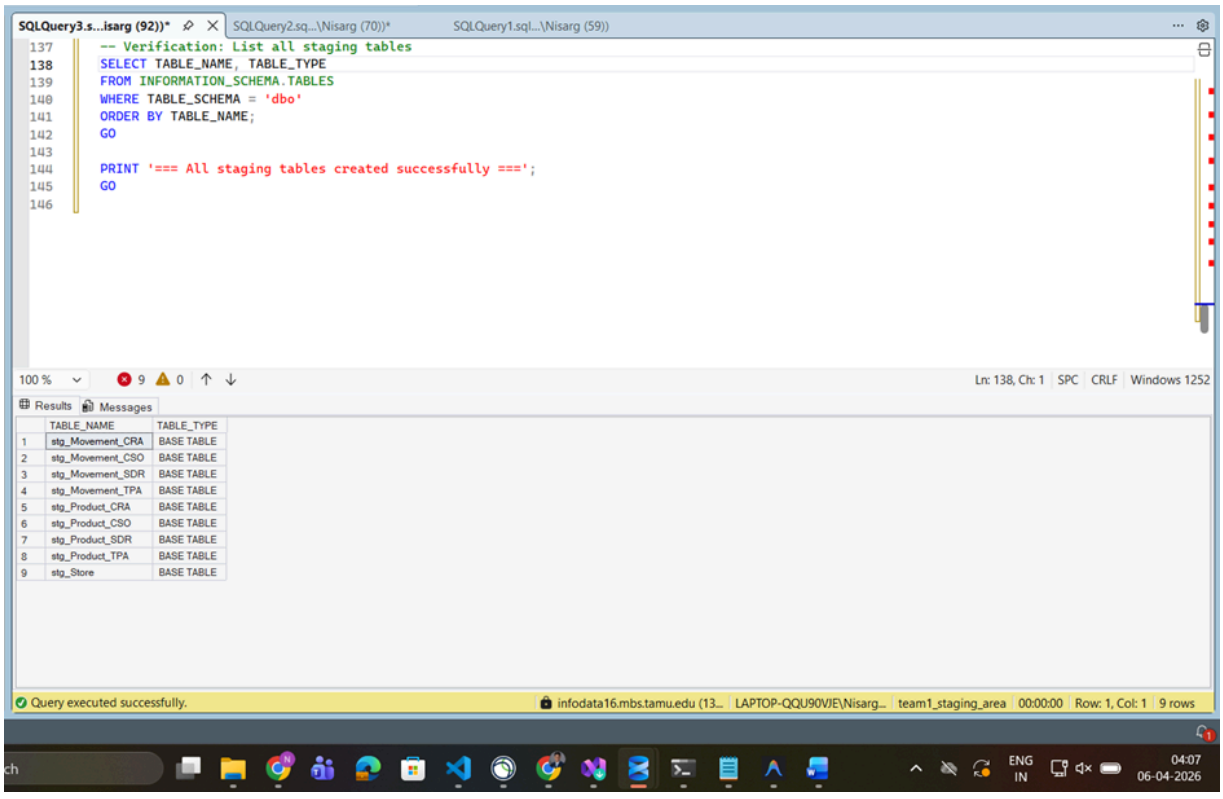

SQLQuery3.s...isarg (92))*
SQLQuery2.sq...\Nisarg (70))*
SQLQuery1.sql...\Nisarg (59))
-- Script 02: Create Staging Tables
USE [team1_staging_area];
GO
-- Movement Staging Tables (one per category), Source: Movement CSV files (comma-delimited, encoding 1252)
-- Columns match the raw CSV structure exactly
-- Soft Drinks (SDR) - 17.7 million rows from wsdr.csv
CREATE TABLE dbo.stg_Movement_SDR (
UPC BIGINT,
STORE INT,
WEEK INT,
MOVE INT,
QTY INT,
PRICE FLOAT,
SALE VARCHAR(5),
PROFIT FLOAT,
OK INT
);
GO
-- Canned Soup (CSO) - 7.0 million rows from WCSO-Done.csv
100 %
Ln: 138, Ch: 1
SPC
CRLF
Windows 1252
Results
Messages
TABLE_NAME
TABLE_TYPE
1 stg_Movement_CRA BASE TABLE
2 stg_Movement_CSO BASE TABLE
3 stg_Movement_SDR BASE TABLE
4 stg_Movement_TPA BASE TABLE
5 stg_Product_CRA BASE TABLE
6 stg_Product_CSO BASE TABLE
7 stg_Product_SDR BASE TABLE
8 stg_Product_TPA BASE TABLE
9 stg_Store BASE TABLE
Query executed successfully.
infodata16.mbs.tamu.edu (13...
LAPTOP-QQU90VJE\Nisarg...
team1_staging_area
00:00:00
Row: 1, Col: 1
9 rows
ENG
IN
04:05
06-04-2026


SQLQuery3.s...isarg (92))*
SQLQuery2.sq...\Nisarg (70))*
SQLQuery1.sql...\Nisarg (59))
-- Canned Soup (CSO) - 7.0 million rows from WCSO-Done.csv
CREATE TABLE dbo.stg_Movement_CSO (
UPC BIGINT,
STORE INT,
WEEK INT,
MOVE INT,
QTY INT,
PRICE FLOAT,
SALE VARCHAR(5),
PROFIT FLOAT,
OK INT
);
GO
-- Toothpaste (TPA) - 6.3 million rows from WTPA_done.csv
CREATE TABLE dbo.stg_Movement_TPA (
UPC BIGINT,
STORE INT,
WEEK INT,
MOVE INT,
QTY INT,
PRICE FLOAT,
100 %
Ln: 138, Ch: 1
SPC
CRLF
Windows 1252
Results
Messages
TABLE_NAME
TABLE_TYPE
1 stg_Movement_CRA BASE TABLE
2 stg_Movement_CSO BASE TABLE
3 stg_Movement_SDR BASE TABLE
4 stg_Movement_TPA BASE TABLE
5 stg_Product_CRA BASE TABLE
6 stg_Product_CSO BASE TABLE
7 stg_Product_SDR BASE TABLE
8 stg_Product_TPA BASE TABLE
9 stg_Store BASE TABLE
Query executed successfully.
infodata16.mbs.tamu.edu (13...
LAPTOP-QQU90VJE\Nisarg...
team1_staging_area
00:00:00
Row: 1, Col: 1
9 rows
ENG
IN
04:05
06-04-2026


SQLQuery3.s...isarg (92))*
SQLQuery2.sq...\Nisarg (70))*
SQLQuery1.sql...\Nisarg (59))
MOVE INT,
QTY INT,
PRICE FLOAT,
SALE VARCHAR(5),
PROFIT FLOAT,
OK INT
);
GO
-- Crackers (CRA) - 3.6 million rows from Done-WCRA.csv
CREATE TABLE dbo.stg_Movement_CRA (
UPC BIGINT,
STORE INT,
WEEK INT,
MOVE INT,
QTY INT,
PRICE FLOAT,
SALE VARCHAR(5),
PROFIT FLOAT,
OK INT
);
GO
100 %
9 0
Ln: 138, Ch: 1 SPC CRLF Windows 1252
Results Messages
TABLE_NAME TABLE_TYPE
1 stg_Movement_CRA BASE TABLE
2 stg_Movement_CSO BASE TABLE
3 stg_Movement_SDR BASE TABLE
4 stg_Movement_TPA BASE TABLE
5 stg_Product_CRA BASE TABLE
6 stg_Product_CSO BASE TABLE
7 stg_Product_SDR BASE TABLE
8 stg_Product_TPA BASE TABLE
9 stg_Store BASE TABLE
Query executed successfully.
infodata16.mbs.tamu.edu (13...
LAPTOP-QQU90VJE\Nisarg...
team1_staging_area
00:00:00
Row: 1, Col: 1
9 rows
ENG
IN
04:06
06-04-2026


SQLQuery3.s...isarg (92))*
SQLQuery2.sq...\Nisarg (70))*
SQLQuery1.sql...\Nisarg (59))
-- Product (UPC) Staging Tables (one per category), Source: UPC CSV files (encoding 1252 / latin-1)
-- Soft Drinks UPC - 1,746 rows from UPCSDR.csv
CREATE TABLE dbo.stg_Product_SDR (
COM_CODE INT,
UPC BIGINT,
DESCRIP VARCHAR(100),
SIZE VARCHAR(30),
CASE_PACK INT,
NITEM BIGINT
);
GO
-- Canned Soup UPC - 453 rows from UPCCSO.csv
CREATE TABLE dbo.stg_Product_CSO (
COM_CODE INT,
UPC BIGINT,
DESCRIP VARCHAR(100),
SIZE VARCHAR(30),
CASE_PACK INT,
NITEM BIGINT
);
GO
100 %
9 0
Ln: 138, Ch: 1 SPC CRLF Windows 1252
Results Messages
TABLE_NAME TABLE_TYPE
1 stg_Movement_CRA BASE TABLE
2 stg_Movement_CSO BASE TABLE
3 stg_Movement_SDR BASE TABLE
4 stg_Movement_TPA BASE TABLE
5 stg_Product_CRA BASE TABLE
6 stg_Product_CSO BASE TABLE
7 stg_Product_SDR BASE TABLE
8 stg_Product_TPA BASE TABLE
9 stg_Store BASE TABLE
Query executed successfully.
infodata16.mbs.tamu.edu (13...
LAPTOP-QQU90VJE\Nisarg...
team1_staging_area
00:00:00
Row: 1, Col: 1
9 rows
ENG
IN
04:06
06-04-2026

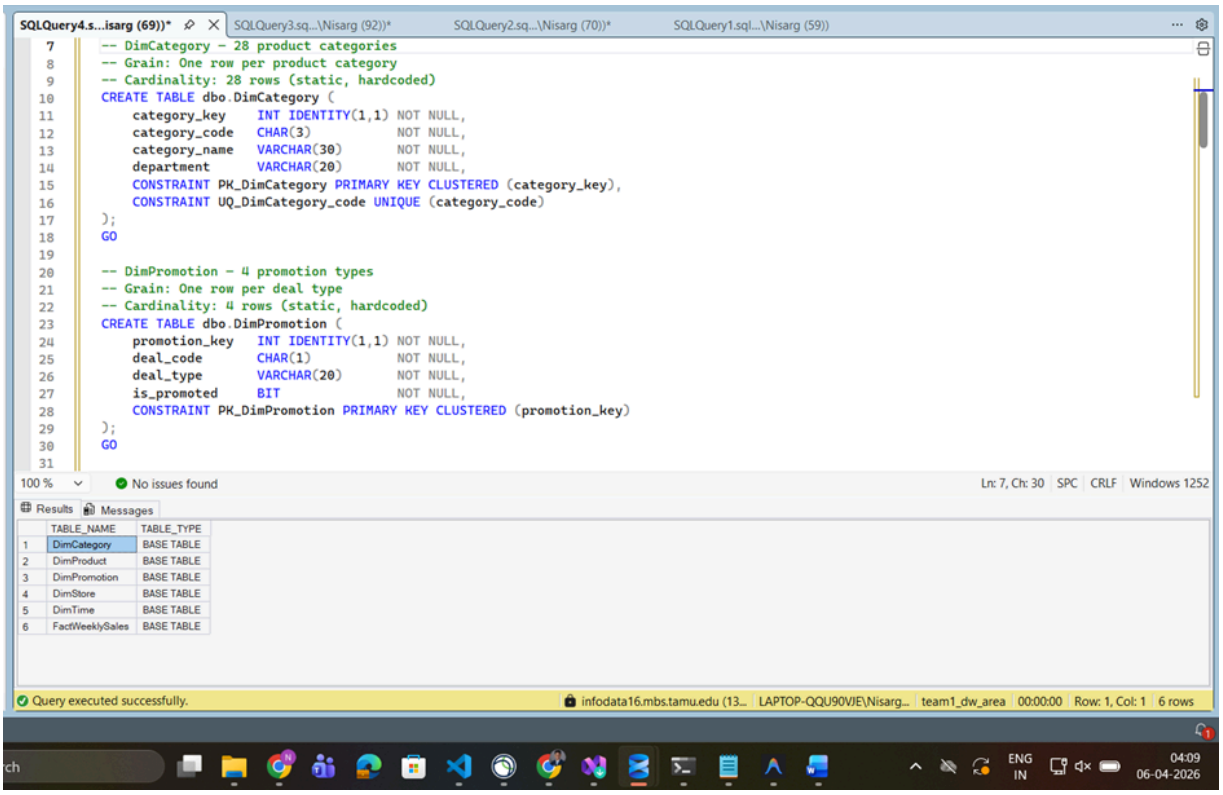

SQLQuery3.s...isarg (92))*
SQLQuery2.sq...\Nisarg (70))*
SQLQuery1.sql...\Nisarg (59))
-- Store Demographics Staging Table, Source: Demographics/DEMO.csv (108 rows, 510 columns)
-- We only stage the columns needed for DimStore
CREATE TABLE dbo.stg_Store (
STORE VARCHAR(10),
NAME VARCHAR(60),
CITY VARCHAR(50),
ZIP VARCHAR(10),
ZONE VARCHAR(10),
URBAN VARCHAR(10),
WEEKVOL VARCHAR(20),
INCOME VARCHAR(20),
EDUC VARCHAR(20),
POVERTY VARCHAR(20),
HSIZEAVG VARCHAR(20),
ETHNIC VARCHAR(20),
DENSITY VARCHAR(20),
AGE9 VARCHAR(20),
AGE60 VARCHAR(20),
WORKWOM VARCHAR(20),
PRICLOW VARCHAR(10),
PRICMED VARCHAR(10),
PRICHIGH VARCHAR(10)
);
GO
100 %
9
0
Ln: 138, Ch: 1
SPC
CRLF
Windows 1252
Results
Messages
TABLE_NAME
TABLE_TYPE
1 stg_Movement_CRA BASE TABLE
2 stg_Movement_CSO BASE TABLE
3 stg_Movement_SDR BASE TABLE
4 stg_Movement_TPA BASE TABLE
5 stg_Product_CRA BASE TABLE
6 stg_Product_CSO BASE TABLE
7 stg_Product_SDR BASE TABLE
8 stg_Product_TPA BASE TABLE
9 stg_Store BASE TABLE
Query executed successfully.
infodata16.mbs.tamu.edu (13...
LAPTOP-QQU90VJE\Nisarg...
team1_staging_area
00:00:00
Row: 1, Col: 1
9 rows
ENG
IN
04:07
06-04-2026

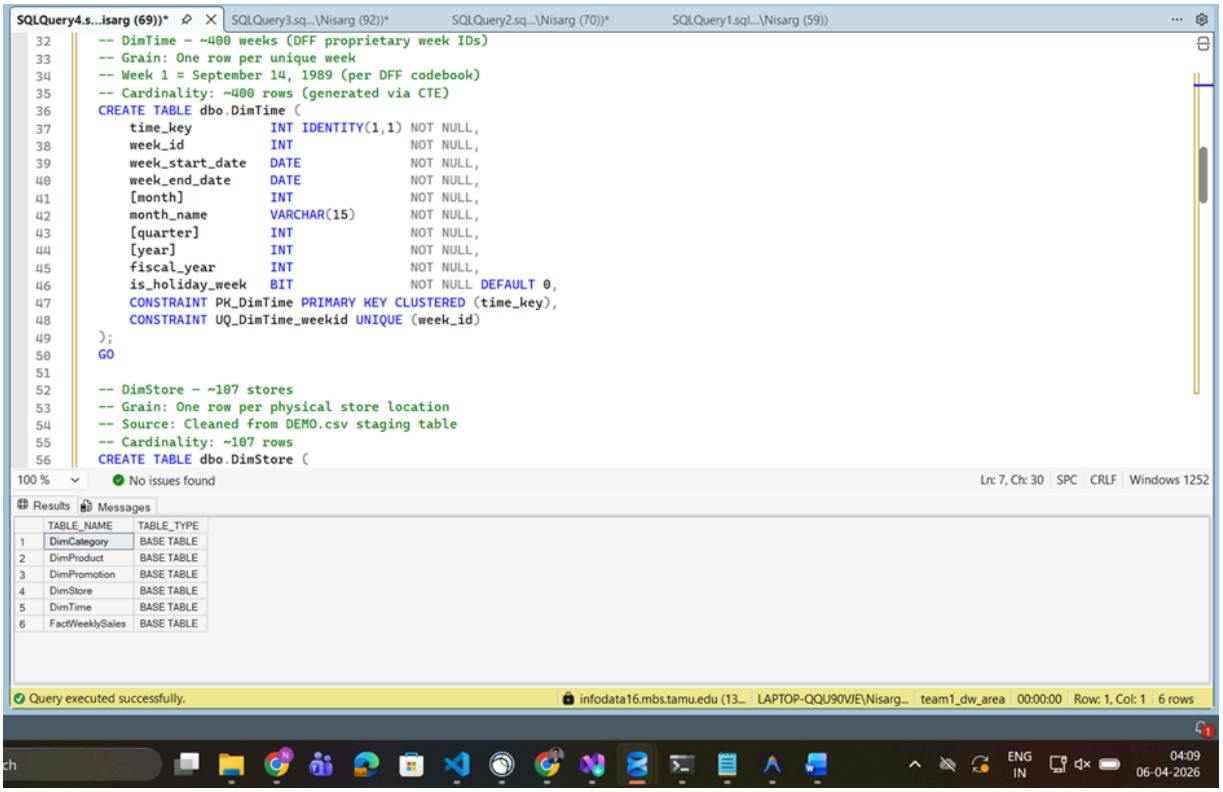

SQLQuery3.s...isarg (92))*
SQLQuery2.sq...\Nisarg (70))*
SQLQuery1.sql...\Nisarg (59))
-- Verification: List all staging tables
SELECT TABLE_NAME, TABLE_TYPE
FROM INFORMATION_SCHEMA.TABLES
WHERE TABLE_SCHEMA = 'dbo'
ORDER BY TABLE_NAME;
GO
PRINT '=== All staging tables created successfully ===';
GO
100 %
9
0
Ln: 138, Ch: 1
SPC
CRLF
Windows 1252
Results
Messages
TABLE_NAME
TABLE_TYPE
1 stg_Movement_CRA BASE TABLE
2 stg_Movement_CSO BASE TABLE
3 stg_Movement_SDR BASE TABLE
4 stg_Movement_TPA BASE TABLE
5 stg_Product_CRA BASE TABLE
6 stg_Product_CSO BASE TABLE
7 stg_Product_SDR BASE TABLE
8 stg_Product_TPA BASE TABLE
9 stg_Store BASE TABLE
Query executed successfully.
infodata16.mbs.tamu.edu (13...
LAPTOP-QQU90VJE\Nisarg...
team1_staging_area
00:00:00
Row: 1, Col: 1
9 rows
ENG
IN
04:07
06-04-2026

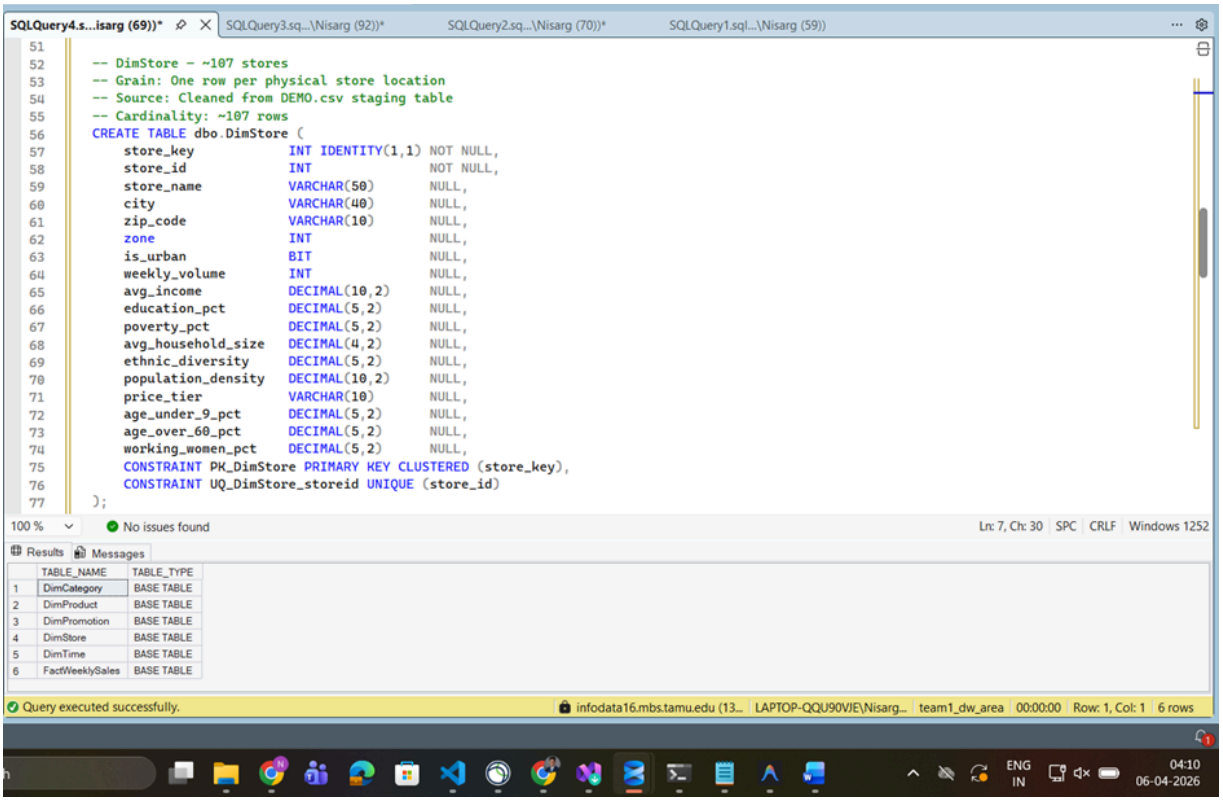
6.3 Data Mart Table Creation

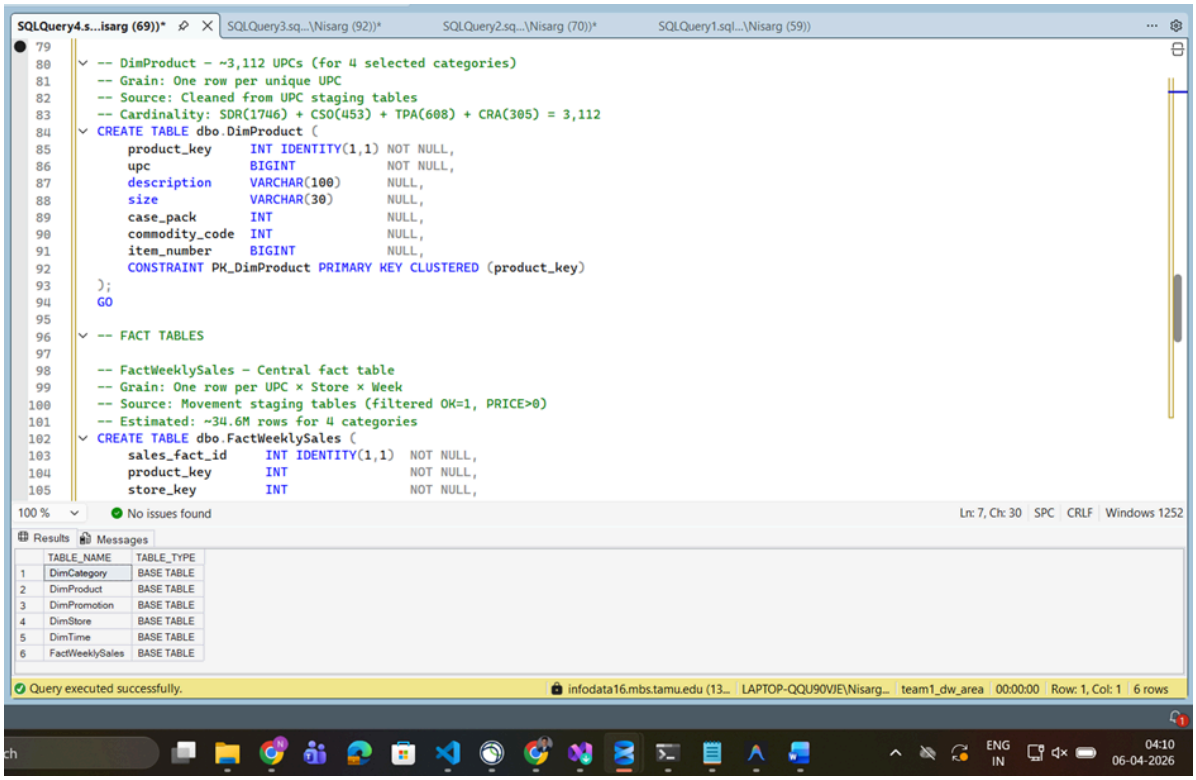
All 5 dimension tables and 1 fact table were created in team1_dw_area, with dimensions created first to enable foreign key constraints. The complete SQL is in Appendix A (03_create_dw_tables.sql).

Screenshot 3: SSMS Object Explorer DW tables listed




SQLQuery4.s...isarg (69))*
SQLQuery3.sq...\Nisarg (92))*
SQLQuery2.sq...\Nisarg (70))*
SQLQuery1.sql...\Nisarg (59))
-- DimTime – ~400 weeks (DFF proprietary week IDs)
-- Grain: One row per unique week
-- Week 1 = September 14, 1989 (per DFF codebook)
-- Cardinality: ~400 rows (generated via CTE)
CREATE TABLE dbo.DimTime (
time_key INT IDENTITY(1,1) NOT NULL,
week_id INT NOT NULL,
week_start_date DATE NOT NULL,
week_end_date DATE NOT NULL,
[month] INT NOT NULL,
month_name VARCHAR(15) NOT NULL,
[quarter] INT NOT NULL,
[year] INT NOT NULL,
fiscal_year INT NOT NULL,
is_holiday_week BIT NOT NULL DEFAULT 0,
CONSTRAINT PK_DimTime PRIMARY KEY CLUSTERED (time_key),
CONSTRAINT UQ_DimTime_weekid UNIQUE (week_id)
);
GO
-- DimStore – ~107 stores
-- Grain: One row per physical store location
-- Source: Cleaned from DEMO.csv staging table
-- Cardinality: ~107 rows
CREATE TABLE dbo.DimStore (
100 %
No issues found
Ln: 7, Ch: 30
SPC
CRLF
Windows 1252
Results
Messages
TABLE_NAME
TABLE_TYPE
1 DimCategory BASE TABLE
2 DimProduct BASE TABLE
3 DimPromotion BASE TABLE
4 DimStore BASE TABLE
5 DimTime BASE TABLE
6 FactWeeklySales BASE TABLE
Query executed successfully.
infodata16.mbs.tamu.edu (13...
LAPTOP-QQU90VJE\Nisarg...
team1_dw_area
00:00:00
Row: 1, Col: 1
6 rows
ENG
IN
04:09
06-04-2026


SQLQuery4.s...isarg (69))*
SQLQuery3.sq...\Nisarg (92))*
SQLQuery2.sq...\Nisarg (70))*
SQLQuery1.sql...\Nisarg (59))
-- DimStore – ~107 stores
-- Grain: One row per physical store location
-- Source: Cleaned from DEMO.csv staging table
-- Cardinality: ~107 rows
CREATE TABLE dbo.DimStore (
store_key INT IDENTITY(1,1) NOT NULL,
store_id INT NOT NULL,
store_name VARCHAR(50) NULL,
city VARCHAR(40) NULL,
zip_code VARCHAR(10) NULL,
zone INT NULL,
is_urban BIT NULL,
weekly_volume INT NULL,
avg_income DECIMAL(10,2) NULL,
education_pct DECIMAL(5,2) NULL,
poverty_pct DECIMAL(5,2) NULL,
avg_household_size DECIMAL(4,2) NULL,
ethnic_diversity DECIMAL(5,2) NULL,
population_density DECIMAL(10,2) NULL,
price_tier VARCHAR(10) NULL,
age_under_9_pct DECIMAL(5,2) NULL,
age_over_60_pct DECIMAL(5,2) NULL,
working_women_pct DECIMAL(5,2) NULL,
CONSTRAINT PK_DimStore PRIMARY KEY CLUSTERED (store_key),
CONSTRAINT UQ_DimStore_storeid UNIQUE (store_id)
);
100 %
No issues found
Ln: 7, Ch: 30
SPC
CRLF
Windows 1252
Results
Messages
TABLE_NAME
TABLE_TYPE
1 DimCategory BASE TABLE
2 DimProduct BASE TABLE
3 DimPromotion BASE TABLE
4 DimStore BASE TABLE
5 DimTime BASE TABLE
6 FactWeeklySales BASE TABLE
Query executed successfully.
infodata16.mbs.tamu.edu (13...
LAPTOP-QQU90VJE\Nisarg...
team1_dw_area
00:00:00
Row: 1, Col: 1
6 rows
ENG
IN
04:10
06-04-2026



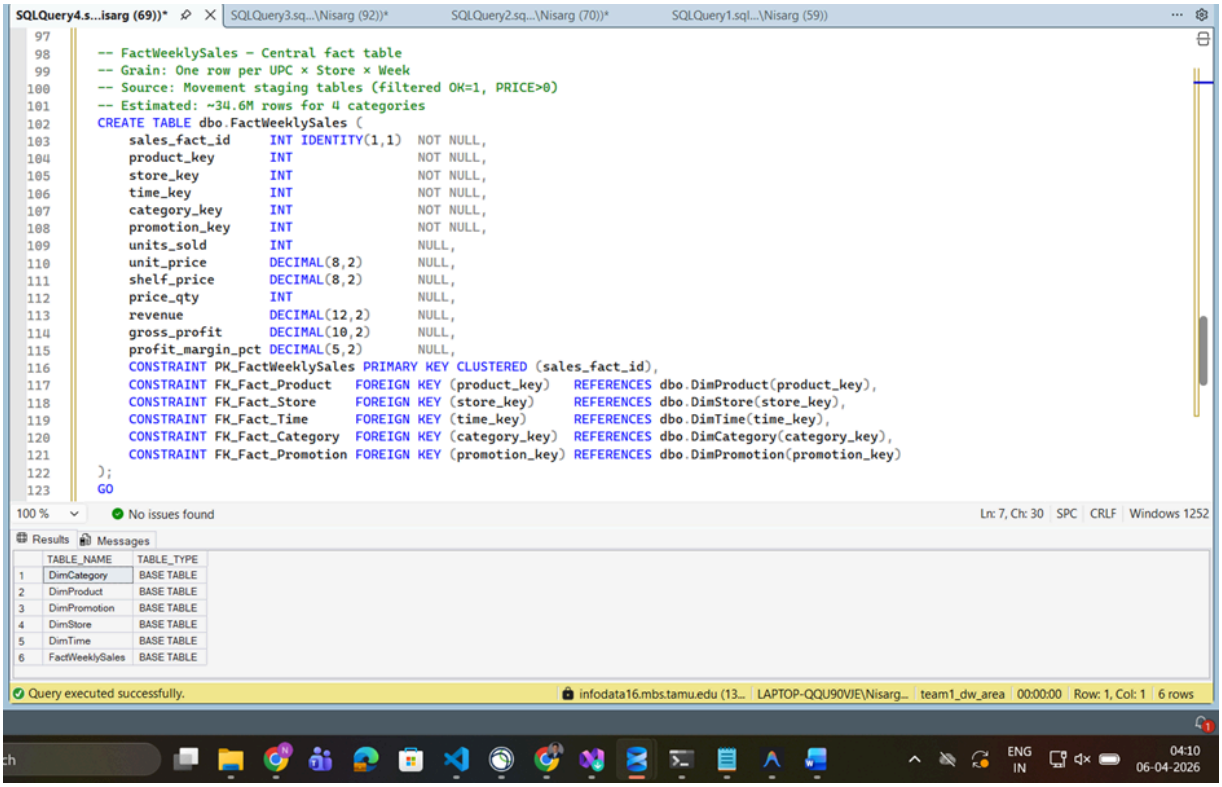
```
-- DimProduct – ~3,112 UPCs (for 4 selected categories)
-- Grain: One row per unique UPC
-- Source: Cleaned from UPC staging tables
-- Cardinality: SDR(1746) + CSO(453) + TPA(608) + CRA(305) = 3,112
CREATE TABLE dbo.DimProduct (
    product_key     INT IDENTITY(1,1) NOT NULL,
    upc             BIGINT            NOT NULL,
    description     VARCHAR(100)      NULL,
    size            VARCHAR(30)       NULL,
    case_pack       INT               NULL,
    commodity_code  INT               NULL,
    item_number     BIGINT            NULL,
    CONSTRAINT PK_DimProduct PRIMARY KEY CLUSTERED (product_key)
);
GO

-- FACT TABLES

-- FactWeeklySales – Central fact table
-- Grain: One row per UPC × Store × Week
-- Source: Movement staging tables (filtered OK=1, PRICE>0)
-- Estimated: ~34.6M rows for 4 categories
CREATE TABLE dbo.FactWeeklySales (
    sales_fact_id   INT IDENTITY(1,1)  NOT NULL,
    product_key     INT                NOT NULL,
    store_key       INT                NOT NULL,
```

100 % No issues found Ln: 7, Ch: 30 SPC CRLF Windows 1252

Results Messages

	TABLE_NAME	TABLE_TYPE
1	DimCategory	BASE TABLE
2	DimProduct	BASE TABLE
3	DimPromotion	BASE TABLE
4	DimStore	BASE TABLE
5	DimTime	BASE TABLE
6	FactWeeklySales	BASE TABLE

Query executed successfully. infodata16.mbs.tamu.edu (13... LAPTOP-QQU90VJE\Nisarg... team1_dw_area 00:00:00 Row: 1, Col: 1 6 rows



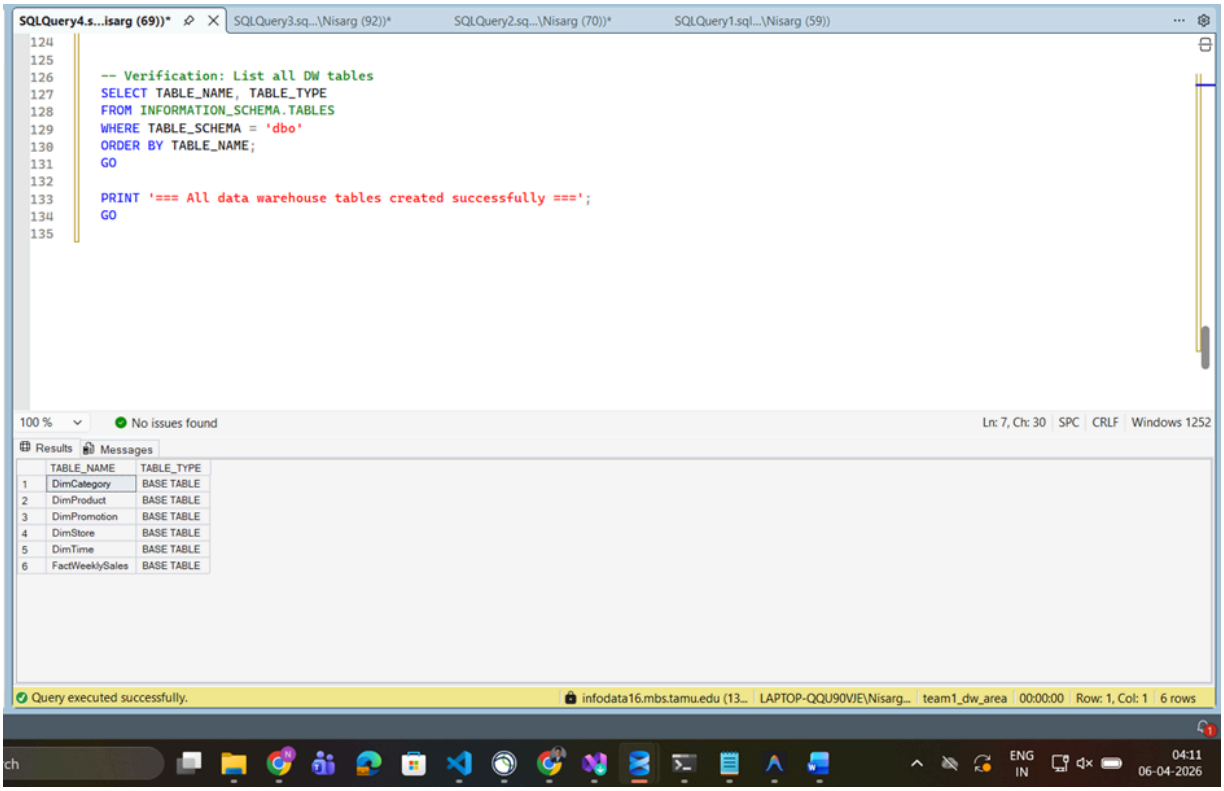
```
-- FactWeeklySales – Central fact table
-- Grain: One row per UPC × Store × Week
-- Source: Movement staging tables (filtered OK=1, PRICE>0)
-- Estimated: ~34.6M rows for 4 categories
CREATE TABLE dbo.FactWeeklySales (
    sales_fact_id   INT IDENTITY(1,1)  NOT NULL,
    product_key     INT                NOT NULL,
    store_key       INT                NOT NULL,
    time_key        INT                NOT NULL,
    category_key    INT                NOT NULL,
    promotion_key   INT                NOT NULL,
    units_sold      INT                NULL,
    unit_price      DECIMAL(8,2)       NULL,
    shelf_price     DECIMAL(8,2)       NULL,
    price_qty       INT                NULL,
    revenue         DECIMAL(12,2)      NULL,
    gross_profit    DECIMAL(10,2)      NULL,
    profit_margin_pct DECIMAL(5,2)     NULL,
    CONSTRAINT PK_FactWeeklySales PRIMARY KEY CLUSTERED (sales_fact_id),
    CONSTRAINT FK_Fact_Product   FOREIGN KEY (product_key)   REFERENCES dbo.DimProduct(product_key),
    CONSTRAINT FK_Fact_Store     FOREIGN KEY (store_key)     REFERENCES dbo.DimStore(store_key),
    CONSTRAINT FK_Fact_Time      FOREIGN KEY (time_key)      REFERENCES dbo.DimTime(time_key),
    CONSTRAINT FK_Fact_Category  FOREIGN KEY (category_key)  REFERENCES dbo.DimCategory(category_key),
    CONSTRAINT FK_Fact_Promotion FOREIGN KEY (promotion_key) REFERENCES dbo.DimPromotion(promotion_key)
);
GO
```

100 % No issues found Ln: 7, Ch: 30 SPC CRLF Windows 1252

Results Messages

	TABLE_NAME	TABLE_TYPE
1	DimCategory	BASE TABLE
2	DimProduct	BASE TABLE
3	DimPromotion	BASE TABLE
4	DimStore	BASE TABLE
5	DimTime	BASE TABLE
6	FactWeeklySales	BASE TABLE

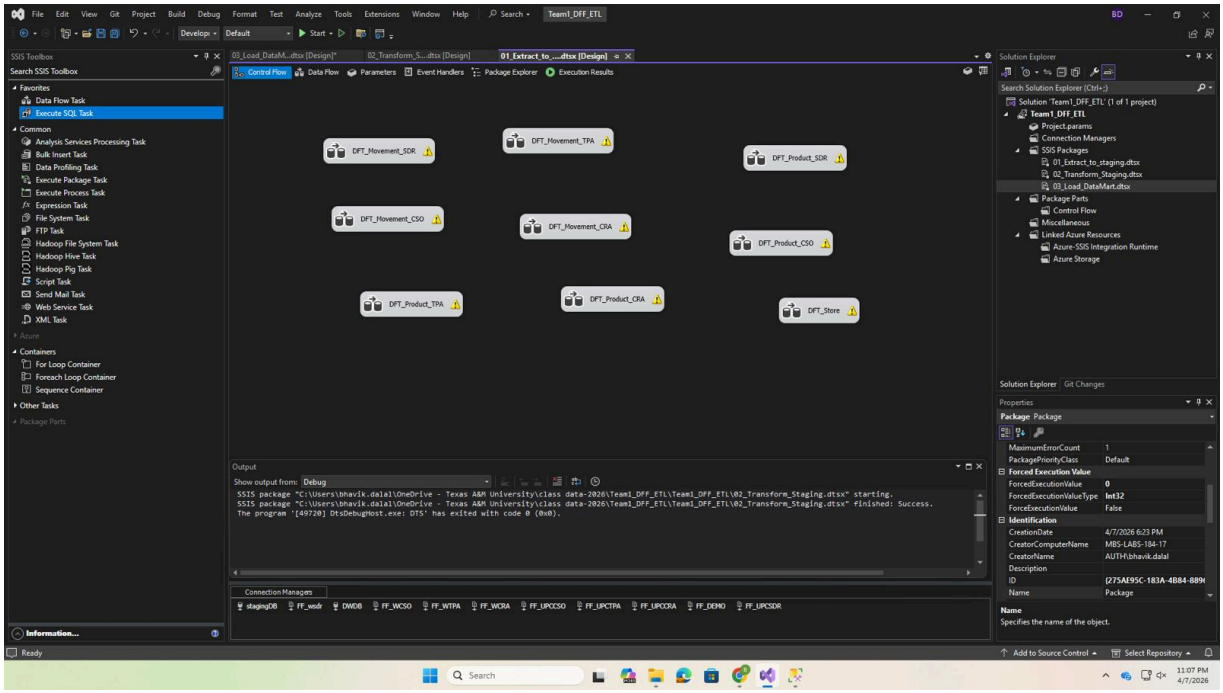
Query executed successfully. infodata16.mbs.tamu.edu (13... LAPTOP-QQU90VJE\Nisarg... team1_dw_area 00:00:00 Row: 1, Col: 1 6 rows



6.4 SSIS Package 1: Extract to Staging

Control Flow: Package 1 contains 9 Data Flow Tasks, each extracting one CSV source file into a corresponding staging table.

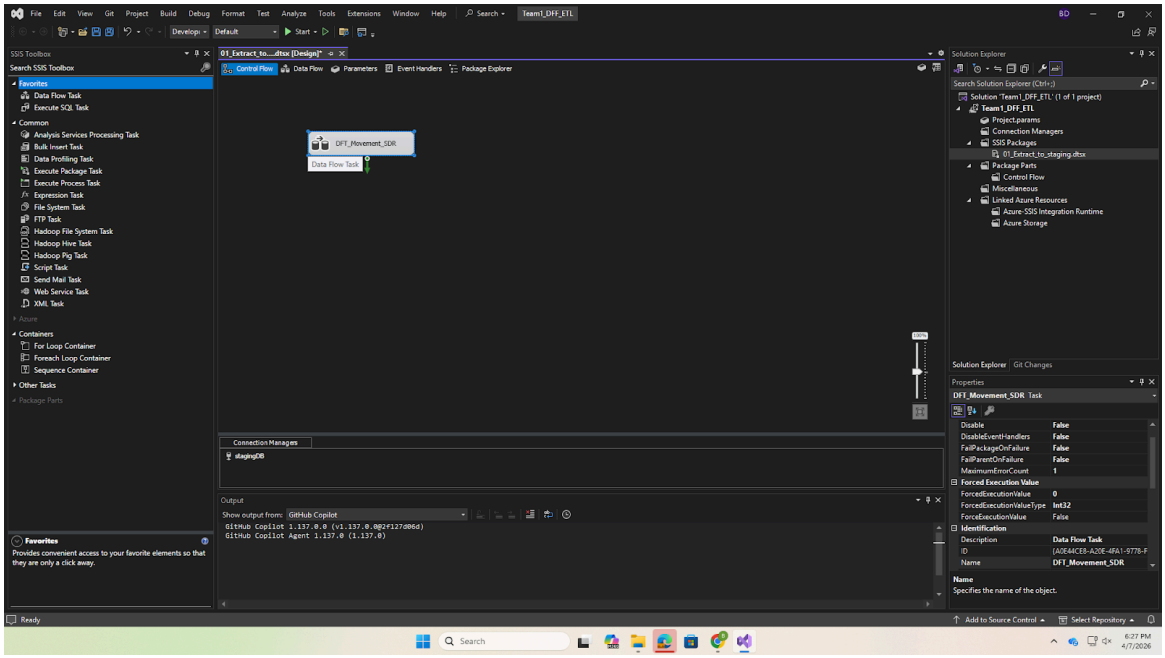
Screenshot 4: SSIS Control Flow Package 1 showing all 9 Data Flow Tasks



Sample Data Flow Task (Soft Drinks Movement): Flat File Source → OLE DB Destination.

This data flow shows how the Soft Drinks movement file is read from a flat file source and loaded into the staging table using an OLE DB destination.

Screenshot 5: SSIS Data Flow DFT_Movement_SDR

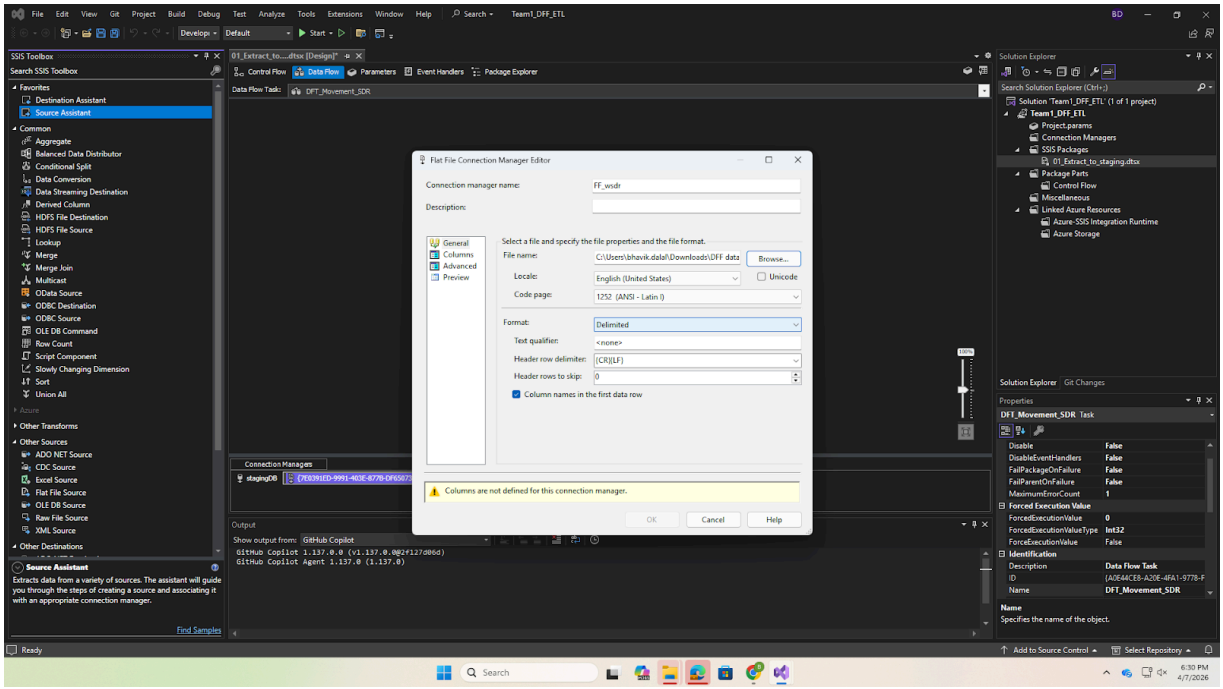


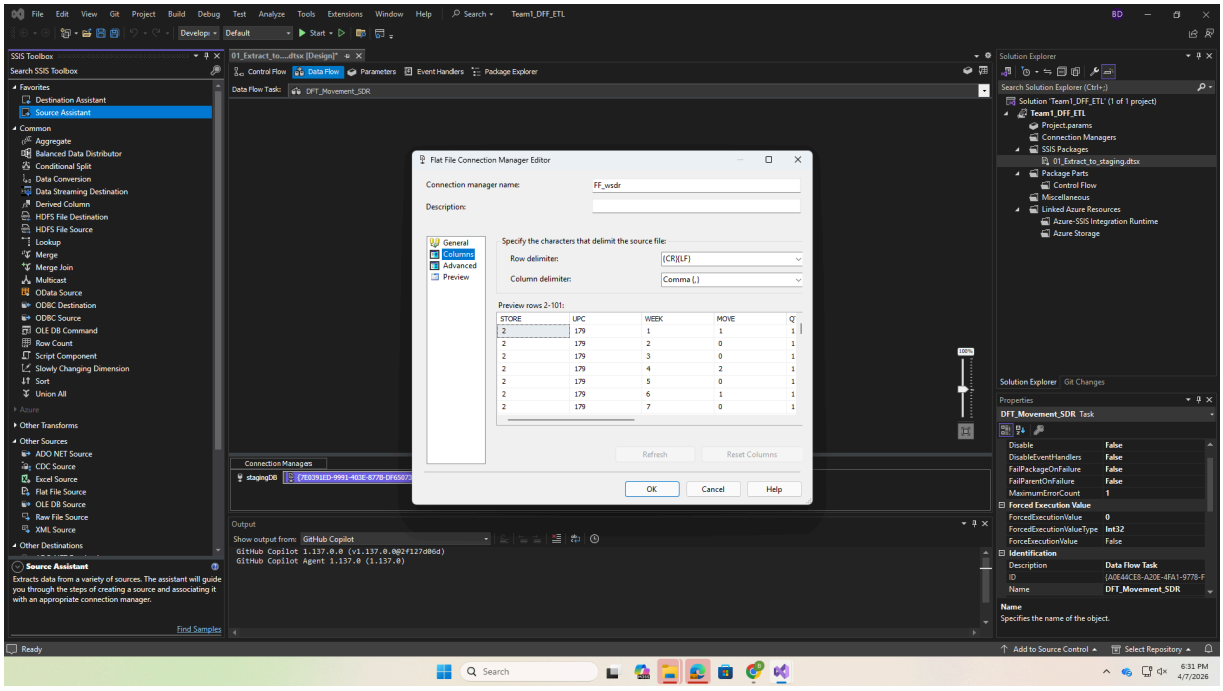
Flat File Connection Settings: Encoding = 1252 (Windows Western), Column Delimiter = Comma, Header Row Delimiter = {CR}{LF}.

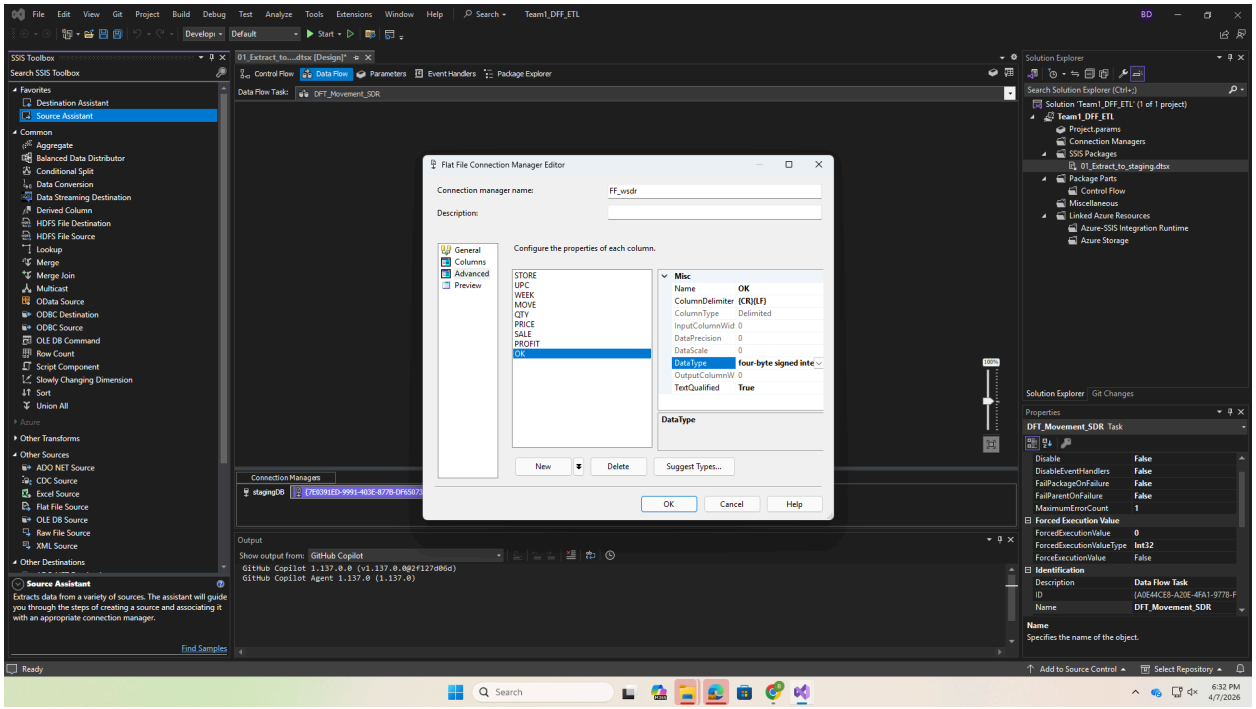
Screenshot 6: Flat File Connection Manager encoding and delimiter settings.

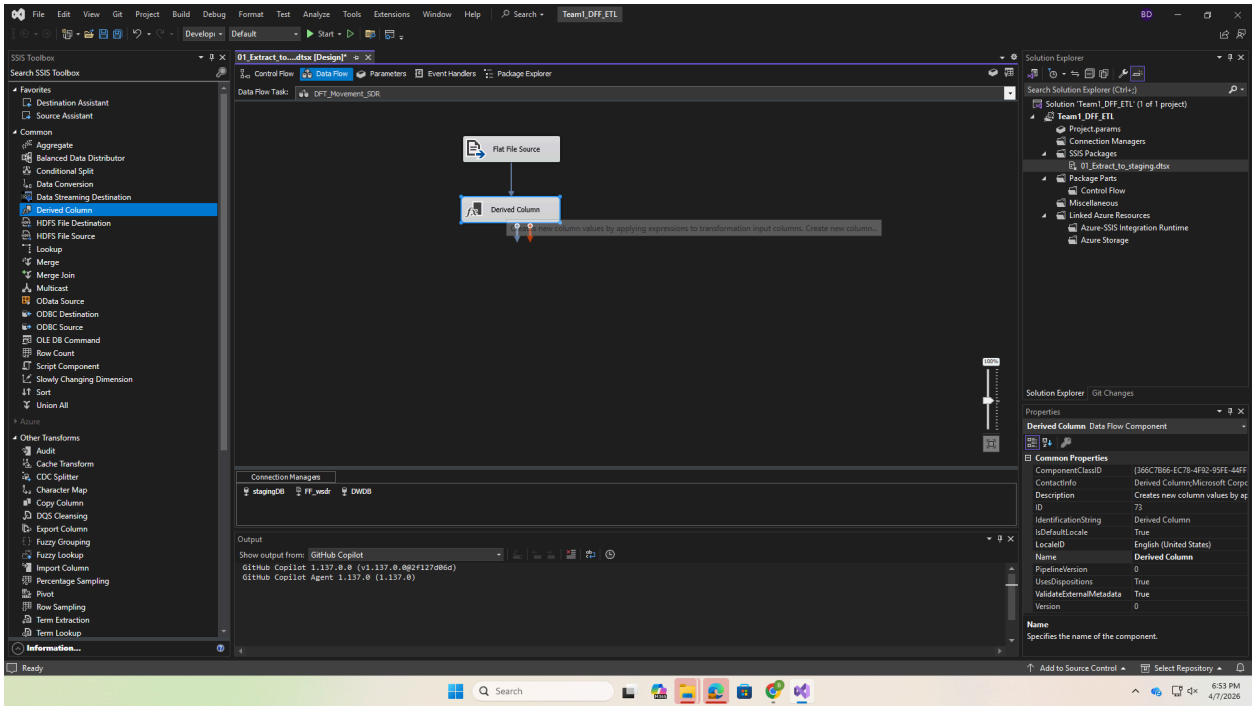
This configuration ensures that the CSV files are correctly parsed using Windows-1252 encoding and comma delimiters.

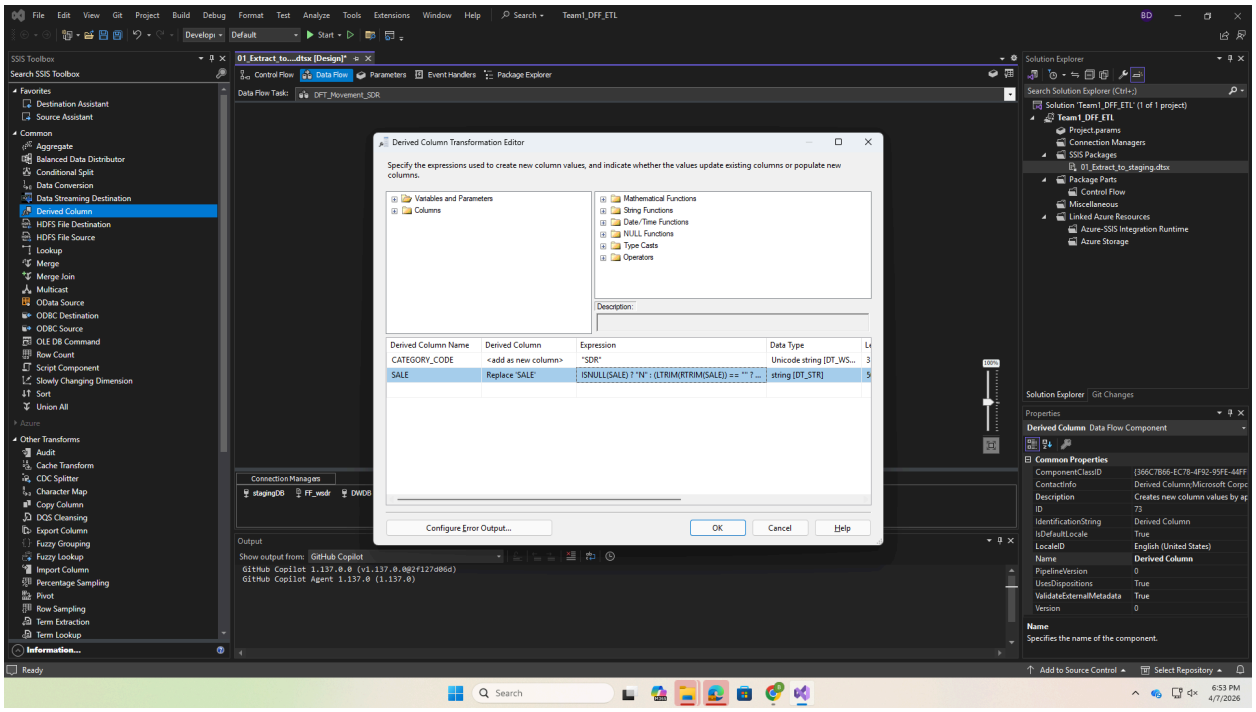
Screenshot 6: Flat File Connection Manager encoding and delimiter settings

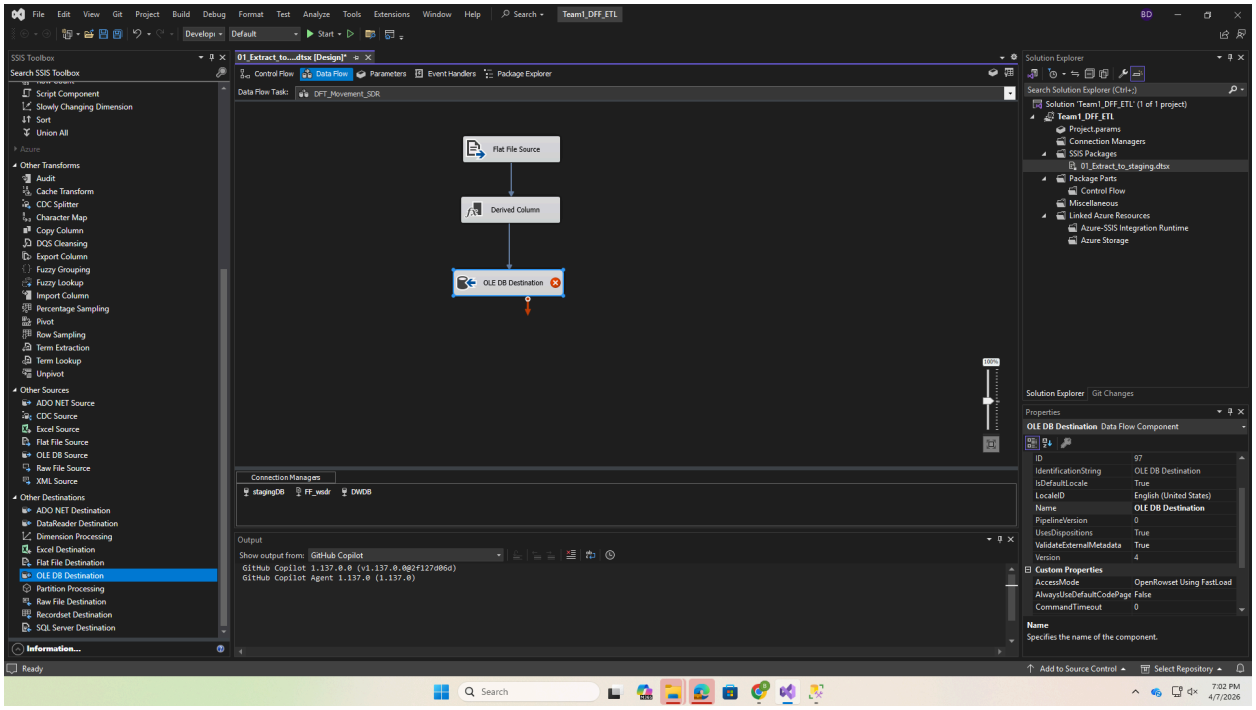


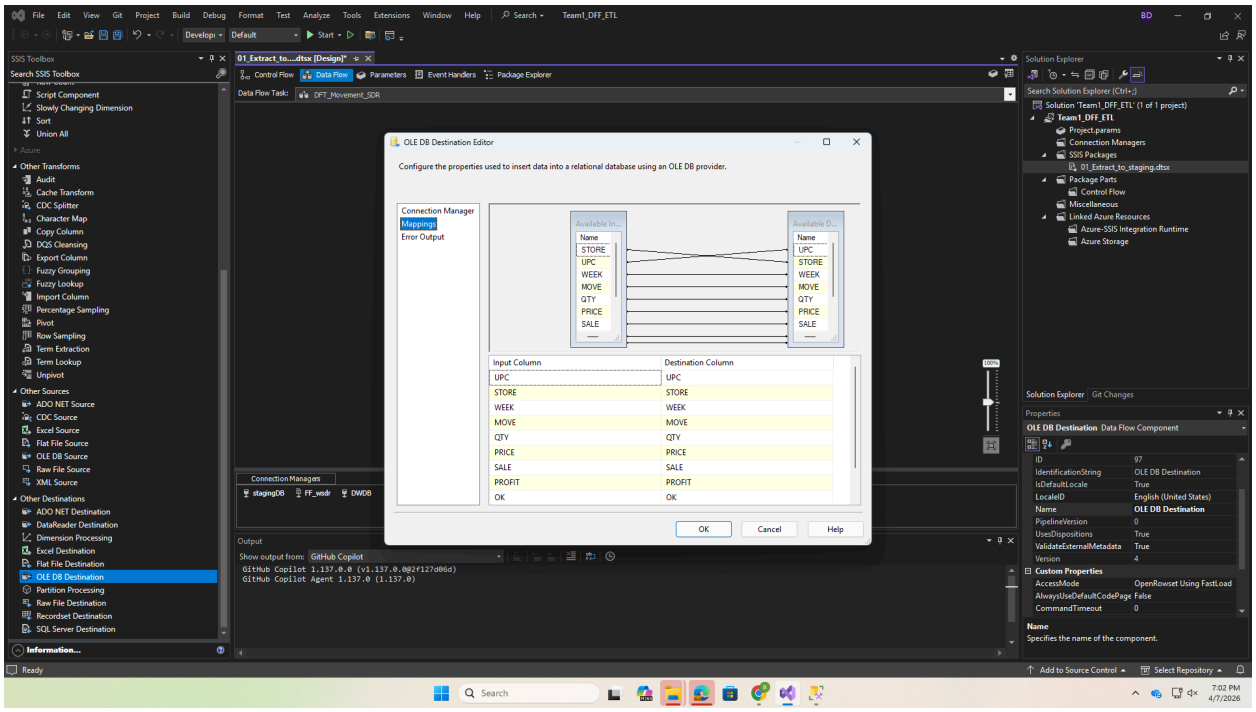

Flat File Connection Manager Editor
Connection manager name:
FF_wsdr
Description:
General
Columns
Advanced
Preview
Specify the characters that delimit the source file:
Row delimiter:
{CR}{LF}
Column delimiter:
Comma (,)
Preview rows 2-101:
STORE
UPC
WEEK
MOVE
2
179
1
1
2
179
2
0
2
179
3
0
2
179
4
2
2
179
5
0
2
179
6
1
2
179
7
0
Refresh
Reset Columns
OK
Cancel
Help

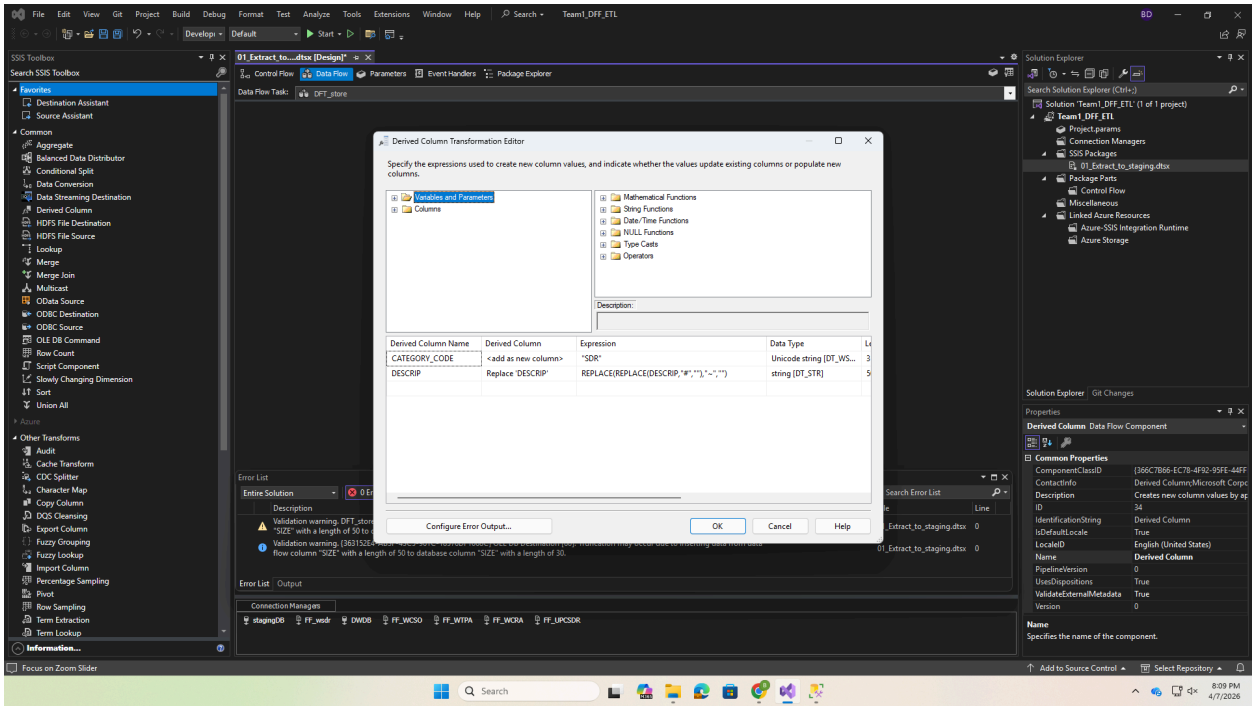

Flat File Connection Manager Editor
Connection manager name:
FF_wsdr
Description:
General
Columns
Advanced
Preview
Configure the properties of each column.
STORE
UPC
WEEK
MOVE
QTY
PRICE
SALE
PROFIT
OK
Misc
Name
OK
ColumnDelimiter
{CR}{LF}
ColumnType
Delimited
DataType
four-byte signed inte
TextQualified
True
New
Delete
Suggest Types...
OK
Cancel
Help

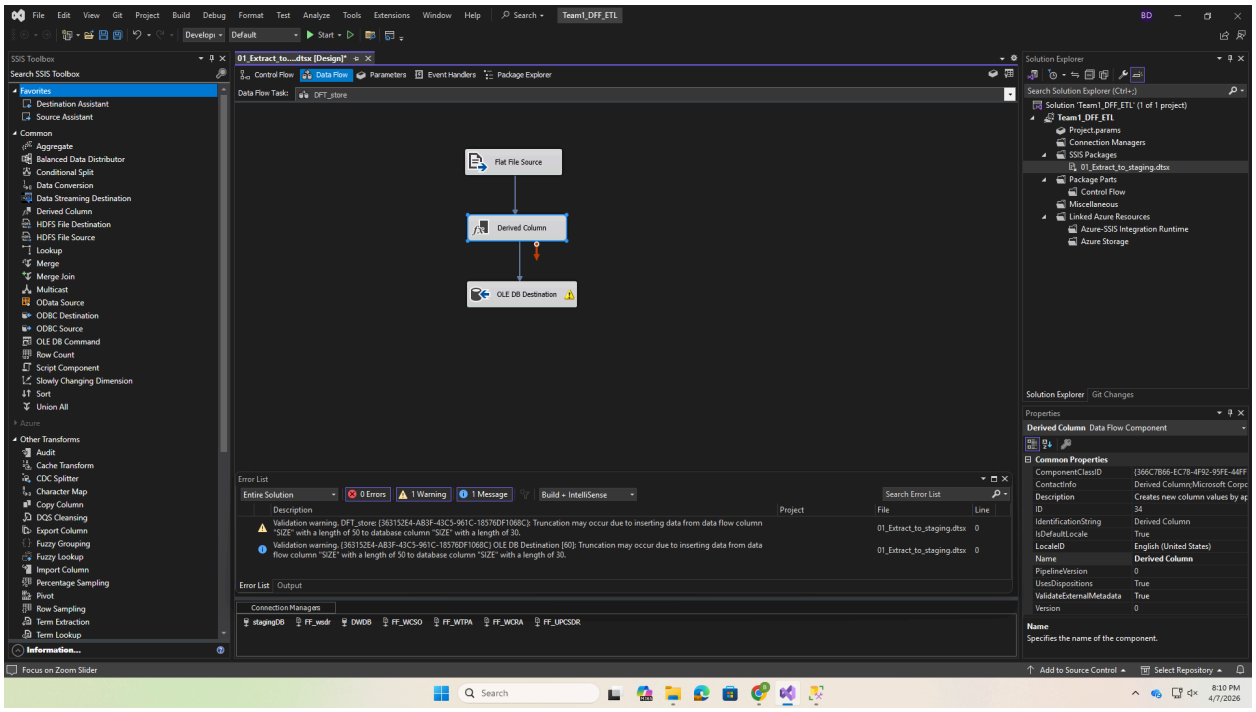

Flat File Source
Derived Column
Connection Managers
stagingDB
FF_wsdr
DWDB
Output
GitHub Copilot 1.137.0 (v1.137.0.0@2f127d06d)
GitHub Copilot Agent 1.137.0 (1.137.0)

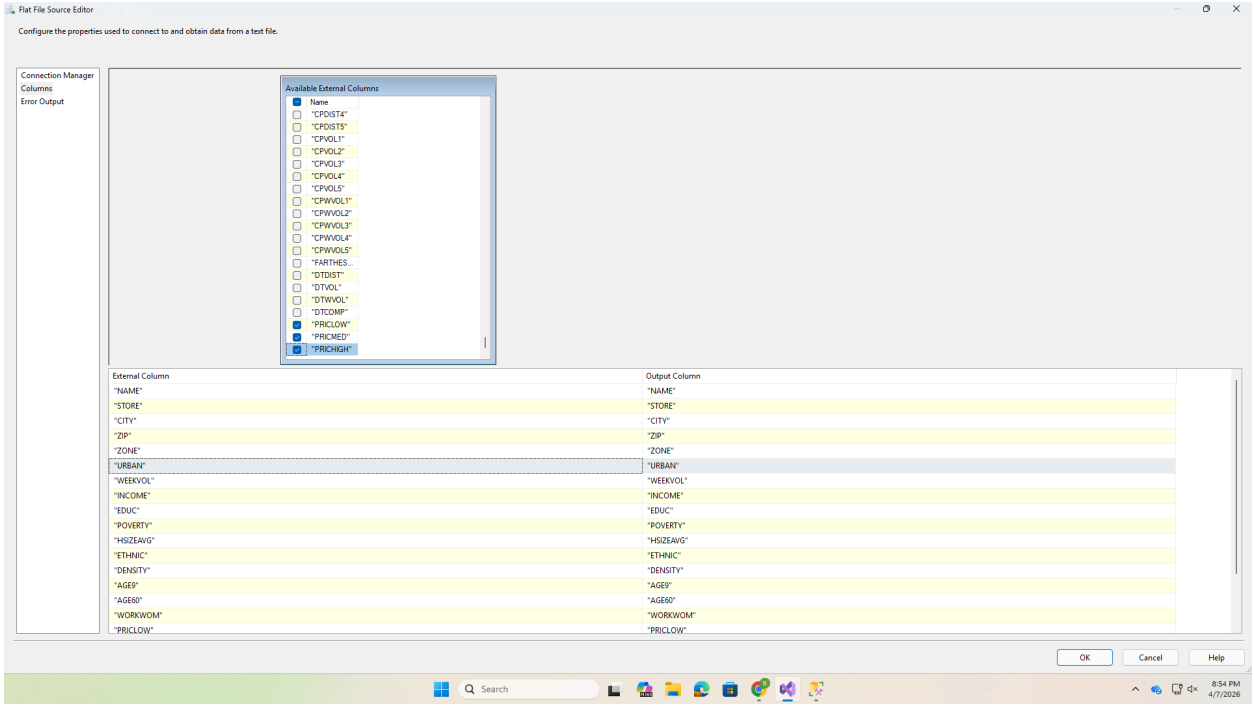

Derived Column Transformation Editor
Specify the expressions used to create new column values, and indicate whether the values update existing columns or populate new columns.
Variables and Parameters
Columns
Mathematical Functions
String Functions
Date/Time Functions
NULL Functions
Type Casts
Operators
Description:
Derived Column Name
Derived Column
Expression
Data Type
CATEGORY_CODE
<add as new column>
"SDR"
Unicode string [DT_WS...
SALE
Replace 'SALE'
ISNULL(SALE) ? "N" : (LTRIM(RTRIM(SALE)) == "" ? ...
string [DT_STR]
Configure Error Output...
OK
Cancel
Help

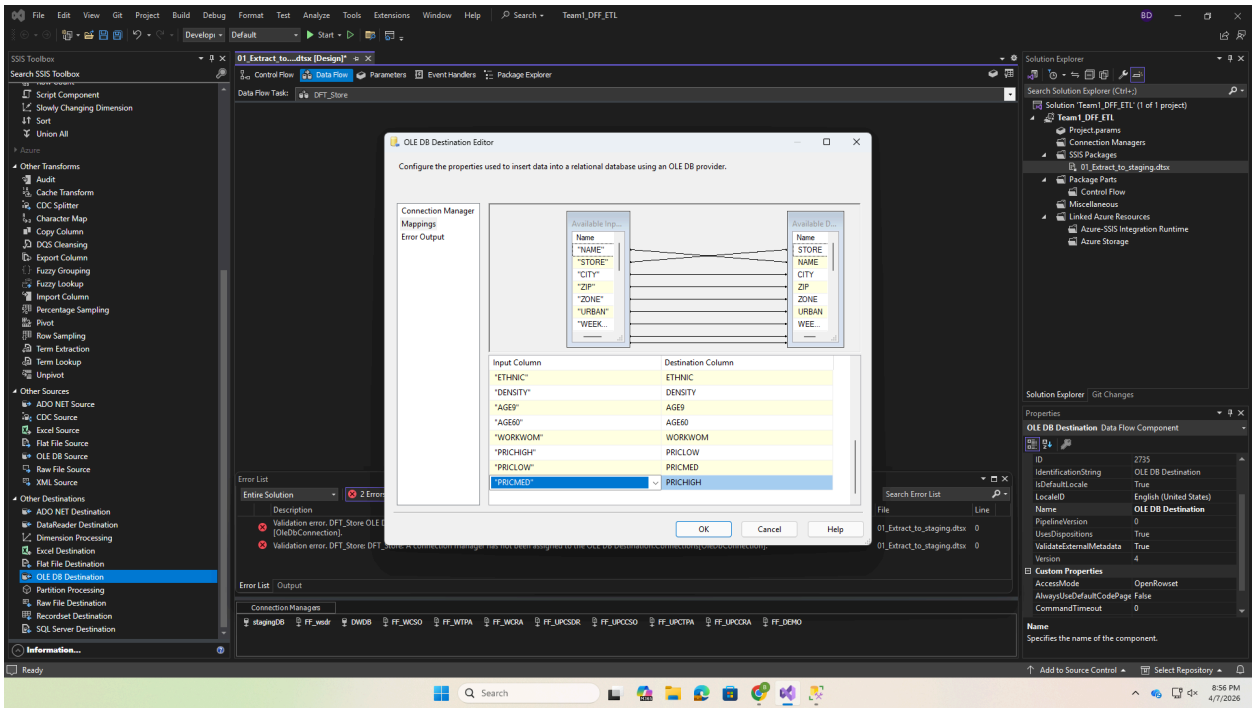

Flat File Source
Derived Column
OLE DB Destination
Connection Managers
stagingDB
FF_wsdr
DWDB
Output
GitHub Copilot 1.137.0.0 (v1.137.0.0@2f127d06d)
GitHub Copilot Agent 1.137.0 (1.137.0)


OLE DB Destination Editor
Configure the properties used to insert data into a relational database using an OLE DB provider.
Connection Manager
Mappings
Error Output
Input Column
Destination Column
UPC
STORE
WEEK
MOVE
QTY
PRICE
SALE
PROFIT
OK
Cancel
Help


Derived Column Transformation Editor
Specify the expressions used to create new column values, and indicate whether the values update existing columns or populate new columns.
Variables and Parameters
Columns
Mathematical Functions
String Functions
Date/Time Functions
NULL Functions
Type Casts
Operators
Description:
Derived Column Name
Derived Column
Expression
Data Type
CATEGORY_CODE
<add as new column>
"SDR"
Unicode string [DT_WS...
DESCRIP
Replace 'DESCRIP'
REPLACE(REPLACE(DESCRIP,"#",""),"-","")
string [DT_STR]
Configure Error Output...
OK
Cancel
Help


Flat File Source
Derived Column
OLE DB Destination

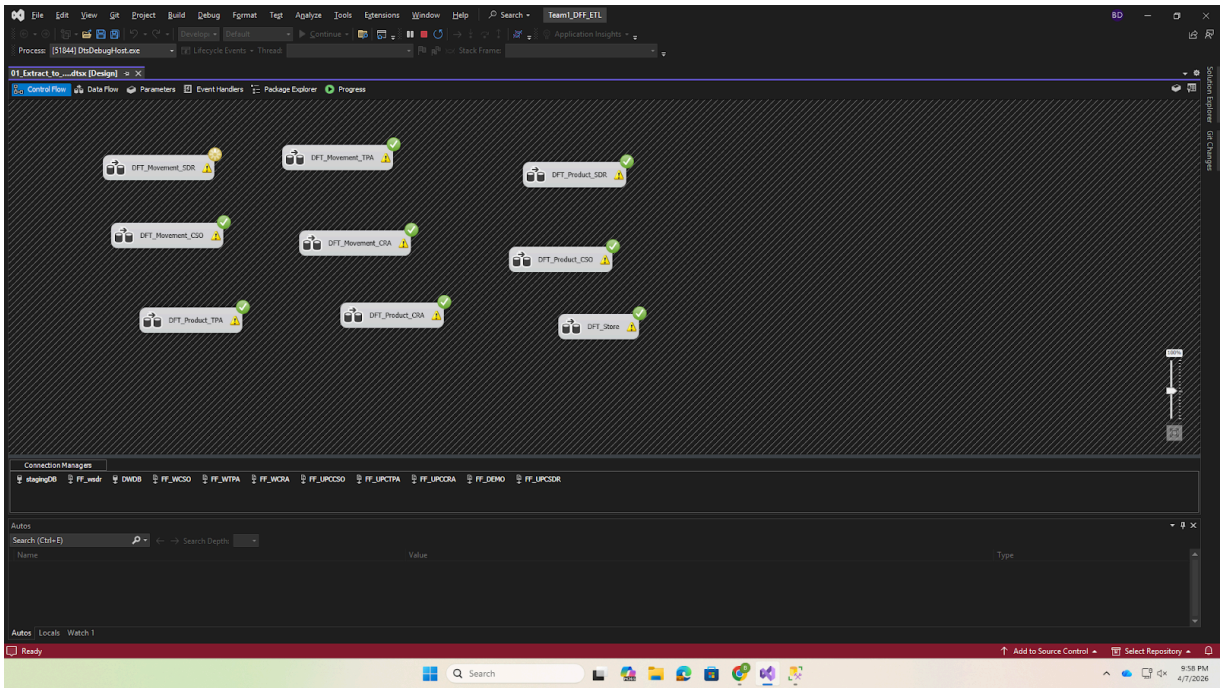




Execution Result:

This result confirms that all extraction tasks completed successfully without errors.

Screenshot 7: SSIS Package 1 execution all green checkmarks

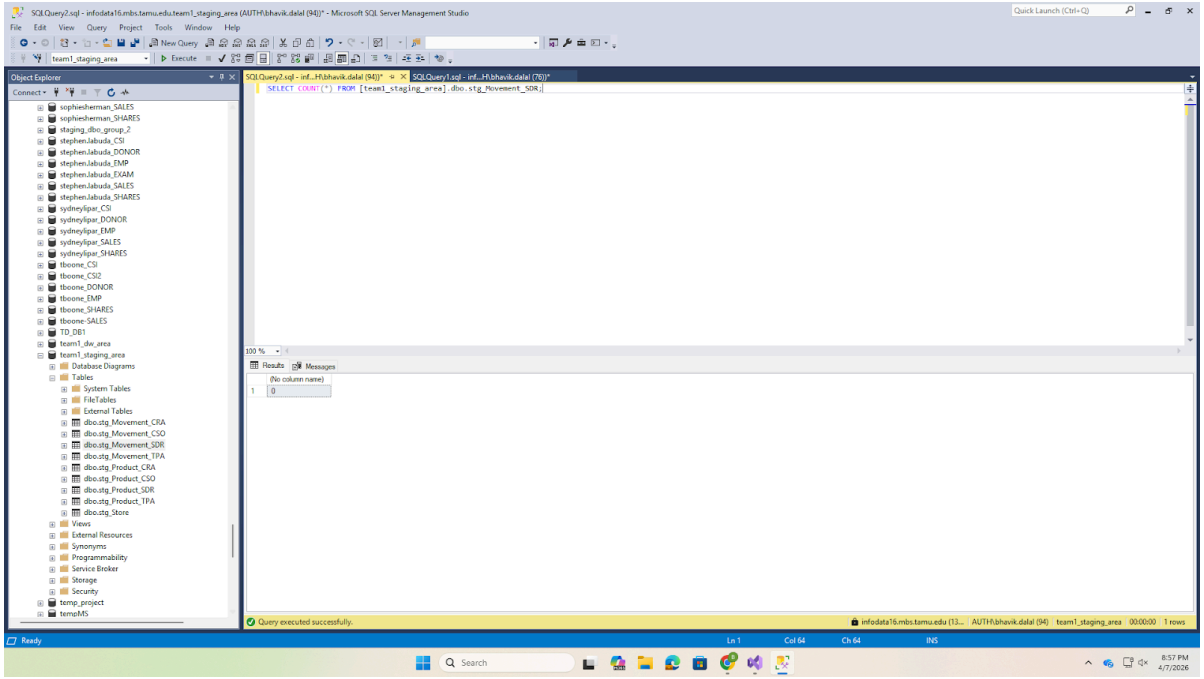


Before Loading (empty staging table):

This query verifies that the staging table is empty before the extraction process begins.

```
SELECT COUNT(*) AS RowCount FROM [team1_staging_area].dbo.stg_Movement_SDR;
-- Result: 0
```

Screenshot 8: SSMS SELECT COUNT showing 0 rows before load

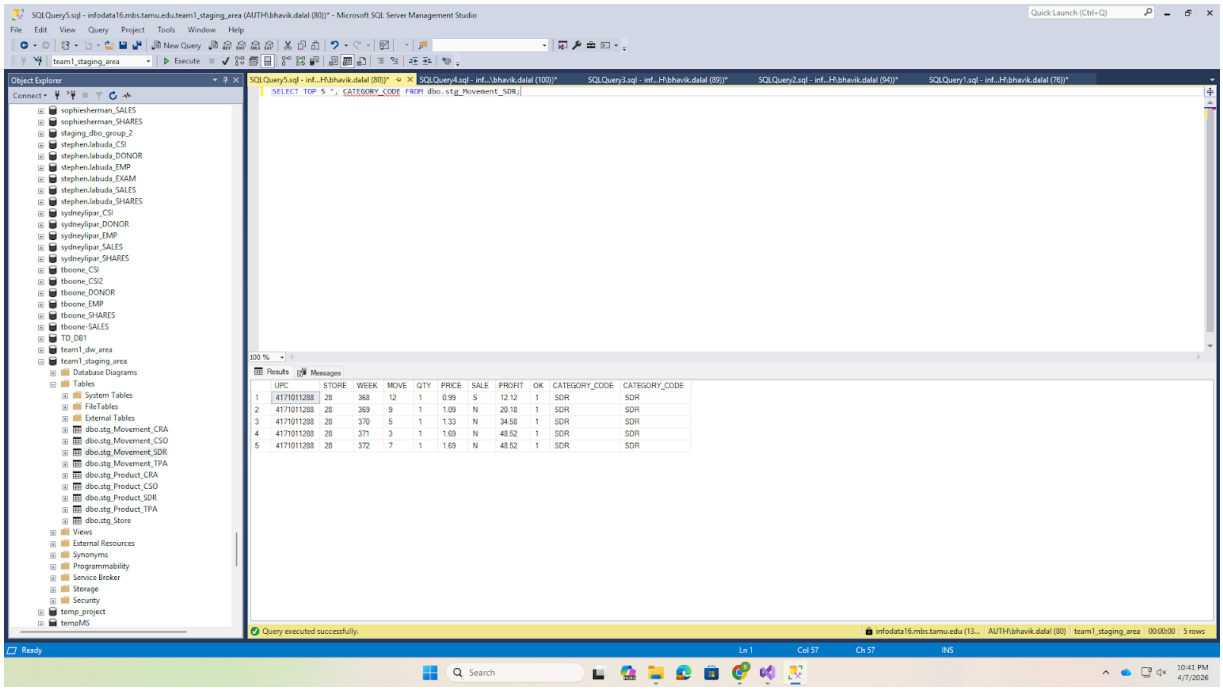


After Loading (staging tables populated):

This output confirms that the staging table has been successfully populated with extracted data.

```
SELECT TOP 5 *, CATEGORY_CODE FROM dbo.stg_Movement_SDR;
```

Screenshot 9: SSMS SELECT TOP 5 showing loaded data

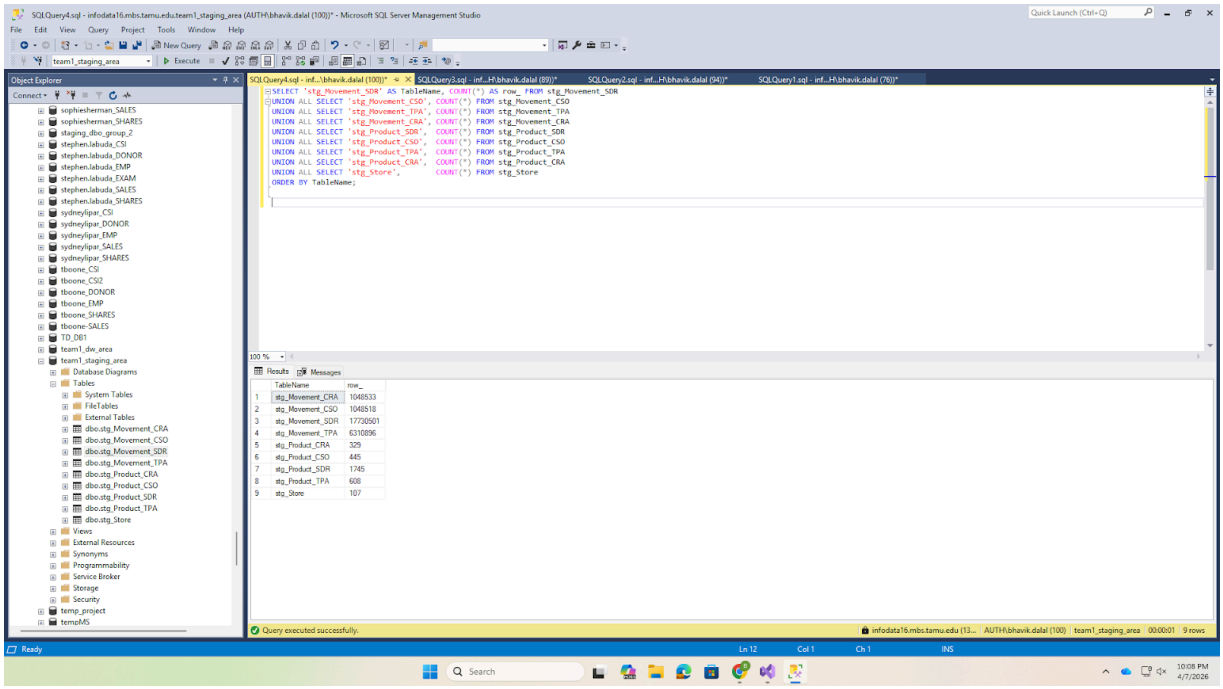


Row Counts After Extraction:

This query validates that all staging tables have been populated with the expected number of rows after extraction.

```
SELECT 'stg_Movement_SDR' AS TableName, COUNT(*) AS RowCount FROM stg_Movement_SDR
UNION ALL SELECT 'stg_Movement_CSO', COUNT(*) FROM stg_Movement_CSO
UNION ALL SELECT 'stg_Movement_TPA', COUNT(*) FROM stg_Movement_TPA
UNION ALL SELECT 'stg_Movement_CRA', COUNT(*) FROM stg_Movement_CRA
UNION ALL SELECT 'stg_Product_SDR',  COUNT(*) FROM stg_Product_SDR
UNION ALL SELECT 'stg_Product_CSO',  COUNT(*) FROM stg_Product_CSO
UNION ALL SELECT 'stg_Product_TPA',  COUNT(*) FROM stg_Product_TPA
UNION ALL SELECT 'stg_Product_CRA',  COUNT(*) FROM stg_Product_CRA
UNION ALL SELECT 'stg_Store',        COUNT(*) FROM stg_Store
ORDER BY TableName;
```

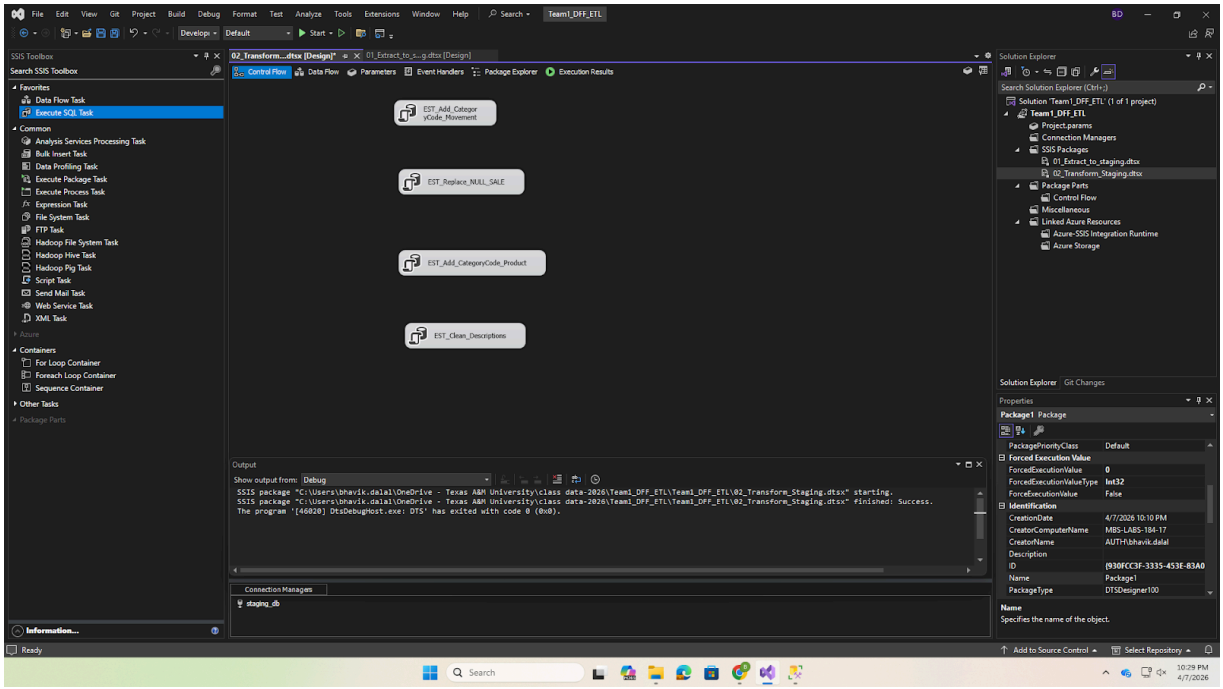
Screenshot 10: SSMS row counts for all staging tables



6.5 SSIS Package 2: Transform Staging

Control Flow: Package 2 contains Execute SQL Tasks for staging-area transformations (T1, T2, T5–T7).

Screenshot 11: SSIS Control Flow Package 2



The SQL transformations executed in this package include:

T1 Add CATEGORY_CODE:

```
ALTER TABLE dbo.stg_Movement_SDR ADD CATEGORY_CODE CHAR(3);
UPDATE dbo.stg_Movement_SDR SET CATEGORY_CODE = 'SDR';
-- (repeated for CSO, TPA, CRA)
```

T2 Replace NULL SALE:

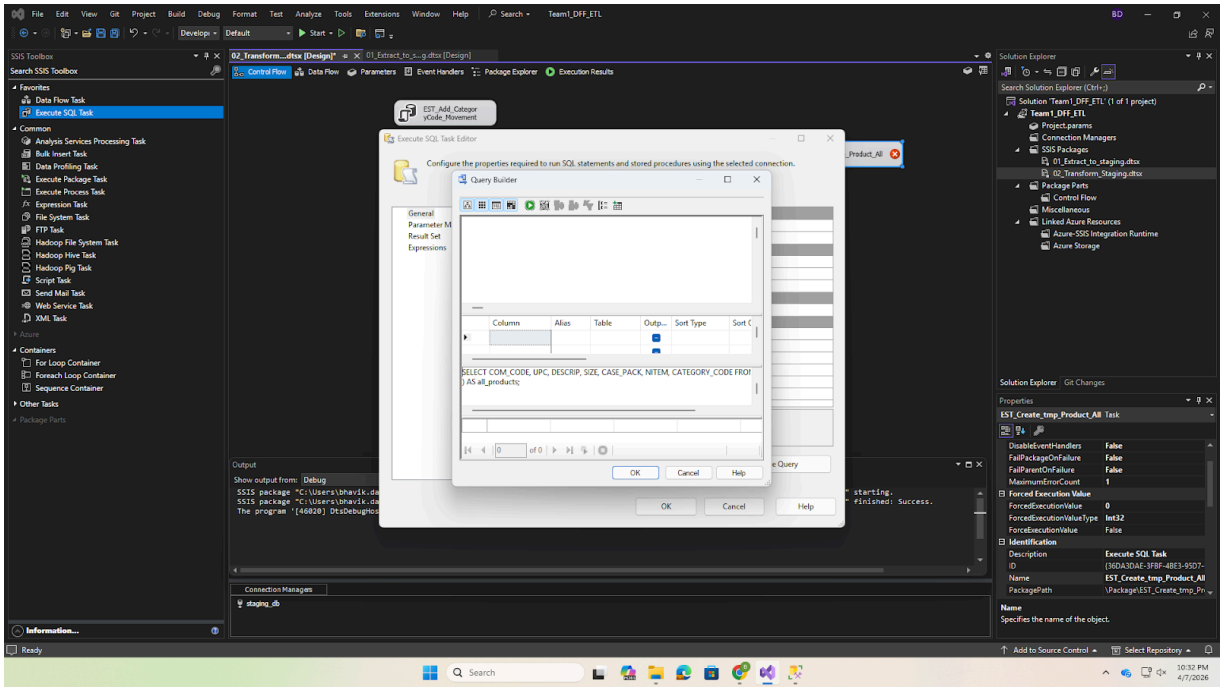
```
UPDATE dbo.stg_Movement_SDR SET SALE = 'N' WHERE SALE IS NULL OR LTRIM(RTRIM(SALE)) = '';
```

T5 Create tmp_Product_All:

```
SELECT COM_CODE, UPC, DESCRIP, SIZE, CASE_PACK, NITEM, CATEGORY_CODE
INTO dbo.tmp_Product_All
FROM (
   SELECT * FROM stg_Product_SDR UNION ALL
   SELECT * FROM stg_Product_CSO UNION ALL
   SELECT * FROM stg_Product_TPA UNION ALL
   SELECT * FROM stg_Product_CRA
) AS all_products;
```

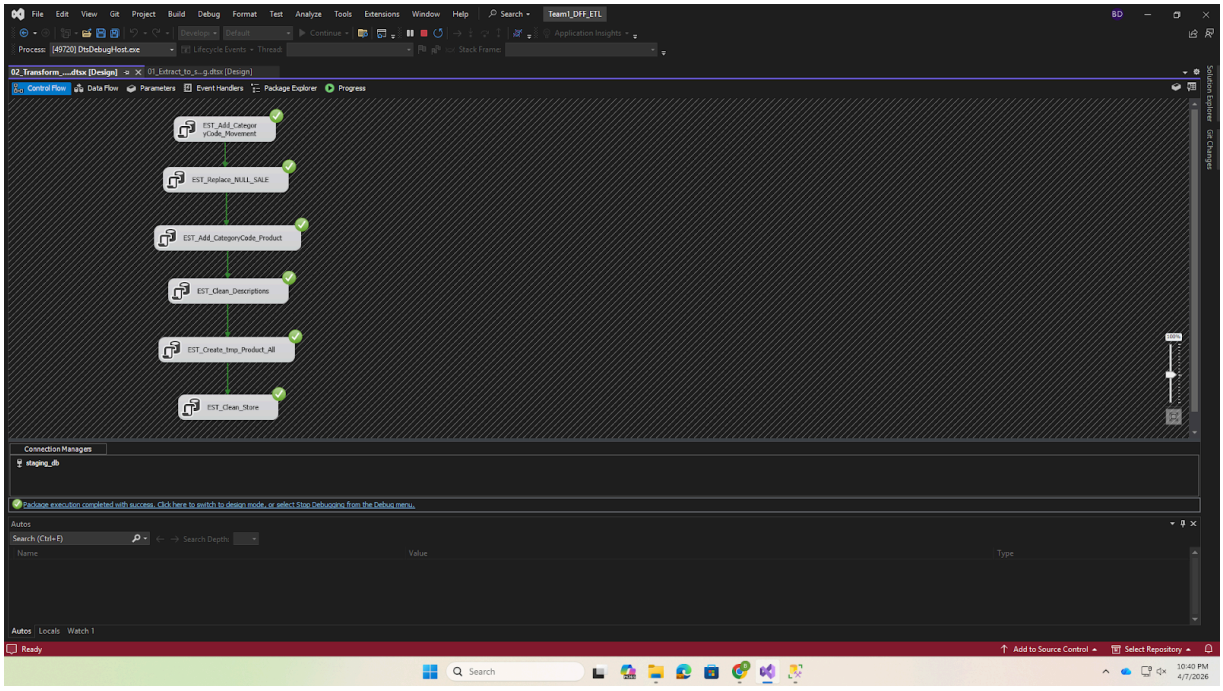
Screenshot 12: SSIS Execute SQL Task showing transformation SQL

This package executes SQL-based transformations in the staging area, including data cleansing, handling missing values, and deriving new attributes according to the defined ETL rules.



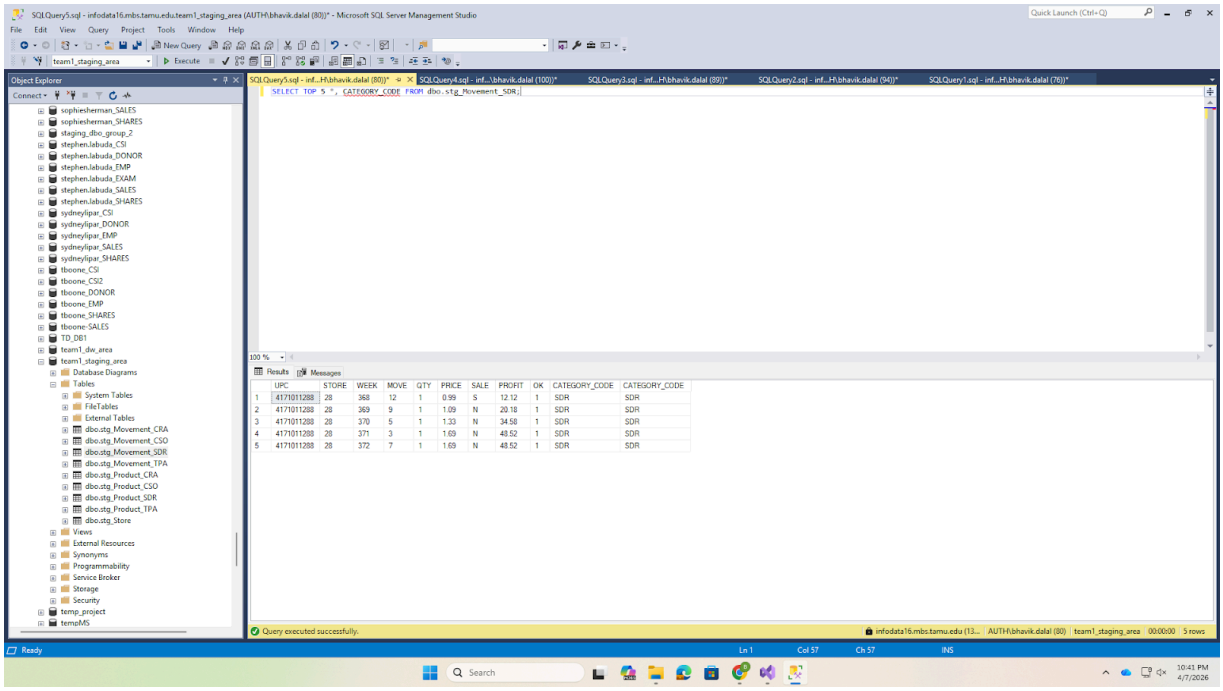
This confirms that all transformation steps were executed successfully.

Screenshot 13: SSIS Package 2 execution all green checkmarks



This verifies that the derived CATEGORY_CODE attribute has been correctly added to the staging tables.

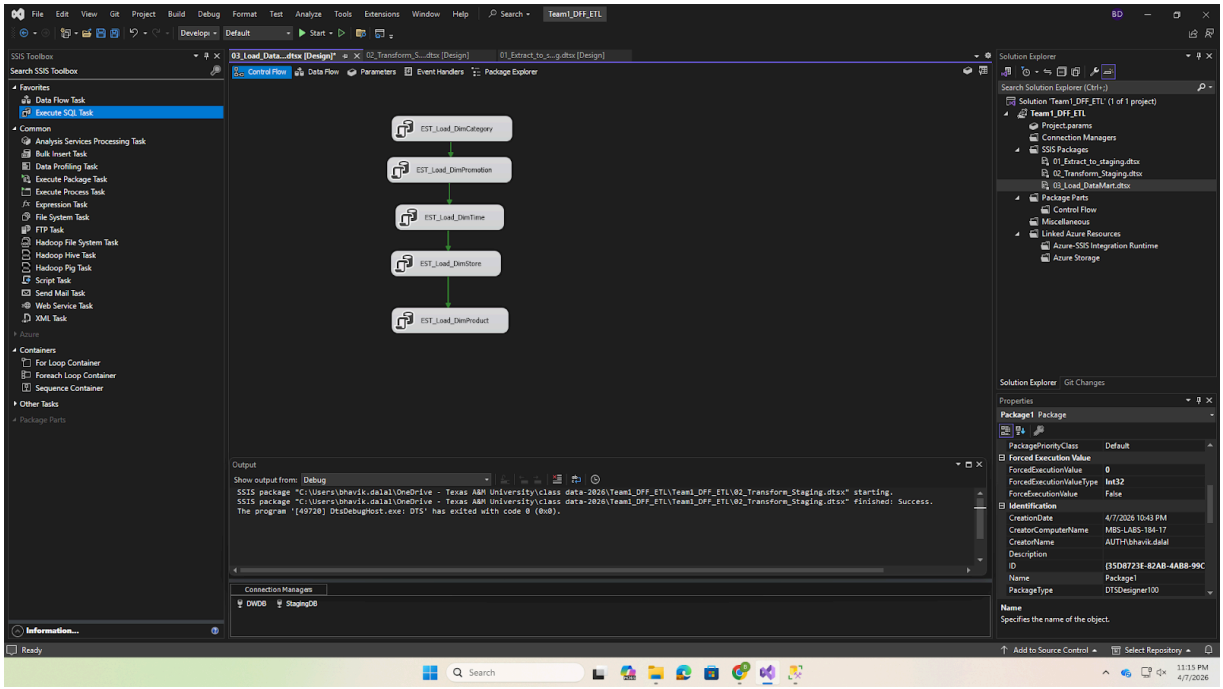
Screenshot 14: SSMS showing CATEGORY_CODE column added to staging table



6.6 SSIS Package 3: Load Data Mart

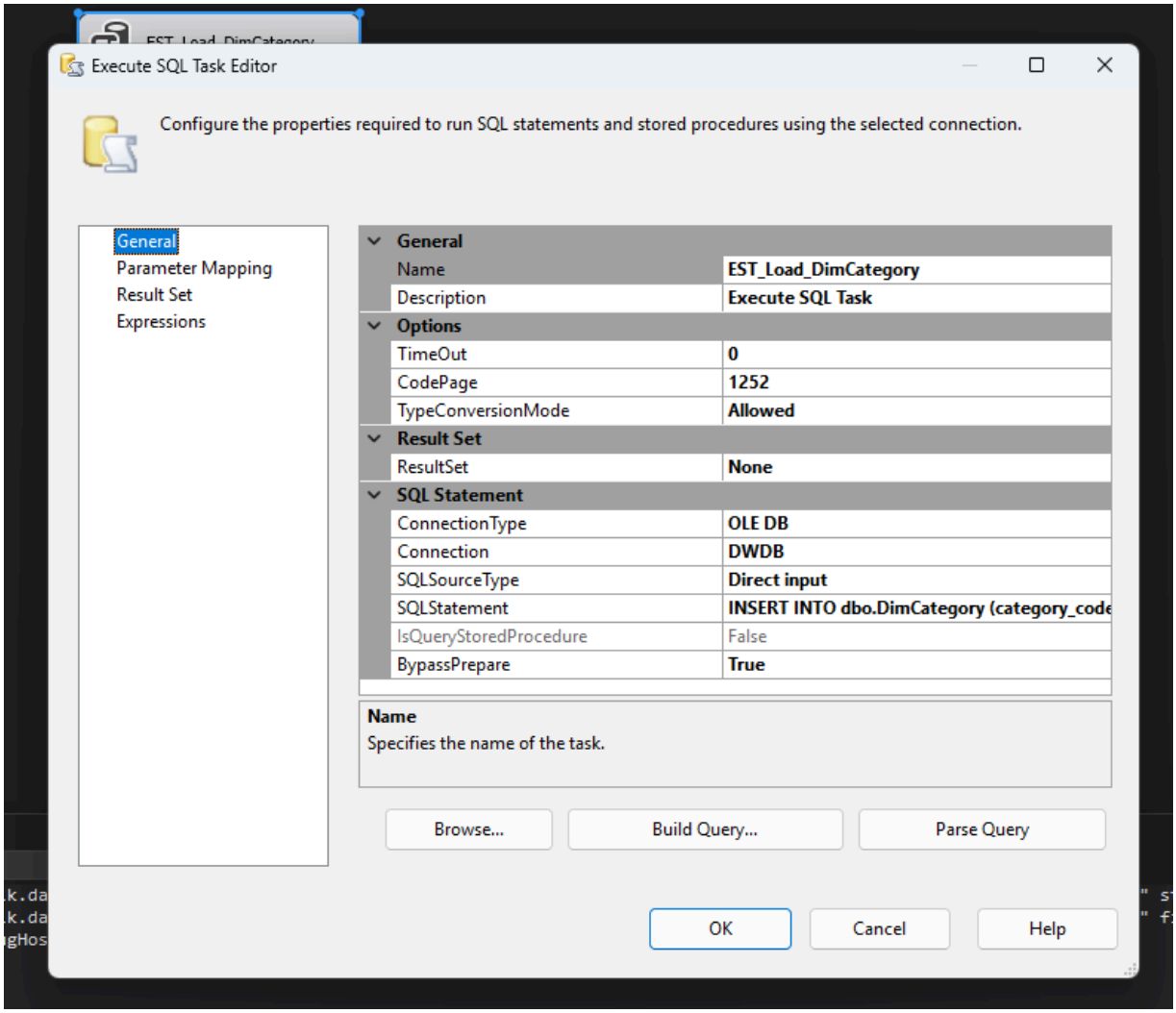
Control Flow: Package 3 loads dimensions first, then the fact table, then drops temporary tables.

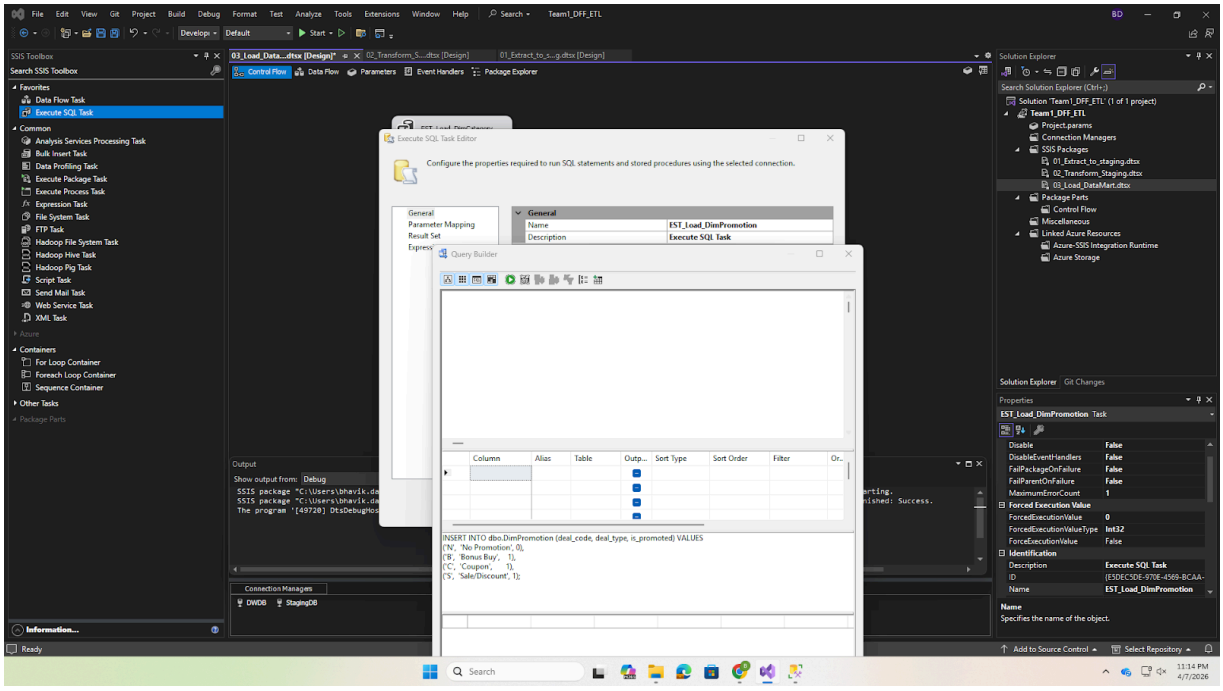
Screenshot 15: SSIS Control Flow Package 3



Sample Execute SQL Task (Dimension Loading):

Screenshot 16: SSIS Package 3 -- Execute SQL Task for dimension loading

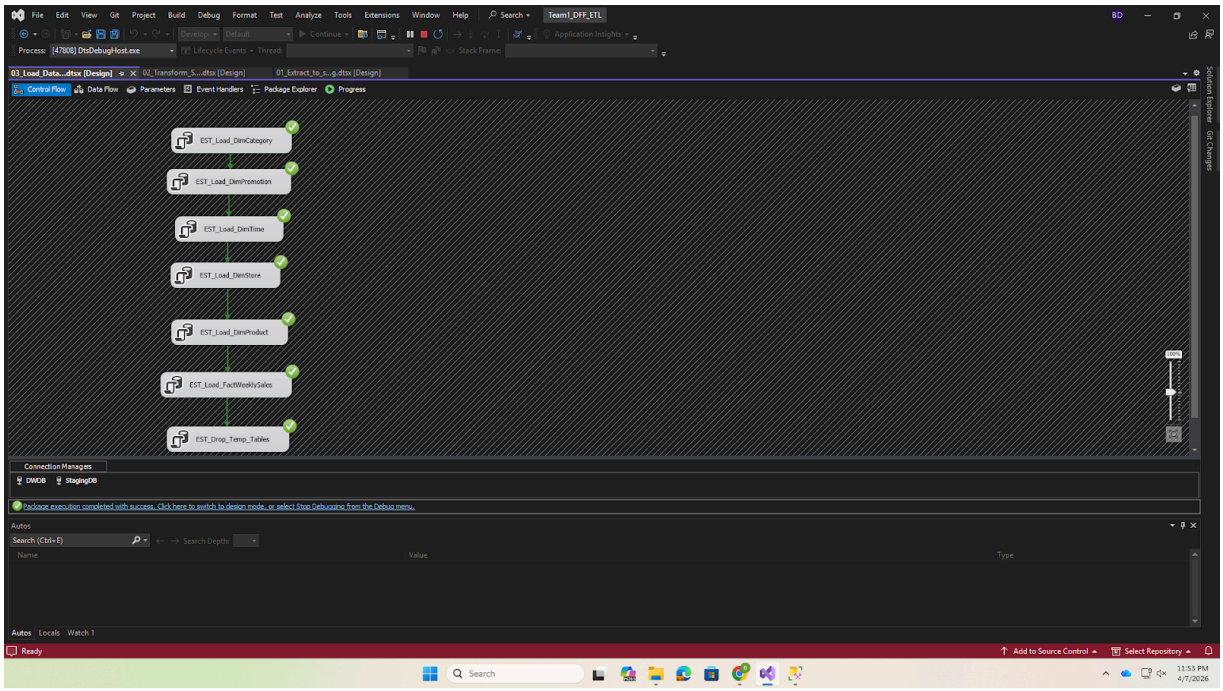

Execute SQL Task Editor
Configure the properties required to run SQL statements and stored procedures using the selected connection.
General
Parameter Mapping
Result Set
Expressions
General
Name
EST_Load_DimCategory
Description
Execute SQL Task
Options
TimeOut
0
CodePage
1252
TypeConversionMode
Allowed
Result Set
ResultSet
None
SQL Statement
ConnectionType
OLE DB
Connection
DWDB
SQLSourceType
Direct input
SQLStatement
INSERT INTO dbo.DimCategory (category_code
IsQueryStoredProcedure
False
BypassPrepare
True
Name
Specifies the name of the task.
Browse...
Build Query...
Parse Query
OK
Cancel
Help



Execution Result:

This confirms that all loading tasks completed successfully without errors.

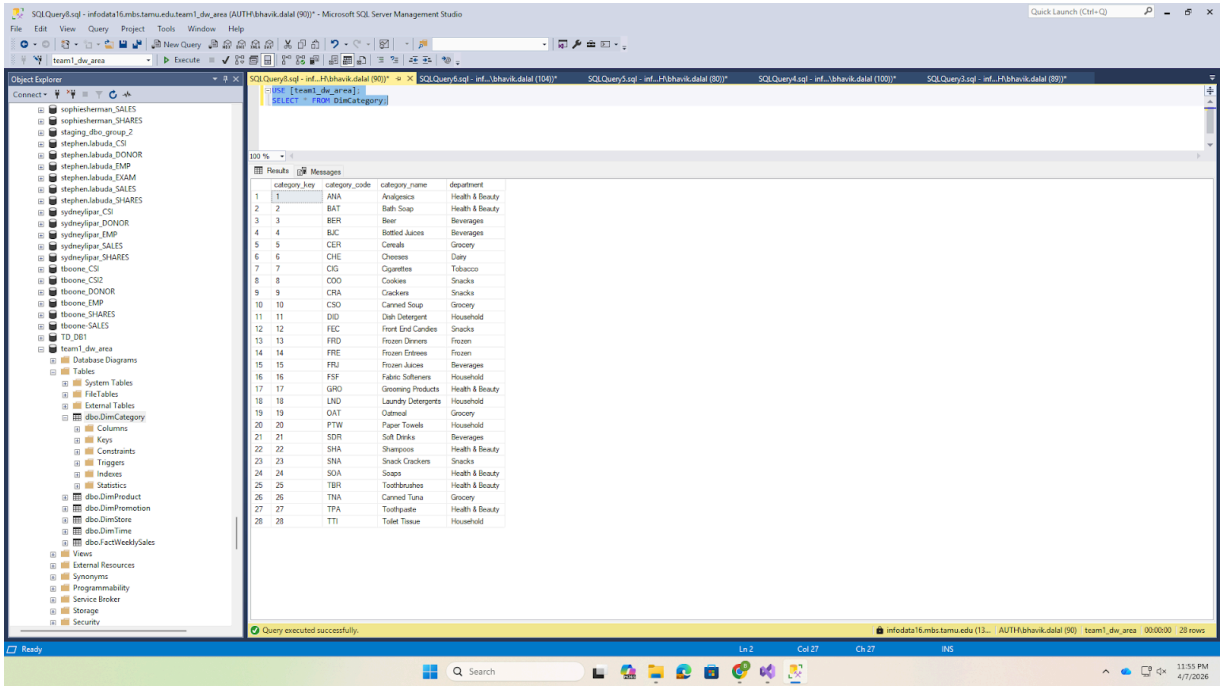
Screenshot 17: SSIS Package 3 execution all green checkmarks



Dimension Loading Results

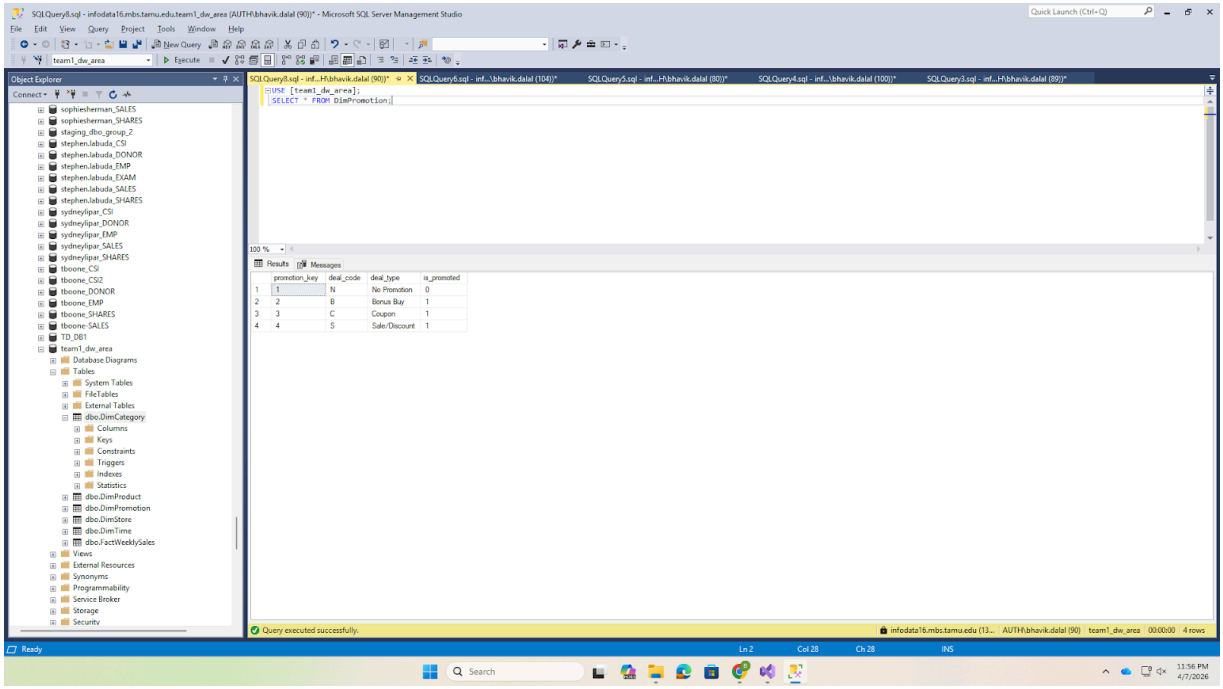
DimCategory (28 rows):

*Screenshot 18: SSMS SELECT * FROM DimCategory showing 28 rows*



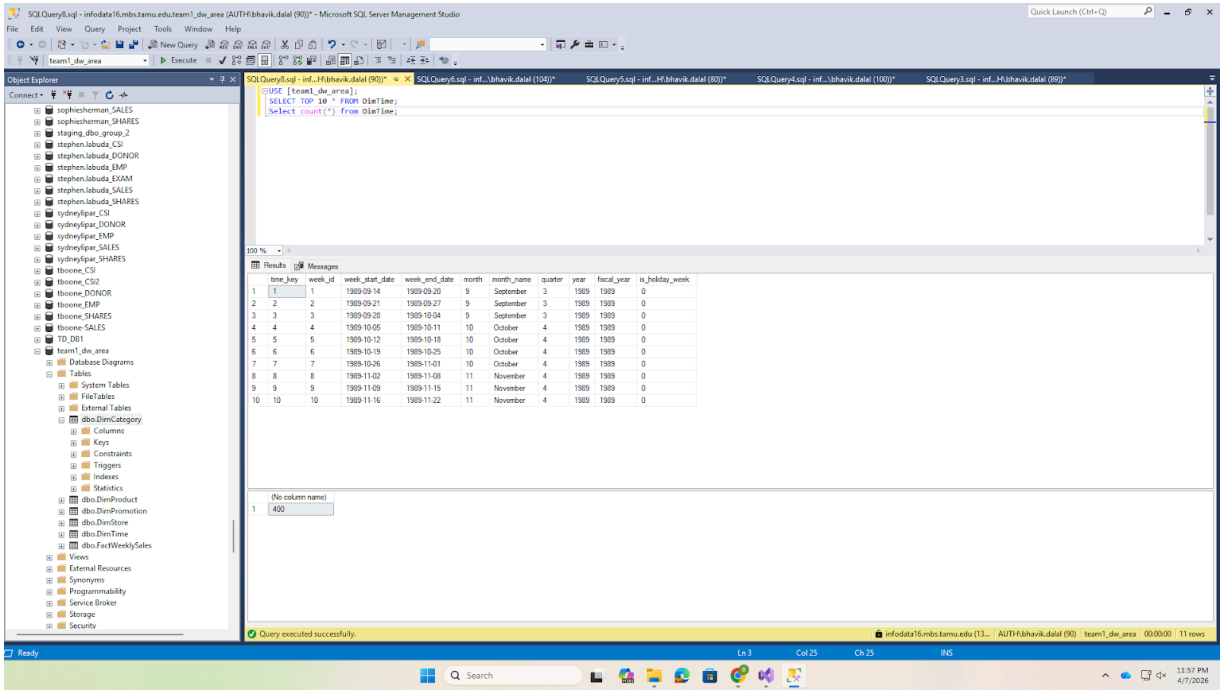
DimPromotion (4 rows):

*Screenshot 19: SSMS SELECT * FROM DimPromotion showing 4 rows*



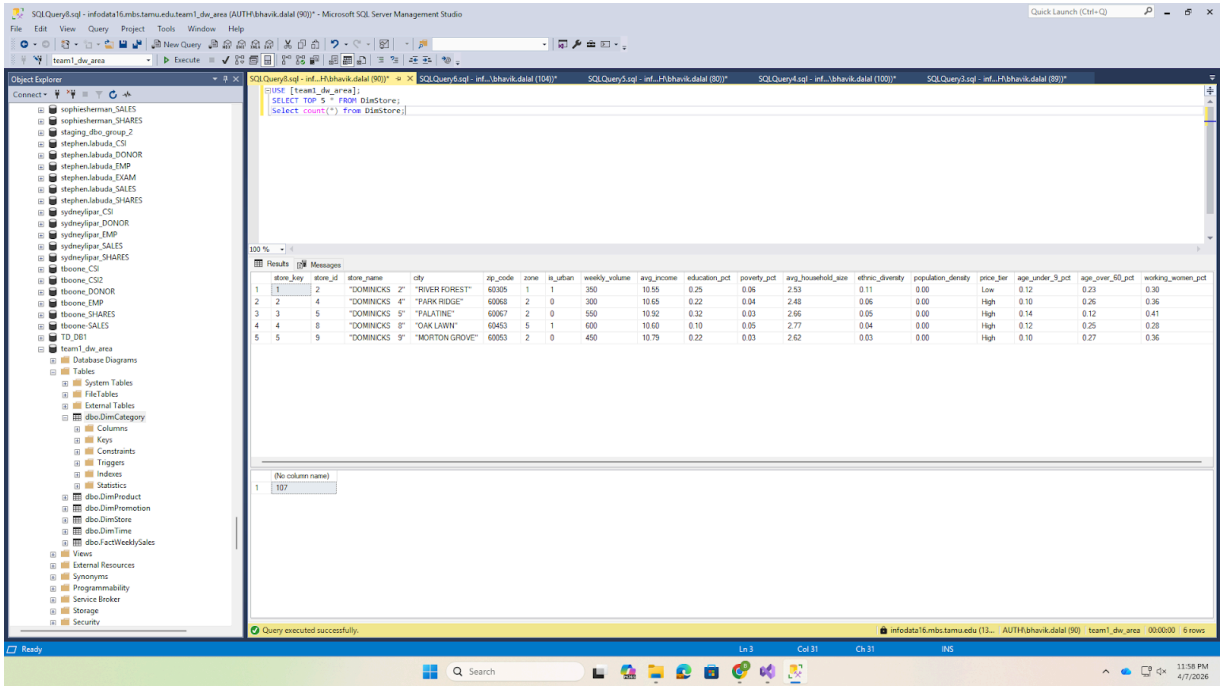
DimTime (~400 rows):

*Screenshot 20: SSMS SELECT TOP 10 * FROM DimTime*



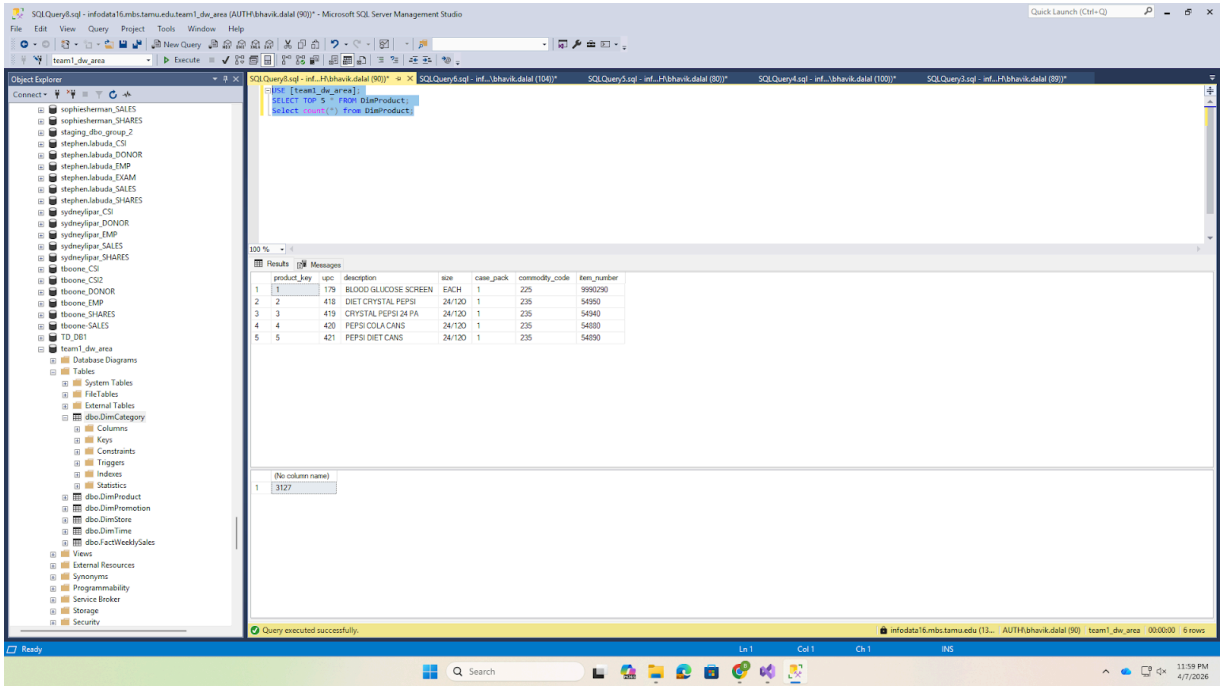
DimStore (~107 rows):

*Screenshot 21: SSMS SELECT TOP 10 * FROM DimStore showing demographics*



DimProduct (~3,112 rows):

*Screenshot 22: SSMS SELECT TOP 10 * FROM DimProduct*



Fact Table Loading Results

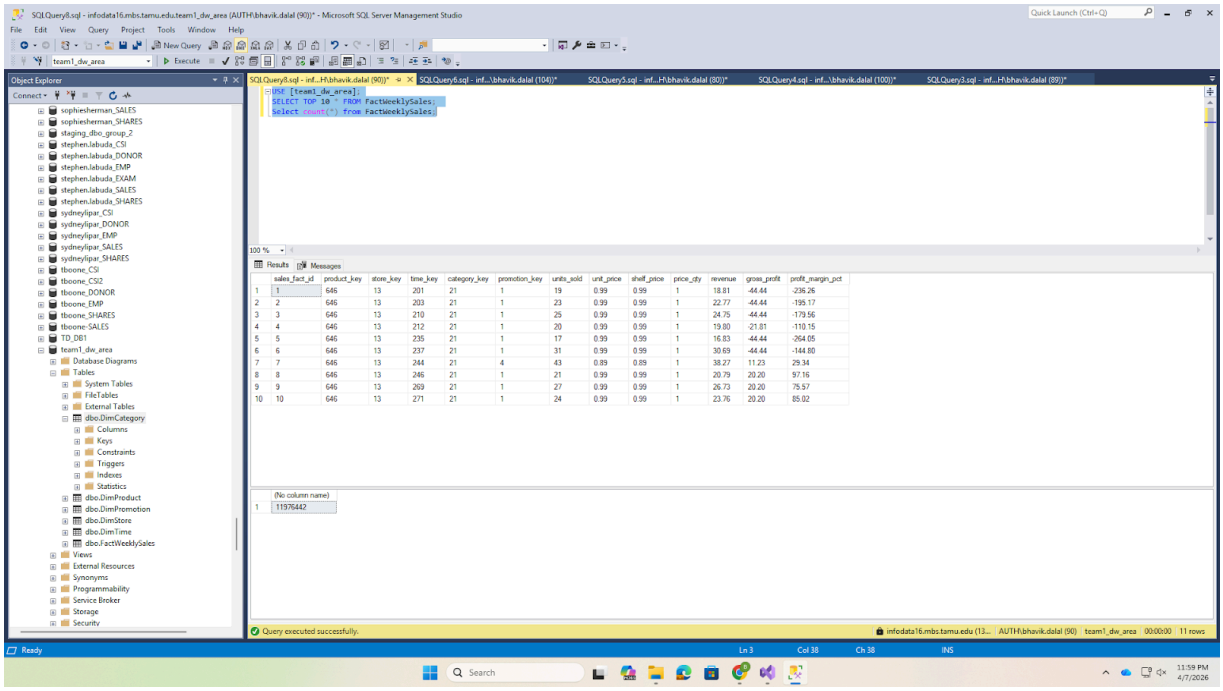
FactWeeklySales (~34.6M rows):

The complete loading SQL JOINs all four movement staging tables to all five dimension tables:

```
INSERT INTO dbo.FactWeeklySales (product_key, store_key, time_key, category_key,
   promotion_key, units_sold, unit_price, shelf_price, price_qty,
   revenue, gross_profit, profit_margin_pct)
SELECT dp.product_key, ds.store_key, dt.time_key, dc.category_key, dpr.promotion_key,
   sm.MOVE, CAST(sm.PRICE / sm.QTY AS DECIMAL(8,2)),
   CAST(sm.PRICE AS DECIMAL(8,2)), sm.QTY,
   CAST(sm.MOVE * (sm.PRICE / sm.QTY) AS DECIMAL(12,2)),
   CAST(sm.PROFIT AS DECIMAL(10,2)),
   CAST((sm.PROFIT / (sm.MOVE * (sm.PRICE / sm.QTY))) * 100 AS DECIMAL(5,2))
FROM (
   SELECT UPC, STORE, WEEK, MOVE, QTY, PRICE, SALE, PROFIT, OK, CATEGORY_CODE
   FROM [team1_staging_area].dbo.stg_Movement_SDR WHERE OK = 1 AND PRICE > 0
   UNION ALL
   SELECT UPC, STORE, WEEK, MOVE, QTY, PRICE, SALE, PROFIT, OK, CATEGORY_CODE
   FROM [team1_staging_area].dbo.stg_Movement_CSO WHERE OK = 1 AND PRICE > 0
   UNION ALL
   SELECT UPC, STORE, WEEK, MOVE, QTY, PRICE, SALE, PROFIT, OK, CATEGORY_CODE
   FROM [team1_staging_area].dbo.stg_Movement_TPA WHERE OK = 1 AND PRICE > 0
   UNION ALL
   SELECT UPC, STORE, WEEK, MOVE, QTY, PRICE, SALE, PROFIT, OK, CATEGORY_CODE
   FROM [team1_staging_area].dbo.stg_Movement_CRA WHERE OK = 1 AND PRICE > 0
) sm
INNER JOIN dbo.DimProduct dp   ON sm.UPC = dp.upc
INNER JOIN dbo.DimStore ds     ON sm.STORE = ds.store_id
INNER JOIN dbo.DimTime dt      ON sm.WEEK = dt.week_id
```

```
INNER JOIN dbo.DimCategory dc  ON sm.CATEGORY_CODE = dc.category_code
INNER JOIN dbo.DimPromotion dpr ON sm.SALE = dpr.deal_code;
```

*Screenshot 23: SSMS SELECT TOP 10 * FROM FactWeeklySales + COUNT*

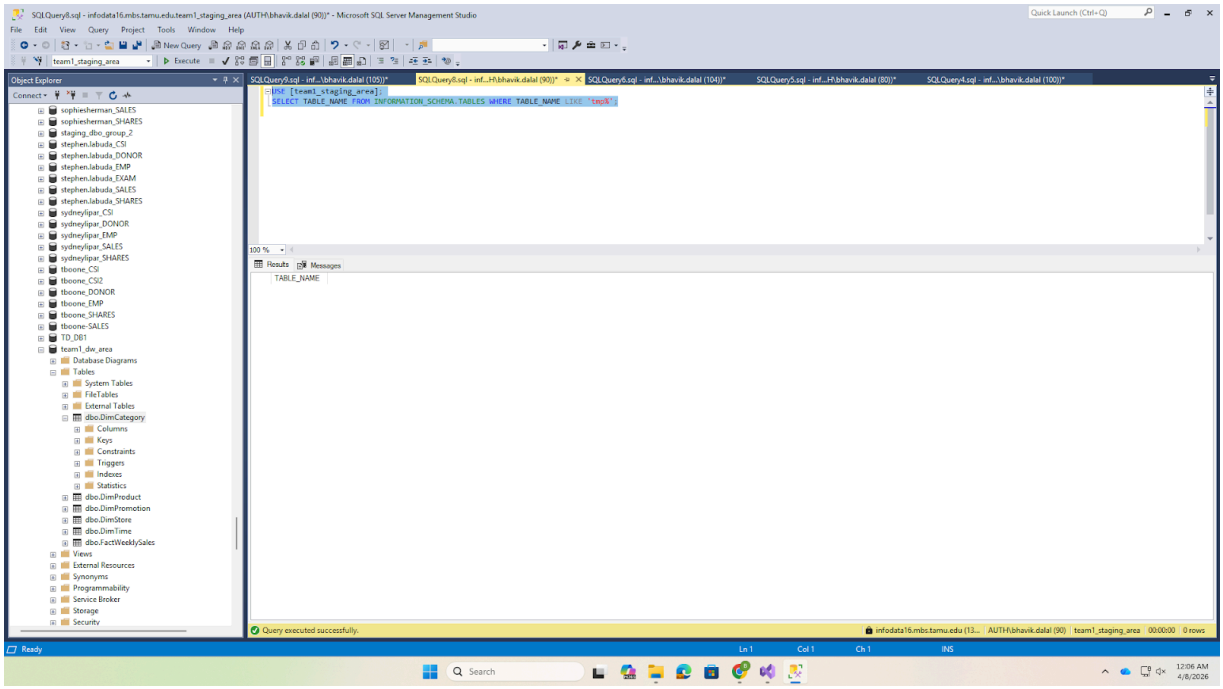


6.7 Temporary Table Cleanup

After all dimension and fact tables were loaded, temporary tables were dropped:

```
DROP TABLE [team1_staging_area].dbo.tmp_Product_All;
```

Screenshot 24: SSMS Object Explorer tmp_Product_All no longer visible



6.8 Granularity Discussion

The **grain** of the FactWeeklySales table is defined as: _"one row per unique combination of UPC, Store, and Week."_ This grain was chosen because:

6. **It matches the source data grain** — each row in the Movement CSV files represents one UPC at one store for one week.
7. **It supports all 5 selected BQs** — BQ2 aggregates by week (roll-up on UPC and store), BQ3/BQ4 slice by promotion type, BQ8 aggregates by store (roll-up on UPC and week), and BQ9 aggregates by UPC and week (roll-up on store).
8. **It preserves maximum analytical flexibility** — the atomic grain allows users to drill down to any combination of product, store, time, category, and promotion.

A coarser grain (e.g., Category × Store × Week) would make BQ9 impossible because individual product rankings require UPC-level data.

6.9 Business Question Verification

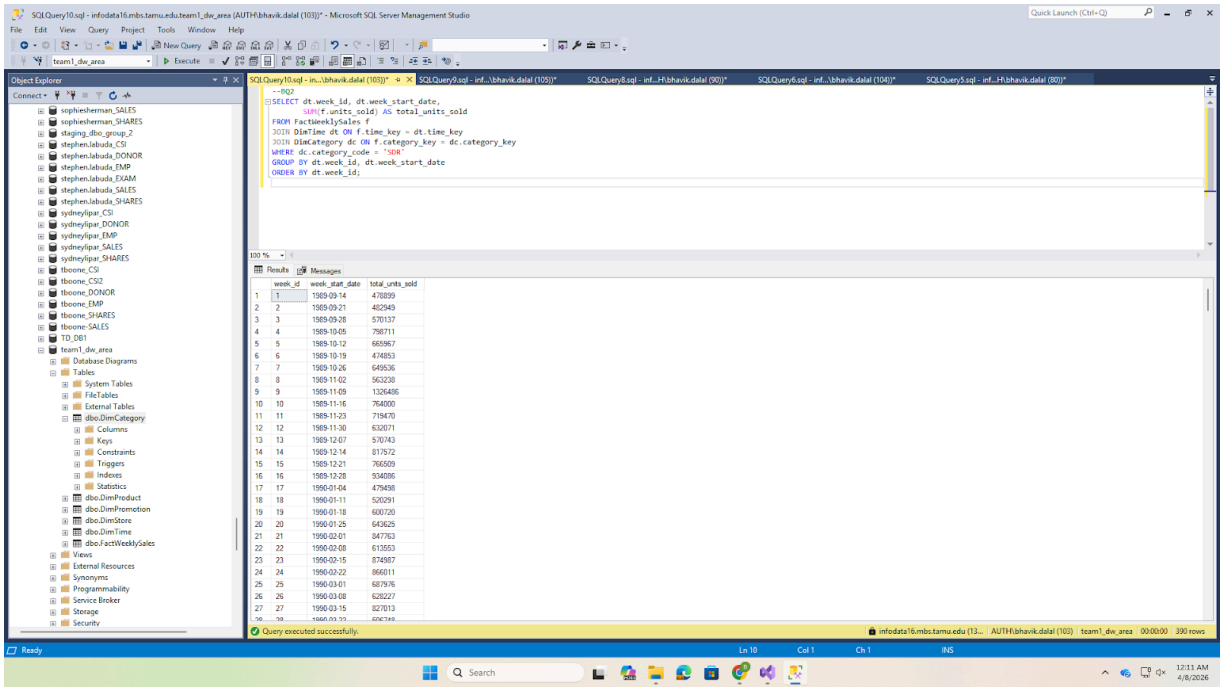
After completing the ETL, we verified the data mart can answer all 5 selected BQs:

BQ2 — Weekly Soft Drink Sales:

```
SELECT dt.week_id, dt.week_start_date,
       SUM(f.units_sold) AS total_units_sold
FROM FactWeeklySales f
JOIN DimTime dt ON f.time_key = dt.time_key
JOIN DimCategory dc ON f.category_key = dc.category_key
WHERE dc.category_code = 'SDR'
```

```
GROUP BY dt.week_id, dt.week_start_date
ORDER BY dt.week_id;
```

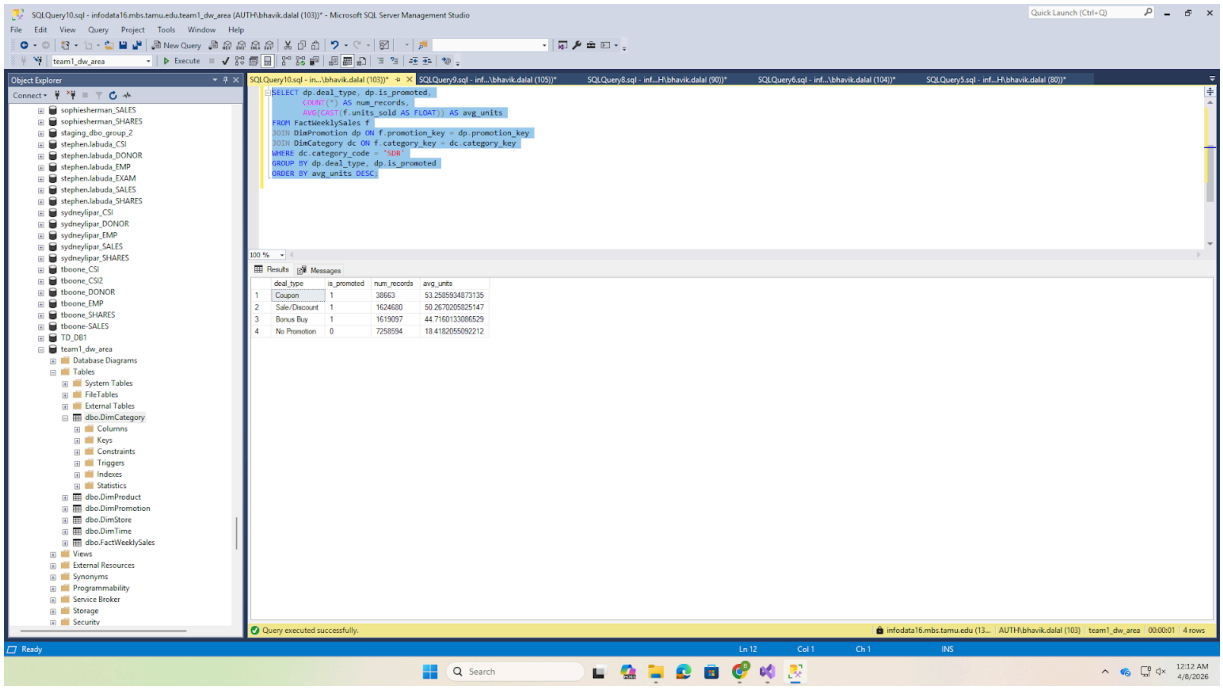
Screenshot 25: SSMS BQ2 query results



BQ3 Promotion vs Non-Promotion:

```
SELECT dp.deal_type, dp.is_promoted,
    COUNT(*) AS num_records,
    AVG(CAST(f.units_sold AS FLOAT)) AS avg_units
FROM FactWeeklySales f
JOIN DimPromotion dp ON f.promotion_key = dp.promotion_key
JOIN DimCategory dc ON f.category_key = dc.category_key
WHERE dc.category_code = 'SDR'
GROUP BY dp.deal_type, dp.is_promoted
ORDER BY avg_units DESC;
```

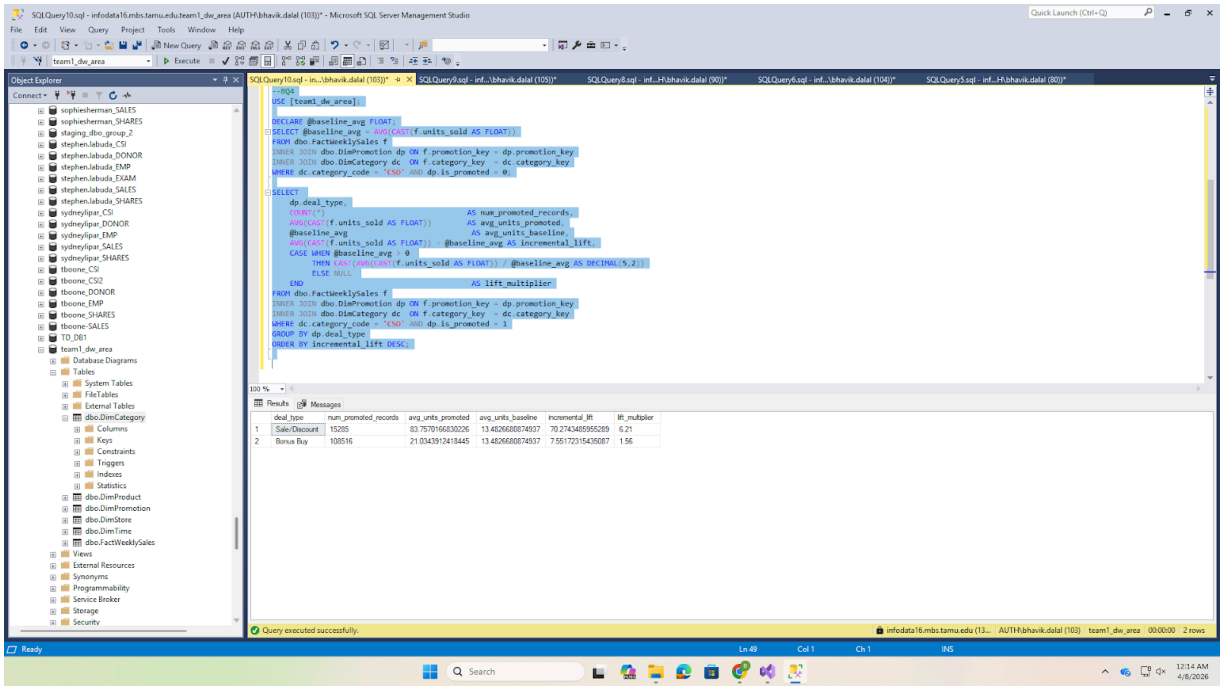
Screenshot 26: SSMS BQ3 query results



BQ4 Promotion Lift by Type (Canned Soup):

```
DECLARE @baseline FLOAT;
SELECT @baseline = AVG(CAST(f.units_sold AS FLOAT))
FROM FactWeeklySales f
JOIN DimPromotion dp ON f.promotion_key = dp.promotion_key
JOIN DimCategory dc ON f.category_key = dc.category_key
WHERE dc.category_code = 'CSO' AND dp.is_promoted = 0;

SELECT dp.deal_type,
   AVG(CAST(f.units_sold AS FLOAT)) AS avg_promoted,
   @baseline AS avg_baseline,
   AVG(CAST(f.units_sold AS FLOAT)) - @baseline AS lift
FROM FactWeeklySales f
JOIN DimPromotion dp ON f.promotion_key = dp.promotion_key
JOIN DimCategory dc ON f.category_key = dc.category_key
WHERE dc.category_code = 'CSO' AND dp.is_promoted = 1
GROUP BY dp.deal_type
ORDER BY lift DESC;
```

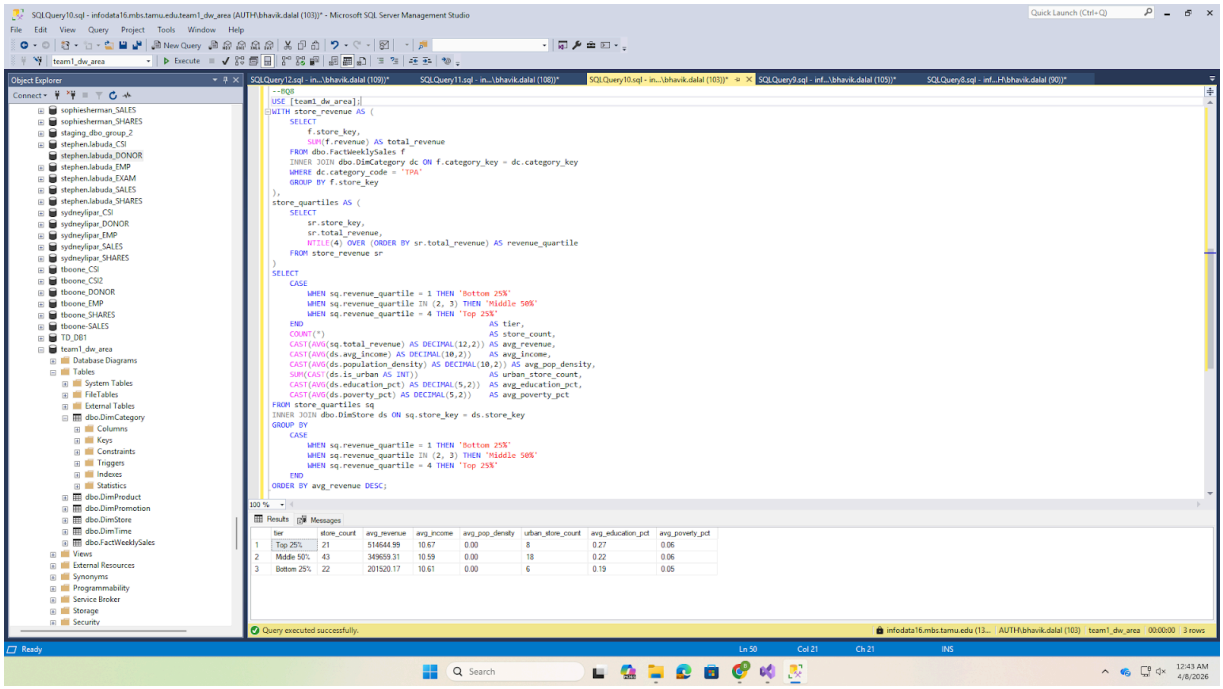
Screenshot 27: SSMS BQ4 query results



BQ8 Store Quartile Tiers (Toothpaste):

```
WITH store_rev AS (
   SELECT f.store_key, SUM(f.revenue) AS total_rev
   FROM FactWeeklySales f
   JOIN DimCategory dc ON f.category_key = dc.category_key
   WHERE dc.category_code = 'TPA'
   GROUP BY f.store_key
),
quartiles AS (
   SELECT store_key, total_rev,
          NTILE(4) OVER (ORDER BY total_rev) AS q
   FROM store_rev
)
SELECT CASE WHEN q=1 THEN 'Bottom 25%' WHEN q IN (2,3) THEN 'Middle 50%'
            ELSE 'Top 25%' END AS tier,
      COUNT(*) AS stores, AVG(total_rev) AS avg_rev,
      AVG(ds.avg_income) AS avg_income, SUM(CAST(ds.is_urban AS INT)) AS urban_cnt
FROM quartiles qt
JOIN DimStore ds ON qt.store_key = ds.store_key
GROUP BY CASE WHEN q=1 THEN 'Bottom 25%' WHEN q IN (2,3) THEN 'Middle 50%'
              ELSE 'Top 25%' END
ORDER BY avg_rev DESC;
```

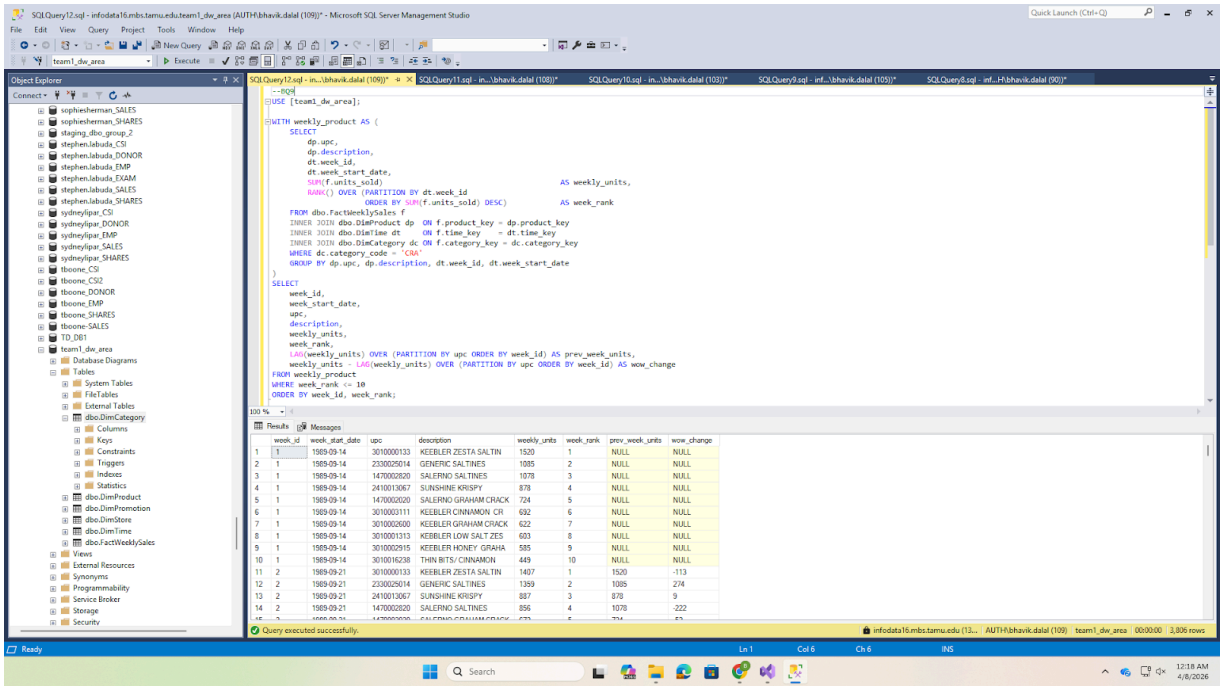
Screenshot 28: SSMS BQ8 query results



BQ9 Top 10 Weekly Cracker Products with WoW Change:

```
WITH weekly AS (
   SELECT dp.upc, dp.description, dt.week_id,
        SUM(f.units_sold) AS wk_units,
        RANK() OVER (PARTITION BY dt.week_id ORDER BY SUM(f.units_sold) DESC) AS rnk
   FROM FactWeeklySales f
   JOIN DimProduct dp ON f.product_key = dp.product_key
   JOIN DimTime dt ON f.time_key = dt.time_key
   JOIN DimCategory dc ON f.category_key = dc.category_key
   WHERE dc.category_code = 'CRA'
   GROUP BY dp.upc, dp.description, dt.week_id
)
SELECT week_id, upc, description, wk_units, rnk,
      LAG(wk_units) OVER (PARTITION BY upc ORDER BY week_id) AS prev_wk,
      wk_units - LAG(wk_units) OVER (PARTITION BY upc ORDER BY week_id) AS wow
FROM weekly WHERE rnk <= 10
ORDER BY week_id, rnk;
```

Screenshot 29: SSMS BQ9 query results



6.10 Final Data Mart Summary

Table	Rows
DimCategory	28
DimPromotion	4
DimTime	400
DimStore	107
DimProduct	3127
FactWeeklySales	14921365

Section 7: References

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11. Hoch, S. J., Drèze, X., & Purk, M. E. (1994). EDLP, Hi-Lo, and Margin Arithmetic. *Journal of Marketing*, 58(4), 16–27.
12. Hoch, S. J., Kim, B. D., Montgomery, A. L., & Rossi, P. E. (1995). Determinants of Store-Level Price Elasticity. *Journal of Marketing Research*, 32(1), 17–29.
13. Kimball, R., & Ross, M. (2013). *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling* (3rd ed.). John Wiley & Sons.

14. Kilts Center for Marketing (2013). *Dominick's Data Manual*. University of Chicago Booth School of Business. Retrieved from https://www.chicagobooth.edu/research/kilts/datasets/dominicks
15. Mehta, S., & Ma, P. (2012). A High Dimensional Data Analysis Approach for Retail Data. *Proceedings of the ACM SIGKDD*.
16. Montgomery, A. L. (1997). Creating Micro-Marketing Pricing Strategies Using Supermarket Scanner Data. *Marketing Science*, 16(4), 315–337.

Section 8: Appendix

Appendix A: Complete SQL Scripts

The complete SQL logic for the implementation is provided below.

01_create_databases.sql

```
-- ============================================================
-- Script 01: Create Databases
-- DFF Data Warehouse Project  Report 3 (ISTM 637, Spring 2026)
-- Run this in SSMS connected to SQL Server 2016
-- ============================================================

-- Step 1: Create the Staging Area database
-- This database holds raw imported CSV data and temporary
-- transformation tables. It is purged after ETL is complete.
IF NOT EXISTS (SELECT name FROM sys.databases WHERE name = 'team1_staging_area')
BEGIN
    CREATE DATABASE [team1_staging_area];
    PRINT 'Database [team1_staging_area] created successfully.';
END
ELSE
    PRINT 'Database [team1_staging_area] already exists.';
GO

-- Step 2: Create the Data Mart / Presentation Server database
-- This database holds the final star schema (dimension + fact tables)
-- that end users query for business intelligence.
IF NOT EXISTS (SELECT name FROM sys.databases WHERE name = 'team1_dw_area')
BEGIN
    CREATE DATABASE [team1_dw_area];
    PRINT 'Database [team1_dw_area] created successfully.';
END
ELSE
    PRINT 'Database [team1_dw_area] already exists.';
GO

-- Verification: List both databases
SELECT name, create_date
FROM sys.databases
WHERE name IN ('team1_staging_area', 'team1_dw_area');
GO
```

02_create_staging_tables.sql

```
-- ============================================================
-- Script 02: Create Staging Tables
-- DFF Data Warehouse Project  Report 3
```

```
-- Run in SSMS after Script 01. These tables receive raw CSV data.
-- ============================================================
USE [team1_staging_area];
GO

-- ------------------------------------------------------------
-- Movement Staging Tables (one per category)
-- Source: Movement CSV files (comma-delimited, encoding 1252)
-- Columns match the raw CSV structure exactly
-- ------------------------------------------------------------

-- Soft Drinks (SDR)  17.7 million rows from wsdr.csv
CREATE TABLE dbo.stg_Movement_SDR (
    UPC          BIGINT,
    STORE        INT,
    WEEK         INT,
    MOVE         INT,
    QTY          INT,
    PRICE        FLOAT,
    SALE         VARCHAR(5),
    PROFIT       FLOAT,
    OK           INT
);
GO

-- Canned Soup (CSO)  7.0 million rows from WCSO-Done.csv
CREATE TABLE dbo.stg_Movement_CSO (
    UPC          BIGINT,
    STORE        INT,
    WEEK         INT,
    MOVE         INT,
    QTY          INT,
    PRICE        FLOAT,
    SALE         VARCHAR(5),
    PROFIT       FLOAT,
    OK           INT
);
GO

-- Toothpaste (TPA)  6.3 million rows from WTPA_done.csv
CREATE TABLE dbo.stg_Movement_TPA (
    UPC          BIGINT,
    STORE        INT,
    WEEK         INT,
    MOVE         INT,
    QTY          INT,
    PRICE        FLOAT,
    SALE         VARCHAR(5),
    PROFIT       FLOAT,
    OK           INT
);
GO

-- Crackers (CRA)  3.6 million rows from Done-WCRA.csv
CREATE TABLE dbo.stg_Movement_CRA (
    UPC          BIGINT,
    STORE        INT,
    WEEK         INT,
    MOVE         INT,
    QTY          INT,
    PRICE        FLOAT,
    SALE         VARCHAR(5),
    PROFIT       FLOAT,
    OK           INT
);
GO
```

```
-- ------------------------------------------------------------
-- Product (UPC) Staging Tables (one per category)
-- Source: UPC CSV files (encoding 1252 / latin-1)
-- ------------------------------------------------------------

-- Soft Drinks UPC  1,746 rows from UPCSDR.csv
CREATE TABLE dbo.stg_Product_SDR (
    COM_CODE      INT,
    UPC           BIGINT,
    DESCRIP       VARCHAR(100),
    SIZE          VARCHAR(30),
    CASE_PACK     INT,
    NITEM         BIGINT
);
GO

-- Canned Soup UPC  453 rows from UPCCSO.csv
CREATE TABLE dbo.stg_Product_CSO (
    COM_CODE      INT,
    UPC           BIGINT,
    DESCRIP       VARCHAR(100),
    SIZE          VARCHAR(30),
    CASE_PACK     INT,
    NITEM         BIGINT
);
GO

-- Toothpaste UPC  608 rows from UPCTPA.csv
CREATE TABLE dbo.stg_Product_TPA (
    COM_CODE      INT,
    UPC           BIGINT,
    DESCRIP       VARCHAR(100),
    SIZE          VARCHAR(30),
    CASE_PACK     INT,
    NITEM         BIGINT
);
GO

-- Crackers UPC  305 rows from UPCCRA.csv
CREATE TABLE dbo.stg_Product_CRA (
    COM_CODE      INT,
    UPC           BIGINT,
    DESCRIP       VARCHAR(100),
    SIZE          VARCHAR(30),
    CASE_PACK     INT,
    NITEM         BIGINT
);
GO

-- ------------------------------------------------------------
-- Store Demographics Staging Table
-- Source: Demographics/DEMO.csv (108 rows, 510 columns)
-- We only stage the columns needed for DimStore
-- ------------------------------------------------------------
CREATE TABLE dbo.stg_Store (
    STORE         VARCHAR(10),
    NAME          VARCHAR(60),
    CITY          VARCHAR(50),
    ZIP           VARCHAR(10),
    ZONE          VARCHAR(10),
    URBAN         VARCHAR(10),
    WEEKVOL       VARCHAR(20),
    INCOME        VARCHAR(20),
    EDUC          VARCHAR(20),
    POVERTY       VARCHAR(20),
```

```
    HSIZEAVG      VARCHAR(20),
    ETHNIC        VARCHAR(20),
    DENSITY       VARCHAR(20),
    AGE9          VARCHAR(20),
    AGE60         VARCHAR(20),
    WORKWOM       VARCHAR(20),
    PRICLOW       VARCHAR(10),
    PRICMED       VARCHAR(10),
    PRICHIGH      VARCHAR(10)
);
GO


-- ------------------------------------------------------------
-- Verification: List all staging tables
-- ------------------------------------------------------------
SELECT TABLE_NAME, TABLE_TYPE
FROM INFORMATION_SCHEMA.TABLES
WHERE TABLE_SCHEMA = 'dbo'
ORDER BY TABLE_NAME;
GO

PRINT '=== All staging tables created successfully ===';
GO
```

03_create_dw_tables.sql

```
-- ================================================================
-- Script 03: Create Data Warehouse (Data Mart) Tables
-- DFF Data Warehouse Project  Report 3
-- Run in SSMS after Script 02. Creates the star schema tables.
-- IMPORTANT: Create DIMENSION tables first, then FACT tables
--            (fact tables have FK references to dimensions).
-- ================================================================
USE [team1_dw_area];
GO

-- ================================
-- DIMENSION TABLES
-- ================================

-- ------------------------------------------------------------
-- DimCategory  28 product categories
-- Grain: One row per product category
-- Cardinality: 28 rows (static, hardcoded)
-- ------------------------------------------------------------
CREATE TABLE dbo.DimCategory (
    category_key    INT IDENTITY(1,1) NOT NULL,
    category_code   CHAR(3)           NOT NULL,
    category_name   VARCHAR(30)       NOT NULL,
    department      VARCHAR(20)       NOT NULL,
    CONSTRAINT PK_DimCategory PRIMARY KEY CLUSTERED (category_key),
    CONSTRAINT UQ_DimCategory_code UNIQUE (category_code)
);
GO

-- ------------------------------------------------------------
-- DimPromotion  4 promotion types
-- Grain: One row per deal type
-- Cardinality: 4 rows (static, hardcoded)
-- ------------------------------------------------------------
CREATE TABLE dbo.DimPromotion (
    promotion_key   INT IDENTITY(1,1) NOT NULL,
    deal_code       CHAR(1)           NOT NULL,
    deal_type       VARCHAR(20)       NOT NULL,
    is_promoted     BIT               NOT NULL,
```

```
    CONSTRAINT PK_DimPromotion PRIMARY KEY CLUSTERED (promotion_key)
);
GO

-- ------------------------------------------------------------
-- DimTime  ~400 weeks (DFF proprietary week IDs)
-- Grain: One row per unique week
-- Week 1 = September 14, 1989 (per DFF codebook)
-- Cardinality: ~400 rows (generated via CTE)
-- ------------------------------------------------------------
CREATE TABLE dbo.DimTime (
    time_key          INT IDENTITY(1,1) NOT NULL,
    week_id           INT               NOT NULL,
    week_start_date   DATE              NOT NULL,
    week_end_date     DATE              NOT NULL,
    [month]           INT               NOT NULL,
    month_name        VARCHAR(15)       NOT NULL,
    [quarter]         INT               NOT NULL,
    [year]            INT               NOT NULL,
    fiscal_year       INT               NOT NULL,
    is_holiday_week   BIT               NOT NULL DEFAULT 0,
    CONSTRAINT PK_DimTime PRIMARY KEY CLUSTERED (time_key),
    CONSTRAINT UQ_DimTime_weekid UNIQUE (week_id)
);
GO

-- ------------------------------------------------------------
-- DimStore  ~107 stores
-- Grain: One row per physical store location
-- Source: Cleaned from DEMO.csv staging table
-- Cardinality: ~107 rows
-- ------------------------------------------------------------
CREATE TABLE dbo.DimStore (
    store_key           INT IDENTITY(1,1) NOT NULL,
    store_id            INT               NOT NULL,
    store_name          VARCHAR(50)       NULL,
    city                VARCHAR(40)       NULL,
    zip_code            VARCHAR(10)       NULL,
    zone                INT               NULL,
    is_urban            BIT               NULL,
    weekly_volume       INT               NULL,
    avg_income          DECIMAL(10,2)     NULL,
    education_pct       DECIMAL(5,2)      NULL,
    poverty_pct         DECIMAL(5,2)      NULL,
    avg_household_size  DECIMAL(4,2)      NULL,
    ethnic_diversity    DECIMAL(5,2)      NULL,
    population_density  DECIMAL(10,2)     NULL,
    price_tier          VARCHAR(10)       NULL,
    age_under_9_pct     DECIMAL(5,2)      NULL,
    age_over_60_pct     DECIMAL(5,2)      NULL,
    working_women_pct   DECIMAL(5,2)      NULL,
    CONSTRAINT PK_DimStore PRIMARY KEY CLUSTERED (store_key),
    CONSTRAINT UQ_DimStore_storeid UNIQUE (store_id)
);
GO

-- ------------------------------------------------------------
-- DimProduct  ~3,112 UPCs (for 4 selected categories)
-- Grain: One row per unique UPC
-- Source: Cleaned from UPC staging tables
-- Cardinality: SDR(1746) + CSO(453) + TPA(608) + CRA(305) = 3,112
-- ------------------------------------------------------------
CREATE TABLE dbo.DimProduct (
    product_key     INT IDENTITY(1,1) NOT NULL,
    upc             BIGINT            NOT NULL,
    description     VARCHAR(100)      NULL,
```

```
    size            VARCHAR(30)       NULL,
    case_pack       INT               NULL,
    commodity_code  INT               NULL,
    item_number     BIGINT            NULL,
    CONSTRAINT PK_DimProduct PRIMARY KEY CLUSTERED (product_key)
);
GO

-- ================================
-- FACT TABLES
-- ================================

-- ------------------------------------------------------------
-- FactWeeklySales  Central fact table
-- Grain: One row per UPC × Store × Week
-- Source: Movement staging tables (filtered OK=1, PRICE>0)
-- Estimated: ~34.6M rows for 4 categories
-- ------------------------------------------------------------
CREATE TABLE dbo.FactWeeklySales (
    sales_fact_id     INT IDENTITY(1,1)  NOT NULL,
    product_key       INT                NOT NULL,
    store_key         INT                NOT NULL,
    time_key          INT                NOT NULL,
    category_key      INT                NOT NULL,
    promotion_key     INT                NOT NULL,
    units_sold        INT                NULL,
    unit_price        DECIMAL(8,2)       NULL,
    shelf_price       DECIMAL(8,2)       NULL,
    price_qty         INT                NULL,
    revenue           DECIMAL(12,2)      NULL,
    gross_profit      DECIMAL(10,2)      NULL,
    profit_margin_pct DECIMAL(5,2)       NULL,
    CONSTRAINT PK_FactWeeklySales PRIMARY KEY CLUSTERED (sales_fact_id),
    CONSTRAINT FK_Fact_Product   FOREIGN KEY (product_key)   REFERENCES dbo.DimProduct(product_key),
    CONSTRAINT FK_Fact_Store     FOREIGN KEY (store_key)     REFERENCES dbo.DimStore(store_key),
    CONSTRAINT FK_Fact_Time      FOREIGN KEY (time_key)      REFERENCES dbo.DimTime(time_key),
    CONSTRAINT FK_Fact_Category  FOREIGN KEY (category_key)  REFERENCES dbo.DimCategory(category_key),
    CONSTRAINT FK_Fact_Promotion FOREIGN KEY (promotion_key) REFERENCES dbo.DimPromotion(promotion_key)
);
GO


-- ------------------------------------------------------------
-- Verification: List all DW tables
-- ------------------------------------------------------------
SELECT TABLE_NAME, TABLE_TYPE
FROM INFORMATION_SCHEMA.TABLES
WHERE TABLE_SCHEMA = 'dbo'
ORDER BY TABLE_NAME;
GO

PRINT '=== All data warehouse tables created successfully ===';
GO
```

04_transform_staging.sql

```
-- ============================================================
-- Script 04: Transform and Clean Staging Data
-- DFF Data Warehouse Project  Report 3
-- Run in SSMS AFTER Package 1 (Extract) has loaded CSV data
-- into staging tables. These transformations prepare the data
-- for loading into the data mart.
-- ============================================================
USE [team1_staging_area];
GO
```

```
-- ================================
-- MOVEMENT TABLE TRANSFORMATIONS
-- ================================

-- T1: Add CATEGORY_CODE column to each movement staging table
-- This column does not exist in the CSV files; it is derived
-- from the filename (e.g., wsdr.csv → 'SDR')

ALTER TABLE dbo.stg_Movement_SDR ADD CATEGORY_CODE CHAR(3);
GO
UPDATE dbo.stg_Movement_SDR SET CATEGORY_CODE = 'SDR';
GO

ALTER TABLE dbo.stg_Movement_CSO ADD CATEGORY_CODE CHAR(3);
GO
UPDATE dbo.stg_Movement_CSO SET CATEGORY_CODE = 'CSO';
GO

ALTER TABLE dbo.stg_Movement_TPA ADD CATEGORY_CODE CHAR(3);
GO
UPDATE dbo.stg_Movement_TPA SET CATEGORY_CODE = 'TPA';
GO

ALTER TABLE dbo.stg_Movement_CRA ADD CATEGORY_CODE CHAR(3);
GO
UPDATE dbo.stg_Movement_CRA SET CATEGORY_CODE = 'CRA';
GO

PRINT 'T1 Complete: CATEGORY_CODE added to all movement tables.';
GO

-- T2: Replace NULL/blank SALE with 'N' (No Promotion)
-- The SALE column is NULL for ~90% of rows; we normalize to 'N'
-- so that promotion lookup joins work correctly.

UPDATE dbo.stg_Movement_SDR SET SALE = 'N' WHERE SALE IS NULL OR LTRIM(RTRIM(SALE)) = '';
UPDATE dbo.stg_Movement_CSO SET SALE = 'N' WHERE SALE IS NULL OR LTRIM(RTRIM(SALE)) = '';
UPDATE dbo.stg_Movement_TPA SET SALE = 'N' WHERE SALE IS NULL OR LTRIM(RTRIM(SALE)) = '';
UPDATE dbo.stg_Movement_CRA SET SALE = 'N' WHERE SALE IS NULL OR LTRIM(RTRIM(SALE)) = '';
GO

PRINT 'T2 Complete: NULL SALE values replaced with N.';
GO

-- Verification: Check SALE distribution after cleaning
SELECT 'SDR' AS Category, SALE, COUNT(*) AS cnt FROM dbo.stg_Movement_SDR GROUP BY SALE
UNION ALL
SELECT 'CSO', SALE, COUNT(*) FROM dbo.stg_Movement_CSO GROUP BY SALE
UNION ALL
SELECT 'TPA', SALE, COUNT(*) FROM dbo.stg_Movement_TPA GROUP BY SALE
UNION ALL
SELECT 'CRA', SALE, COUNT(*) FROM dbo.stg_Movement_CRA GROUP BY SALE
ORDER BY 1, 2;
GO

-- ================================
-- PRODUCT (UPC) TABLE TRANSFORMATIONS
-- ================================

-- T3: Add CATEGORY_CODE column to each UPC staging table

ALTER TABLE dbo.stg_Product_SDR ADD CATEGORY_CODE CHAR(3);
GO
UPDATE dbo.stg_Product_SDR SET CATEGORY_CODE = 'SDR';
GO
```

```
ALTER TABLE dbo.stg_Product_CSO ADD CATEGORY_CODE CHAR(3);
GO
UPDATE dbo.stg_Product_CSO SET CATEGORY_CODE = 'CSO';
GO

ALTER TABLE dbo.stg_Product_TPA ADD CATEGORY_CODE CHAR(3);
GO
UPDATE dbo.stg_Product_TPA SET CATEGORY_CODE = 'TPA';
GO

ALTER TABLE dbo.stg_Product_CRA ADD CATEGORY_CODE CHAR(3);
GO
UPDATE dbo.stg_Product_CRA SET CATEGORY_CODE = 'CRA';
GO

PRINT 'T3 Complete: CATEGORY_CODE added to all product tables.';
GO

-- T4: Clean product descriptions  strip leading # and ~ characters
UPDATE dbo.stg_Product_SDR SET DESCRIP = LTRIM(REPLACE(REPLACE(DESCRIP, '#', ''), '~', ''));
UPDATE dbo.stg_Product_CSO SET DESCRIP = LTRIM(REPLACE(REPLACE(DESCRIP, '#', ''), '~', ''));
UPDATE dbo.stg_Product_TPA SET DESCRIP = LTRIM(REPLACE(REPLACE(DESCRIP, '#', ''), '~', ''));
UPDATE dbo.stg_Product_CRA SET DESCRIP = LTRIM(REPLACE(REPLACE(DESCRIP, '#', ''), '~', ''));
GO

PRINT 'T4 Complete: Product descriptions cleaned.';
GO

-- T5: Create unified tmp_Product_All table (UNION of 4 category tables)
-- This temporary table is used to load DimProduct
IF OBJECT_ID('dbo.tmp_Product_All', 'U') IS NOT NULL
   DROP TABLE dbo.tmp_Product_All;
GO

SELECT COM_CODE, UPC, DESCRIP, SIZE, CASE_PACK, NITEM, CATEGORY_CODE
INTO dbo.tmp_Product_All
FROM (
   SELECT COM_CODE, UPC, DESCRIP, SIZE, CASE_PACK, NITEM, CATEGORY_CODE FROM dbo.stg_Product_SDR
   UNION ALL
   SELECT COM_CODE, UPC, DESCRIP, SIZE, CASE_PACK, NITEM, CATEGORY_CODE FROM dbo.stg_Product_CSO
   UNION ALL
   SELECT COM_CODE, UPC, DESCRIP, SIZE, CASE_PACK, NITEM, CATEGORY_CODE FROM dbo.stg_Product_TPA
   UNION ALL
   SELECT COM_CODE, UPC, DESCRIP, SIZE, CASE_PACK, NITEM, CATEGORY_CODE FROM dbo.stg_Product_CRA
) AS all_products;
GO

PRINT 'T5 Complete: tmp_Product_All created.';
SELECT 'tmp_Product_All' AS TableName, COUNT(*) AS RowCount FROM dbo.tmp_Product_All;
GO

-- ================================
-- STORE (DEMOGRAPHICS) TRANSFORMATIONS
-- ================================

-- T6: Remove the '.' placeholder row (first row of DEMO.csv)
-- The first row in DEMO.csv has STORE = '.' for all fields
DELETE FROM dbo.stg_Store WHERE STORE = '.' OR ISNUMERIC(STORE) = 0;
GO

-- T7: Replace NULL/blank NAME and CITY with 'UNKNOWN'
UPDATE dbo.stg_Store SET NAME = 'UNKNOWN' WHERE NAME IS NULL OR LTRIM(RTRIM(NAME)) = '';
UPDATE dbo.stg_Store SET CITY = 'UNKNOWN' WHERE CITY IS NULL OR LTRIM(RTRIM(CITY)) = '';
GO

PRINT 'T6-T7 Complete: Store data cleaned.';
```

```
SELECT COUNT(*) AS StoreCount FROM dbo.stg_Store;
GO

-- ================================
-- FINAL VERIFICATION
-- ================================
PRINT '========================================';
PRINT '  STAGING TRANSFORMATION COMPLETE';
PRINT '========================================';

SELECT 'stg_Movement_SDR' AS TableName, COUNT(*) AS RowCount FROM dbo.stg_Movement_SDR
UNION ALL SELECT 'stg_Movement_CSO', COUNT(*) FROM dbo.stg_Movement_CSO
UNION ALL SELECT 'stg_Movement_TPA', COUNT(*) FROM dbo.stg_Movement_TPA
UNION ALL SELECT 'stg_Movement_CRA', COUNT(*) FROM dbo.stg_Movement_CRA
UNION ALL SELECT 'stg_Product_SDR',  COUNT(*) FROM dbo.stg_Product_SDR
UNION ALL SELECT 'stg_Product_CSO',  COUNT(*) FROM dbo.stg_Product_CSO
UNION ALL SELECT 'stg_Product_TPA',  COUNT(*) FROM dbo.stg_Product_TPA
UNION ALL SELECT 'stg_Product_CRA',  COUNT(*) FROM dbo.stg_Product_CRA
UNION ALL SELECT 'stg_Store',        COUNT(*) FROM dbo.stg_Store
UNION ALL SELECT 'tmp_Product_All',  COUNT(*) FROM dbo.tmp_Product_All
ORDER BY TableName;
GO
```

05_load_dimensions.sql

```
-- ============================================================
-- Script 05: Load Dimension Tables
-- DFF Data Warehouse Project  Report 3
-- Run AFTER Script 04 (Transform). Loads dimensions BEFORE facts.
-- Order: DimCategory → DimPromotion → DimTime → DimStore → DimProduct
-- ============================================================
USE [team1_dw_area];
GO

-- ================================
-- 1. DimCategory (28 rows  hardcoded)
-- ================================
-- These are the 28 product categories from the DFF dataset.
-- Category codes are derived from the 3-letter filename abbreviations.
-- Department groupings are based on standard grocery store organization.

INSERT INTO dbo.DimCategory (category_code, category_name, department) VALUES
('ANA', 'Analgesics',         'Health & Beauty'),
('BAT', 'Bath Soap',          'Health & Beauty'),
('BER', 'Beer',               'Beverages'),
('BJC', 'Bottled Juices',     'Beverages'),
('CER', 'Cereals',            'Grocery'),
('CHE', 'Cheeses',            'Dairy'),
('CIG', 'Cigarettes',         'Tobacco'),
('COO', 'Cookies',            'Snacks'),
('CRA', 'Crackers',           'Snacks'),
('CSO', 'Canned Soup',        'Grocery'),
('DID', 'Dish Detergent',     'Household'),
('FEC', 'Front End Candies',  'Snacks'),
('FRD', 'Frozen Dinners',     'Frozen'),
('FRE', 'Frozen Entrees',     'Frozen'),
('FRJ', 'Frozen Juices',      'Beverages'),
('FSF', 'Fabric Softeners',   'Household'),
('GRO', 'Grooming Products',  'Health & Beauty'),
('LND', 'Laundry Detergents', 'Household'),
('OAT', 'Oatmeal',            'Grocery'),
('PTW', 'Paper Towels',       'Household'),
('SDR', 'Soft Drinks',        'Beverages'),
('SHA', 'Shampoos',           'Health & Beauty'),
('SNA', 'Snack Crackers',     'Snacks'),
('SOA', 'Soaps',              'Health & Beauty'),
```

```
('TBR', 'Toothbrushes',        'Health & Beauty'),
('TNA', 'Canned Tuna',         'Grocery'),
('TPA', 'Toothpaste',          'Health & Beauty'),
('TTI', 'Toilet Tissue',       'Household');
GO

PRINT '1/5 DimCategory loaded:';
SELECT COUNT(*) AS RowCount FROM dbo.DimCategory;
SELECT * FROM dbo.DimCategory ORDER BY category_key;
GO

-- ================================
-- 2. DimPromotion (4 rows  hardcoded)
-- ================================
-- Maps the SALE column values from Movement files to descriptive labels.
-- NULL/blank/'N' → 'No Promotion', B → 'Bonus Buy', C → 'Coupon', S → 'Sale/Discount'
-- deal_code for 'No Promotion' is stored as 'N' (transformed from NULL/blank during staging).

INSERT INTO dbo.DimPromotion (deal_code, deal_type, is_promoted) VALUES
('N',  'No Promotion', 0),
('B',  'Bonus Buy',    1),
('C',  'Coupon',       1),
('S',  'Sale/Discount', 1);
GO

PRINT '2/5 DimPromotion loaded:';
SELECT * FROM dbo.DimPromotion;
GO

-- ================================
-- 3. DimTime (~400 rows  generated via CTE)
-- ================================
-- DFF uses proprietary week IDs (WEEK column in Movement files).
-- Week 1 corresponds to September 14, 1989 per the DFF codebook.
-- Each subsequent week_id increments by 7 days.
-- We generate week_id 1 through 400 to cover the full dataset period.

;WITH WeekCTE AS (
   SELECT 1 AS week_id
   UNION ALL
   SELECT week_id + 1 FROM WeekCTE WHERE week_id < 400
)
INSERT INTO dbo.DimTime
   (week_id, week_start_date, week_end_date, [month], month_name,
    [quarter], [year], fiscal_year, is_holiday_week)
SELECT
   week_id,
   DATEADD(WEEK, week_id - 1, '1989-09-14')     AS week_start_date,
   DATEADD(DAY, 6, DATEADD(WEEK, week_id - 1, '1989-09-14')) AS week_end_date,
   MONTH(DATEADD(WEEK, week_id - 1, '1989-09-14'))           AS [month],
   DATENAME(MONTH, DATEADD(WEEK, week_id - 1, '1989-09-14')) AS month_name,
   DATEPART(QUARTER, DATEADD(WEEK, week_id - 1, '1989-09-14')) AS [quarter],
   YEAR(DATEADD(WEEK, week_id - 1, '1989-09-14'))           AS [year],
   YEAR(DATEADD(WEEK, week_id - 1, '1989-09-14'))           AS fiscal_year,
   -- Mark common US holiday weeks (Thanksgiving week ~47-48, Christmas ~52, July 4th ~27)
   CASE
      WHEN DATEPART(WEEK, DATEADD(WEEK, week_id - 1, '1989-09-14')) IN (47, 48, 52, 1, 27)
      THEN 1 ELSE 0
   END AS is_holiday_week
FROM WeekCTE
OPTION (MAXRECURSION 400);
GO

PRINT '3/5 DimTime loaded:';
SELECT COUNT(*) AS RowCount FROM dbo.DimTime;
SELECT TOP 10 * FROM dbo.DimTime ORDER BY time_key;
```

```
GO

-- ================================
-- 4. DimStore (~107 rows  from staging)
-- ================================
-- Loaded from cleaned stg_Store in the staging database.
-- Transformations applied during INSERT:
--   - CAST string columns to proper data types
--   - Derive price_tier from binary flags (PRICLOW/PRICMED/PRICHIGH)
--   - Replace NULL NAME/CITY with 'UNKNOWN' (already done in Script 04)

INSERT INTO dbo.DimStore
   (store_id, store_name, city, zip_code, zone, is_urban, weekly_volume,
    avg_income, education_pct, poverty_pct, avg_household_size,
    ethnic_diversity, population_density, price_tier,
    age_under_9_pct, age_over_60_pct, working_women_pct)
SELECT
   CAST(s.STORE AS INT)                    AS store_id,
   s.NAME                                  AS store_name,
   s.CITY                                  AS city,
   s.ZIP                                   AS zip_code,
   CAST(s.ZONE AS INT)                     AS zone,
   CAST(s.URBAN AS BIT)                    AS is_urban,
   CAST(s.WEEKVOL AS INT)                  AS weekly_volume,
   CAST(s.INCOME AS DECIMAL(10,2))         AS avg_income,
   CAST(s.EDUC AS DECIMAL(5,2))            AS education_pct,
   CAST(s.POVERTY AS DECIMAL(5,2))         AS poverty_pct,
   CAST(s.HSIZEAVG AS DECIMAL(4,2))        AS avg_household_size,
   CAST(s.ETHNIC AS DECIMAL(5,2))          AS ethnic_diversity,
   CAST(s.DENSITY AS DECIMAL(10,2))        AS population_density,
   CASE
      WHEN s.PRICLOW = '1'  THEN 'Low'
      WHEN s.PRICMED = '1'  THEN 'Medium'
      WHEN s.PRICHIGH = '1' THEN 'High'
      ELSE 'Unknown'
   END                                     AS price_tier,
   CAST(s.AGE9 AS DECIMAL(5,2))            AS age_under_9_pct,
   CAST(s.AGE60 AS DECIMAL(5,2))           AS age_over_60_pct,
   CAST(s.WORKWOM AS DECIMAL(5,2))         AS working_women_pct
FROM [team1_staging_area].dbo.stg_Store s
WHERE ISNUMERIC(s.STORE) = 1;
GO

PRINT '4/5 DimStore loaded:';
SELECT COUNT(*) AS RowCount FROM dbo.DimStore;
SELECT TOP 10 * FROM dbo.DimStore ORDER BY store_key;
GO

-- ================================
-- 5. DimProduct (~3,112 rows  from staging)
-- ================================
-- Loaded from tmp_Product_All in staging (UNION of 4 category tables).

INSERT INTO dbo.DimProduct
   (upc, description, size, case_pack, commodity_code, item_number)
SELECT
   p.UPC,
   p.DESCRIP                               AS description,
   p.SIZE                                  AS size,
   p.CASE_PACK                             AS case_pack,
   p.COM_CODE                              AS commodity_code,
   p.NITEM                                 AS item_number
FROM [team1_staging_area].dbo.tmp_Product_All p;
GO

PRINT '5/5 DimProduct loaded:';
```

```
SELECT COUNT(*) AS RowCount FROM dbo.DimProduct;
SELECT TOP 10 * FROM dbo.DimProduct ORDER BY product_key;
GO

-- ================================
-- DIMENSION LOAD VERIFICATION
-- ================================
PRINT '========================================';
PRINT '  ALL DIMENSION TABLES LOADED';
PRINT '========================================';

SELECT 'DimCategory'  AS TableName, COUNT(*) AS RowCount FROM dbo.DimCategory
UNION ALL SELECT 'DimPromotion', COUNT(*) FROM dbo.DimPromotion
UNION ALL SELECT 'DimTime',      COUNT(*) FROM dbo.DimTime
UNION ALL SELECT 'DimStore',     COUNT(*) FROM dbo.DimStore
UNION ALL SELECT 'DimProduct',   COUNT(*) FROM dbo.DimProduct
ORDER BY TableName;
GO
```

06_load_facts.sql

```
-- ============================================================
-- Script 06: Load Fact Tables
-- DFF Data Warehouse Project  Report 3
-- Run AFTER Script 05 (Load Dimensions).
-- Dimensions must be populated first because fact tables
-- reference dimension surrogate keys via foreign keys.
-- ============================================================
USE [team1_dw_area];
GO

-- ================================
-- 1. FactWeeklySales
-- ================================
-- Grain: One row per UPC × Store × Week
-- Sources: 4 movement staging tables (SDR, CSO, TPA, CRA)
-- Filters: OK = 1 (valid data), PRICE > 0 (avoid zero-price rows)
-- Derived columns:
--   unit_price       = PRICE / QTY
--   revenue          = MOVE × (PRICE / QTY)
--   profit_margin_pct = (PROFIT / revenue) × 100

PRINT 'Loading FactWeeklySales... (this may take several minutes for ~34M rows)';
GO

INSERT INTO dbo.FactWeeklySales
   (product_key, store_key, time_key, category_key, promotion_key,
    units_sold, unit_price, shelf_price, price_qty, revenue,
    gross_profit, profit_margin_pct)
SELECT
   dp.product_key,
   ds.store_key,
   dt.time_key,
   dc.category_key,
   dpr.promotion_key,
   sm.MOVE                                          AS units_sold,
   CASE WHEN sm.QTY > 0
        THEN CAST(sm.PRICE / sm.QTY AS DECIMAL(8,2))
        ELSE NULL
   END                                              AS unit_price,
   CAST(sm.PRICE AS DECIMAL(8,2))                   AS shelf_price,
   sm.QTY                                           AS price_qty,
   CASE WHEN sm.QTY > 0
        THEN CAST(sm.MOVE * (sm.PRICE / sm.QTY) AS DECIMAL(12,2))
        ELSE 0
   END                                              AS revenue,
```

```
    CAST(sm.PROFIT AS DECIMAL(10,2))                    AS gross_profit,
    CASE WHEN sm.MOVE > 0 AND sm.QTY > 0 AND sm.PRICE > 0
         THEN CAST(
             (sm.PROFIT / (sm.MOVE * (sm.PRICE / sm.QTY))) * 100
             AS DECIMAL(5,2))
         ELSE NULL
    END                                                 AS profit_margin_pct
FROM (
    -- UNION ALL of the 4 category movement staging tables
    SELECT UPC, STORE, WEEK, MOVE, QTY, PRICE, SALE, PROFIT, OK, CATEGORY_CODE
    FROM [team1_staging_area].dbo.stg_Movement_SDR
    WHERE OK = 1 AND PRICE > 0
    UNION ALL
    SELECT UPC, STORE, WEEK, MOVE, QTY, PRICE, SALE, PROFIT, OK, CATEGORY_CODE
    FROM [team1_staging_area].dbo.stg_Movement_CSO
    WHERE OK = 1 AND PRICE > 0
    UNION ALL
    SELECT UPC, STORE, WEEK, MOVE, QTY, PRICE, SALE, PROFIT, OK, CATEGORY_CODE
    FROM [team1_staging_area].dbo.stg_Movement_TPA
    WHERE OK = 1 AND PRICE > 0
    UNION ALL
    SELECT UPC, STORE, WEEK, MOVE, QTY, PRICE, SALE, PROFIT, OK, CATEGORY_CODE
    FROM [team1_staging_area].dbo.stg_Movement_CRA
    WHERE OK = 1 AND PRICE > 0
) sm
-- Join to dimension tables to get surrogate keys
INNER JOIN dbo.DimProduct dp
    ON sm.UPC = dp.upc
INNER JOIN dbo.DimStore ds
    ON sm.STORE = ds.store_id
INNER JOIN dbo.DimTime dt
    ON sm.WEEK = dt.week_id
INNER JOIN dbo.DimCategory dc
    ON sm.CATEGORY_CODE = dc.category_code
INNER JOIN dbo.DimPromotion dpr
    ON sm.SALE = dpr.deal_code;
GO

PRINT 'FactWeeklySales loaded:';
SELECT COUNT(*) AS RowCount FROM dbo.FactWeeklySales;
SELECT TOP 10 * FROM dbo.FactWeeklySales ORDER BY sales_fact_id;
GO

-- Row count by category (verification)
SELECT dc.category_name, COUNT(*) AS FactRows
FROM dbo.FactWeeklySales f
JOIN dbo.DimCategory dc ON f.category_key = dc.category_key
GROUP BY dc.category_name
ORDER BY FactRows DESC;
GO

-- ================================
-- FACT TABLE VERIFICATION
-- ================================
PRINT '========================================';
PRINT '  FACT TABLE LOADED';
PRINT '========================================';

SELECT 'FactWeeklySales' AS TableName, COUNT(*) AS RowCount FROM dbo.FactWeeklySales;
GO

-- Quick data quality check: any NULL keys?
SELECT 'NULL product_key'   AS Issue, COUNT(*) AS Cnt FROM dbo.FactWeeklySales WHERE product_key IS NULL
UNION ALL SELECT 'NULL store_key',   COUNT(*) FROM dbo.FactWeeklySales WHERE store_key IS NULL
UNION ALL SELECT 'NULL time_key',    COUNT(*) FROM dbo.FactWeeklySales WHERE time_key IS NULL
UNION ALL SELECT 'NULL category_key', COUNT(*) FROM dbo.FactWeeklySales WHERE category_key IS NULL
```

```
UNION ALL SELECT 'NULL promotion_key', COUNT(*) FROM dbo.FactWeeklySales WHERE promotion_key IS NULL;
GO
```

07_drop_temp_tables.sql

```
-- ============================================================
-- Script 07: Drop Temporary Tables from Staging Area
-- DFF Data Warehouse Project  Report 3
-- Run AFTER all dimension and fact tables have been loaded.
-- The assignment requires: "Once loading of each table is done,
-- remove all temp tables from the staging area."
-- ============================================================
USE [team1_staging_area];
GO

-- ================================
-- LIST OF TEMPORARY TABLES
-- ================================
-- The following temporary tables were created in the staging area
-- during the transformation phase (Script 04) and are no longer
-- needed after the data mart has been populated:
--
--   1. tmp_Product_All   UNION of 4 UPC category staging tables
--                        Used to load DimProduct
--
-- Note: The original staging tables (stg_Movement_*, stg_Product_*,
-- stg_Store) are also temporary but may be
-- retained for audit/verification purposes until the project is
-- fully validated.

PRINT 'Dropping temporary tables from staging area...';
GO

-- Drop tmp_Product_All
IF OBJECT_ID('dbo.tmp_Product_All', 'U') IS NOT NULL
BEGIN
   DROP TABLE dbo.tmp_Product_All;
   PRINT 'DROPPED: tmp_Product_All';
END
GO

-- ================================
-- VERIFICATION
-- ================================
-- Show remaining tables in staging area
SELECT TABLE_NAME
FROM INFORMATION_SCHEMA.TABLES
WHERE TABLE_SCHEMA = 'dbo' AND TABLE_TYPE = 'BASE TABLE'
ORDER BY TABLE_NAME;
GO

PRINT '========================================';
PRINT '  TEMPORARY TABLES REMOVED';
PRINT '  Remaining tables are base staging tables';
PRINT '  retained for audit/verification.';
PRINT '========================================';
GO
```

08_verify_bq_queries.sql

```
-- ============================================================
-- Script 08: Verify Business Questions Against Loaded DW
-- DFF Data Warehouse Project  Report 3
-- Run AFTER all tables are loaded to validate the ETL.
-- Each query corresponds to one of the 5 selected BQs.
-- ============================================================
```

```
USE [team1_dw_area];
GO

-- ================================
-- BQ2: Total weekly unit sales of Soft Drinks across all stores
-- Difficulty: Easy | OLAP Operation: Roll-up
-- ================================
PRINT '=== BQ2: Weekly Soft Drink Unit Sales ===';

SELECT
    dt.[year],
    dt.[month],
    dt.week_id,
    dt.week_start_date,
    SUM(f.units_sold)          AS total_units_sold,
    SUM(f.revenue)             AS total_revenue,
    COUNT(DISTINCT f.store_key) AS num_stores
FROM dbo.FactWeeklySales f
INNER JOIN dbo.DimTime dt      ON f.time_key     = dt.time_key
INNER JOIN dbo.DimCategory dc  ON f.category_key  = dc.category_key
WHERE dc.category_code = 'SDR'
GROUP BY dt.[year], dt.[month], dt.week_id, dt.week_start_date
ORDER BY dt.week_id;
GO

-- ================================
-- BQ3: Promotion vs Non-Promotion weeks  sales comparison
-- Difficulty: Easy | OLAP Operation: Dice (Slice by promotion status)
-- ================================
PRINT '=== BQ3: Promoted vs Non-Promoted Sales Volume ===';

SELECT
    dp.deal_type               AS promotion_status,
    dp.is_promoted,
    COUNT(*)                   AS num_records,
    SUM(f.units_sold)          AS total_units_sold,
    AVG(CAST(f.units_sold AS FLOAT)) AS avg_units_per_record,
    SUM(f.revenue)             AS total_revenue
FROM dbo.FactWeeklySales f
INNER JOIN dbo.DimPromotion dp ON f.promotion_key = dp.promotion_key
INNER JOIN dbo.DimCategory dc  ON f.category_key  = dc.category_key
WHERE dc.category_code = 'SDR'
GROUP BY dp.deal_type, dp.is_promoted
ORDER BY avg_units_per_record DESC;
GO

-- ================================
-- BQ4: Which promotion type has highest incremental sales lift in Canned Soup?
-- Difficulty: Medium | OLAP Operation: Dice (filter by category + promo type)
-- ================================
PRINT '=== BQ4: Promotion Lift by Deal Type  Canned Soup ===';

-- First get the baseline (no promotion average)
DECLARE @baseline_avg FLOAT;
SELECT @baseline_avg = AVG(CAST(f.units_sold AS FLOAT))
FROM dbo.FactWeeklySales f
INNER JOIN dbo.DimPromotion dp ON f.promotion_key = dp.promotion_key
INNER JOIN dbo.DimCategory dc  ON f.category_key  = dc.category_key
WHERE dc.category_code = 'CSO' AND dp.is_promoted = 0;

SELECT
    dp.deal_type,
    COUNT(*)                           AS num_promoted_records,
    AVG(CAST(f.units_sold AS FLOAT))   AS avg_units_promoted,
    @baseline_avg                      AS avg_units_baseline,
    AVG(CAST(f.units_sold AS FLOAT)) - @baseline_avg AS incremental_lift,
```

```
    CASE WHEN @baseline_avg > 0
         THEN CAST(AVG(CAST(f.units_sold AS FLOAT)) / @baseline_avg AS DECIMAL(5,2))
         ELSE NULL
    END                                 AS lift_multiplier
FROM dbo.FactWeeklySales f
INNER JOIN dbo.DimPromotion dp ON f.promotion_key = dp.promotion_key
INNER JOIN dbo.DimCategory dc  ON f.category_key  = dc.category_key
WHERE dc.category_code = 'CSO' AND dp.is_promoted = 1
GROUP BY dp.deal_type
ORDER BY incremental_lift DESC;
GO

-- ================================
-- BQ8: Store quartile tiers by Toothpaste revenue + demographics
-- Difficulty: Hard | OLAP Operation: NTILE + Drill-down
-- ================================
PRINT '=== BQ8: Store Revenue Quartiles  Toothpaste ===';

WITH store_revenue AS (
    SELECT
        f.store_key,
        SUM(f.revenue) AS total_revenue
    FROM dbo.FactWeeklySales f
    INNER JOIN dbo.DimCategory dc ON f.category_key = dc.category_key
    WHERE dc.category_code = 'TPA'
    GROUP BY f.store_key
),
store_quartiles AS (
    SELECT
        sr.store_key,
        sr.total_revenue,
        NTILE(4) OVER (ORDER BY sr.total_revenue) AS revenue_quartile
    FROM store_revenue sr
)
SELECT
    CASE
        WHEN sq.revenue_quartile = 1 THEN 'Bottom 25%'
        WHEN sq.revenue_quartile IN (2, 3) THEN 'Middle 50%'
        WHEN sq.revenue_quartile = 4 THEN 'Top 25%'
    END                                     AS tier,
    COUNT(*)                                AS store_count,
    CAST(AVG(sq.total_revenue) AS DECIMAL(12,2)) AS avg_revenue,
    CAST(AVG(ds.avg_income) AS DECIMAL(10,2))    AS avg_income,
    CAST(AVG(ds.population_density) AS DECIMAL(10,2)) AS avg_pop_density,
    SUM(ds.is_urban)                        AS urban_store_count,
    CAST(AVG(ds.education_pct) AS DECIMAL(5,2))  AS avg_education_pct,
    CAST(AVG(ds.poverty_pct) AS DECIMAL(5,2))    AS avg_poverty_pct
FROM store_quartiles sq
INNER JOIN dbo.DimStore ds ON sq.store_key = ds.store_key
GROUP BY
    CASE
        WHEN sq.revenue_quartile = 1 THEN 'Bottom 25%'
        WHEN sq.revenue_quartile IN (2, 3) THEN 'Middle 50%'
        WHEN sq.revenue_quartile = 4 THEN 'Top 25%'
    END
ORDER BY avg_revenue DESC;
GO

-- ================================
-- BQ9: Weekly top 10 Cracker products by unit sales with WoW change
-- Difficulty: Hard | OLAP Operation: RANK + LAG
-- ================================
PRINT '=== BQ9: Weekly Top 10 Cracker Products with WoW Change ===';

WITH weekly_product AS (
    SELECT
```

```
        dp.upc,
        dp.description,
        dt.week_id,
        dt.week_start_date,
        SUM(f.units_sold)                                   AS weekly_units,
        RANK() OVER (PARTITION BY dt.week_id
                     ORDER BY SUM(f.units_sold) DESC)       AS week_rank
    FROM dbo.FactWeeklySales f
    INNER JOIN dbo.DimProduct dp  ON f.product_key = dp.product_key
    INNER JOIN dbo.DimTime dt     ON f.time_key    = dt.time_key
    INNER JOIN dbo.DimCategory dc ON f.category_key = dc.category_key
    WHERE dc.category_code = 'CRA'
    GROUP BY dp.upc, dp.description, dt.week_id, dt.week_start_date
)
SELECT
    week_id,
    week_start_date,
    upc,
    description,
    weekly_units,
    week_rank,
    LAG(weekly_units) OVER (PARTITION BY upc ORDER BY week_id) AS prev_week_units,
    weekly_units - LAG(weekly_units) OVER (PARTITION BY upc ORDER BY week_id) AS wow_change
FROM weekly_product
WHERE week_rank <= 10
ORDER BY week_id, week_rank;
GO

-- ================================
-- OVERALL VALIDATION SUMMARY
-- ================================
PRINT '========================================';
PRINT '  BUSINESS QUESTION VERIFICATION COMPLETE';
PRINT '========================================';

-- Final row count summary
SELECT 'DimCategory'         AS TableName, COUNT(*) AS RowCount FROM dbo.DimCategory
UNION ALL SELECT 'DimPromotion',      COUNT(*) FROM dbo.DimPromotion
UNION ALL SELECT 'DimTime',           COUNT(*) FROM dbo.DimTime
UNION ALL SELECT 'DimStore',          COUNT(*) FROM dbo.DimStore
UNION ALL SELECT 'DimProduct',        COUNT(*) FROM dbo.DimProduct
UNION ALL SELECT 'FactWeeklySales',   COUNT(*) FROM dbo.FactWeeklySales
ORDER BY TableName;
GO
```

Appendix B: Mapping Tables (Excel)

Attached with the submission in canvas

Appendix C: SSIS Screenshot Index

#	Description
1	SSMS Object Explorer both databases visible
2	SSMS staging tables list
3	SSMS DW tables list
4	SSIS Package 1 Control Flow 9 Data Flow Tasks
5	SSIS Package 1 sample Data Flow (Flat File Source → OLE DB Dest)
6	Flat File Connection Manager encoding/delimiter settings
7	SSIS Package 1 execution all green checkmarks
8	SSMS stg_Movement_SDR COUNT = 0 (before load)
9	SSMS stg_Movement_SDR TOP 10 (after load)
10	SSMS all staging table row counts

11	SSIS Package 2 Control Flow Execute SQL Tasks
12	SSIS transformation SQL visible in task
13	SSIS Package 2 execution all green
14	SSMS CATEGORY CODE column visible in staging
15	SSIS Package 3 Control Flow
16	SSIS Package 3 -- Execute SQL Task for dimension loading
17	SSIS Package 3 execution all green
18	SSMS DimCategory (28 rows)
19	SSMS DimPromotion (4 rows)
20	SSMS DimTime TOP 10
21	SSMS DimStore TOP 10
22	SSMS DimProduct TOP 10
23	SSMS FactWeeklySales TOP 10 + COUNT
24	SSMS Object Explorer temp tables removed
25	SSMS BQ2 verification query results
26	SSMS BQ3 verification query results
27	SSMS BQ4 verification query results
28	SSMS BQ8 verification query results
29	SSMS BQ9 verification query results